$\mathbb{Z}$ is Diophantine in $\mathbb{Q}$

November 4, 2026

Abstract

We construct an integer-coefficient polynomial in 292 890 231 905 variables whose rational zero projection is exactly $\mathbb{Z}$. Its degree is less than $2 \uparrow\uparrow 7$, and the absolute value of every coefficient is less than $2 \uparrow\uparrow 8$.

Well-known consequences of $\mathbb{Z}$ being Diophantine in $\mathbb{Q}$ include a refutation of Mazur's conjecture that the topological closure of the rational points of a variety over $\mathbb{Q}$ has finitely many components, and existence of a universal rational Diophantine polynomial. Moreover, the set $\mathrm{HTP}(\mathbb{Q})$ of integer polynomials with a rational zero is many-one equivalent to the halting set $0'$.

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1 The main theorem and proof structure

This construction builds on OpenAI's preprint *Hilbert's tenth problem over the rational numbers* [34].

We use $\mathbb{N} = \{0, 1, 2, \ldots\}$. For a reduced rational number a/b, with $b > 0$, put $h(a/b) = \log \max(|a|, b)$. Define $2 \uparrow\uparrow 1 = 2$ and $2 \uparrow\uparrow (r+1) = 2^{2 \uparrow\uparrow r}$; every tower is associated to the right.

Theorem 1.1 (A polynomial definition of the integers)**.** *There exists one polynomial*

$$P \in \mathbb{Z}[X_0, X_1, \ldots, X_{292890231904}]$$

such that

$$\begin{aligned}
&\deg P < 2 \uparrow\uparrow 7,\\
&|[X^\alpha]P| < 2 \uparrow\uparrow 8 \qquad \textit{for every multi-index } \alpha,\\
&q \in \mathbb{Z} \iff \exists (w_1, \ldots, w_{292890231904}) \in \mathbb{Q}^{292890231904}, \quad P(q, w_1, \ldots, w_{292890231904}) = 0 \quad (q \in \mathbb{Q}).
\end{aligned}$$

All conclusions concern the same polynomial. Here $q \in \mathbb{Z}$ means the ordinary embedded integers. The ambient variable count does not assert that every variable occurs essentially.

The file `ordinary-integer-definition.json` specifies the polynomial P_*. The argument first represents integer indices by points on an elliptic curve over $\mathbb{Q}(\sqrt{2})$. A separate value relation identifies the ordinary rational value of an index. The point coordinates and the integer index have distinct roles throughout.

Theorem 1.1 refutes Mazur's conjecture that the closure of the rational points of a variety in its real locus has finitely many connected components. It also gives a universal rational Diophantine polynomial, with HTP($\mathbb{Q}$) many-one equivalent to $0'$. Section 13.1 proves these three consequences.

The final composition uses the following precise statements.

G — Canonical point codes form exactly the infinite cyclic group $2E(\mathbb{Q}(\sqrt{2})) = \mathbb{Z}P_0$. The membership polynomial types arbitrary inputs; raw addition describes the group law on canonical codes.

M — The actual guarded multiplication template has rational witnesses exactly for triples with indices a, b, ab.

H — The six-contact height theorem holds with the constants used by the precision computation and, separately, by multiplication.

PC — The explicit pole polynomial is soluble for every input in Theorem 5.2.

C — The actual five-input natural polynomial and its rational translation have the exact computation and substitution semantics of Section 10, including the power template.

B — The templates, fixed constructors and private witness substitutions specify the same saved expression P_*. The quantitative syntax calculation applies to that expression.

Lemma 1.2 (Assembly)**.** *Assume* $\mathsf{G}, \mathsf{M}, \mathsf{H}, \mathsf{PC}, \mathsf{C}, \mathsf{B}$. *Then* P_* *satisfies Theorem 1.1.*

Proof. Section 11 proves the exact ordinary-value relation using one shared shift and disjoint private witness blocks. The literal substitutions in B identify its existential polynomial with P_*. Section 12 proves the stated bounds for that same expression. □

Proof of Theorem 1.1. Section 3, including Proposition 3.1, proves G with its canonical membership and addition formulas. Theorem 4.1 proves H for each of the two sets of constants used below. Sections 7–9, with Appendices E–G, prove the restricted converse in Theorem 5.3. The simultaneous descent in Section 5 then proves PC. Using these results, Section 6 proves M on all rational input codes, including the finite false-triple guard.

Section 10 proves C for the fixed five-input natural polynomial and its rational model translation. Its witness variables remain natural before that translation. Sections 2 and 12 and Appendix C prove B, including the fixed scalars and private witness substitutions. All six statements therefore hold, and Lemma 1.2 applies to the specified P_*. □

2 A finite specification of the polynomial

The saved specification is a 23 180 023-byte JSON expression with SHA256

`a86790bbf1095d6b1cea9c66067ad8432dd35adc320254eecde21f9f98574714.`

It contains 23 library templates and seven named recipes. The root recipe is `integers`; its only parameter is the rational variable at index zero. Every other variable is existentially quantified over $\mathbb{Q}$. Sources S01–S03 fix the construction and its semantics. Stable paths and hashes are in the companion `SOURCE_MAP.json`.

2.1 Literal operations and witness scope

A scalar node is a variable, integer constant, sum, product, or a call to an earlier template. A call substitutes its parameter expressions and a private consecutive witness block into the callee's polynomial. Shared nodes represent repeated subexpressions, not new equations. A recipe sums squares of constraint polynomials or multiplies branch polynomials. Over $\mathbb{Q}$,

$$\sum_i f_i^2 = 0 \iff \bigwedge_i f_i = 0, \qquad \prod_i f_i = 0 \iff \bigvee_i f_i = 0.$$

Private witness blocks can be moved to the front because the rational domain is nonempty. A whole inactive product-call block may receive arbitrary values. This does not discard fixed equations inside an active callee. A rational constant p/q, $q \neq 0$, is represented by a fresh c and the globally conjoined equation $qc = p$. In particular the cached coordinates of P_0 satisfy

$$8g_0 = 1641, \qquad 4g_1 = -15465, \qquad 32g_2 = 87483.$$

These equations remain active outside internal disjunctions in the multiplication, signed-encoding and exception templates. Inverses are equations $uv = 1$, never partial evaluation operations.

Strict positivity has the polynomial expression

$$s > 0 \iff \exists z_1, z_2, z_3, z_4 \in \mathbb{Q}, \quad s \sum_i z_i^2 = 1.$$

Indeed, if $1/s = a/b > 0$, express ab as four integer squares and divide their coordinates by b. The reverse direction follows from the order on $\mathbb{Q}$. Absolute value has the exact expression

$$a = |x| \iff \exists z_1, \ldots, z_4 \in \mathbb{Q}, \quad a = \sum_i z_i^2, \quad a^2 = x^2.$$

2.2 Fixed finite constructors

Three node types abbreviate finite polynomials. A `fixed_integer` inserts a fixed integer expression; `psi_component` inserts one coordinate of a division polynomial; `finite_guard_product` inserts a product over a fixed box of integers. Their bounds never depend on the free rational input.

For example, on $y^2 = x^3 + Ax$, write $\psi_n = f_n$ for odd n and $\psi_n = 2yf_n$ for even n. The bases are

$$f_0 = 0, \qquad f_1 = f_2 = 1,$$
$$f_3 = 3x^4 + 6Ax^2 - A^2, \qquad f_4 = 2(x^6 + 5Ax^4 - 5A^2x^2 - A^3).$$

For $m \geq 3$, $f_{2m} = f_m(f_{m+2}f_{m-1}^2 - f_{m-2}f_{m+1}^2)$. For $m \geq 2$, put $L = f_{m+2}f_m^3$, $R = f_{m-1}f_{m+1}^3$, and $D_0 = 16(x^3 + Ax)^2$. Then $f_{2m+1} = D_0L - R$ for even m and $L - D_0R$ for odd m. Negative indices satisfy $\psi_{-n} = -\psi_n$. This is a terminating recursion.

The fixed prime catalogue consists of the first 22 primes above 324 in each unit residue class modulo $W = 4\prod_{p\leq 324} p$. Dirichlet's theorem ensures termination. Put $N_s = (\prod_{p\leq 324} p)H_{\mathrm{cat}}$, where H_{cat} is the catalogue product. Assign inverse abscissa zero at the identity. For the normalized representation $1/x([m]P_0) = (a + bs)/c$ with $a, b, c \in \mathbb{Z}$, $c > 0$, and $\gcd(a, b, c) = 1$, define M_0 to be the least $m \geq 1$ such that $N_s^4 \mid a, b$ and $\gcd(c, N_s) = 1$ (S31). This is a fixed integer, with a successful index bounded in Section 12. The fixed point coordinates in the guard have nonzero denominators by the infinite order of P_0. Thus finite expansion gives one integer polynomial without requiring a monomial expansion. Its arithmetic zero set remains a separate issue. In particular, bounding the integer T does not prove that it excludes all false multiplication triples.

3 The elliptic model of integer indices

Let

$$K_0 = \mathbb{Q}(s), \quad s^2 = 2, \quad d = 75 - 53s, \quad E : y^2 = x^3 - d^2x,$$

and fix

$$P_0 = \left(\frac{1641 - 1160s}{8}, \frac{-123720 + 87483s}{32}\right).$$

The identity has code $(1, 0, 0, 0, 0)$. An affine point has code $(0, x_0, x_1, y_0, y_1)$ with $x = x_0 + x_1s$, $y = y_0 + y_1s$. Linear independence of $1, s$ makes this unique. Write $\iota(n)$ for the code of $[n]P_0$.

Proposition 3.1 (Generator certificate)**.** *The point P_0 has infinite order and $2E(K_0) = \mathbb{Z}P_0$.*

Proof. We give the argument of S05–S07, making its finite certificate explicit. The ring of integers $\mathbb{Z}[s]$ has class number one because its Minkowski bound is $\sqrt{2} < 2$. It has four unit squareclasses. The sign vectors of $-1, 1 + s$ are independent, so every totally positive unit is a square.

The x-squareclass map in the two-isogeny descent on E has image contained in the eight classes generated by $-1, 1 + s, d$. Away from (d), odd $v(x)$ makes the minimum of $3v(x)$ and $v(x) + 2v(d)$ odd, impossible for y^2. The torsion points give $1, -1, d, -d$. Put $\epsilon = 1 + s$, $\delta = 3 - s$. The change $(x, y) \mapsto (\epsilon^4x, \epsilon^6y)$ takes E to $C : y^2 = x^3 - \delta^2x$, and $R = (1 + s, 2s)$ provides the remaining unit class. Thus the first image has order eight.

On $E' : y^2 = x^3 + 4d^2x$, a nonzero x is positive in both real embeddings. Its valuations are even. At (s) the two terms have valuations $3m$ and $m + 4$, with equality only at $m = 2$. At (d) the only odd case would require $u^2 = -4$ in $\mathbb{F}_7$. Thus the second image is trivial. The exact two-isogeny sequence gives $[E(K_0) : 2E(K_0)] = 8$; its possible extra kernel term vanishes since a nonkernel rational two-torsion point maps to the dual kernel. Mordell–Weil gives rank one.

The good reductions at $(17, s-6)$ and $(23, s-5)$ have 20 and 24 points. Prime-to-residue-characteristic torsion injects, so the torsion group has order at most four, hence is exactly the displayed full two-torsion. The x-class of R has opposite real signs, whereas all torsion classes have equal signs. Its free index is therefore odd.

The duplication forms on C are

$$F_2(a,b) = (a^2+\delta^2 b^2)^2, \qquad G_2(a,b) = 4ab(a^2-\delta^2 b^2).$$

At either real embedding their maximum lies between $\max(|a|,|b|)^4$ and $676\max(|a|,|b|)^4$. For primitive a, b their common ideal has valuations at most six at (s) and four at (δ), and none elsewhere. For the latter bound use that -1 is nonsquare modulo seven. For the former, a local unit is $A+Bs$ with A odd, and $v_s((A+Bs)^2+1)$ is two or three according to the parity of B. Hence

$$4h_x(T) - \log 392 \le h_x(2T) \le 4h_x(T) + \log 676.$$

With $2\widehat{h}(T) = \lim_m 4^{-m} h_x(2^m T)$ this gives $h_x(T) \le 2\widehat{h}(T) + \log 392/3$. Direct doubling gives $x(2R) = (57-28s)/8$, with height $\log 41$. If $R = mS + T$ for torsion T and $|m| \ge 3$, it follows that

$$h_x(S) \le \frac{\log(41^3 \cdot 676)}{108} + \frac{\log 392}{3} < \log 9.$$

The strict inequality is the exact integer comparison $41^3 \cdot 676 \cdot 392^{36} < 9^{108}$.

A nonrational $x \in K_0$ in this height range has primitive polynomial aX^2+bX+c with $1 \le a \le 81$, $|b| \le 162$, $|c| \le 81$, and discriminant $2k^2$, $1 \le k \le 162$. Both roots are included. Rational possibilities have coprime numerator of absolute value at most nine and denominator from one to nine. The exact finite square test of $x^3 - \delta^2 x$ gives precisely

$$0, \quad \delta, \quad -\delta, \quad 1+s, \quad 23-17s, \quad 2+4s, \quad (5-4s)/2.$$

The certificate covers 7,004 primitive quadratics, their 14,008 roots, and 111 rational candidates. To test whether $u+vs$ is a square, take both signs of a rational square root t of u^2-2v^2, test $y_0^2 = (u+t)/2$, then set $y_1 = v/(2y_0)$ if $y_0 \ne 0$. The $y_0 = 0$ case requires $v = 0$ and $y_1^2 = u/2$. Substitution verifies each result. This is a necessary and sufficient test, including zero. The last four listed coordinates are those of $R+T$ for $T \in C[2]$. Every surviving nontorsion point therefore has height $\widehat{h}(R)$, contradicting $\widehat{h}(R) = m^2\widehat{h}(S)$. This finite computation is an explicit part of the proof, with a recorded independent replay; it is not replaced by a rank oracle.

Thus $2C(K_0) = \mathbb{Z}(2R)$. The inverse coordinate change sends $2R$ to P_0, as asserted. □

3.1 Polynomial membership and addition

Apart from the identity branch, membership in $2E(K_0)$ is expressed by choosing $X, Y, H \in K_0$, $u \in \mathbb{Q}$, and imposing

$$Y^2 = X^3 - d^2X, \quad u\,\mathrm{N}(Y) = 1, \quad 2YH = 3X^2 - d^2, \quad x = H^2 - 2X, \quad y = H(X-x) - Y.$$

Every nonidentity double has a preimage with nonzero ordinate. Each field equation is two rational equations. Membership is an all-input statement. Raw addition has the narrower exact contract

$$\mathrm{Add}_{\mathrm{raw}}(\iota(a), \iota(b), R) \quad \Longleftrightarrow \quad R = \iota(a+b).$$

Its branches cover both identities, opposite points, distinct abscissas and doubling with nonzero ordinate. Every later use supplies canonical input codes. Multiplication first imposes membership of all three roots and is sound on arbitrary rational tuples. In the natural predicate, four initially untyped tuples enter multiplication calls, so are typed before their squared values are added. Lagrange's theorem then gives exactly $\iota(\mathbb{N})$. This uses multiplication completeness; the generator proposition alone does not supply the natural predicate.

4 The six-contact height estimate

Set

$$\mathcal{C} = \{0, -15, -20, -24, -30, -60\}, \qquad F(u) = u(u+15)(u+20)(u+24)(u+30)(u+60),$$

and $G(u) = 7u^2 + 300u + 3600$. For a finite prime set S containing all primes at most 324 and $u \notin \mathcal{C}$, define

$$M_S(u) = \prod_{\substack{p \notin S:\ v_p(u-c)>0 \text{ odd} \\ \text{for some } c \in \mathcal{C}}} p.$$

Negative valuations of u do not enter the product. Put

$$\begin{aligned} F_0 &= 324!, \qquad D = F_0\, 3^{256F_0^4}, \qquad G_0 = 116064000000, \\ E_0 &= 10(DG_0)^2, \qquad Q_R = R^{RE_0}, \\ B_{\mathrm{cm}} &= D^2\, 3^{21000D^2} + 1, \\ C_{\mathrm{cm}} &= 2B_{\mathrm{cm}} + 4096(1028 + 5D + 84000D^2), \\ H_6(S;R) &= 2 \cdot 10^{13} \max(1, C_{\mathrm{cm}}, Q_R^{17}), \\ R &\geq \max(D, \max S, 10^{12}). \end{aligned}$$

When the second argument is omitted, we use $H_6(S) = H_6(S; \max(D, \max S, 10^{12}))$. The proof follows the corrected geometric route in S09, with the constants of S08. It does not use the older alternative modularity route.

Theorem 4.1 (Six-contact estimate)**.** *For every such S, R and every $u \in \mathbb{Q} \setminus \mathcal{C}$,*

$$h(u) \leq H_6(S;R) M_S(u)^{24}.$$

More precisely the external arithmetic-geometric inputs below give $h(u) \leq C(S) M_S(u)^{24}$, where

$$C(S) = 10^{13} \max((8D)^{288D}, 6^{100}) \Big(\prod_{p \in S} p \Big)^{24}.$$

Proof. Take $B = (2, -3)$, the maximal order $\mathcal{O} = \mathbb{Z}[i, (1+j)/2]$, and $\mu = 2j + ij$. With its positive sign choice, the form $\mathrm{Trd}(\mu^{-1} x \overline{y})$ has matrix

$$\begin{pmatrix} 0 & 0 & -1 & 1 \\ 0 & 0 & 1 & -2 \\ 1 & -1 & 0 & 0 \\ -1 & 2 & 0 & 0 \end{pmatrix}$$

and determinant one. This is the principal-polarization datum whose normalizer quotient is $\mathbf{P}^1_j$, with branch values $0, \infty, -16/27$ and indices $4, 6, 2$ [1, 3]. The full $\mathcal{O}$-linear level-three cover Y_3 has total degree at most $4|\mathcal{O}/3\mathcal{O}| = 324$, retaining all determinant components. Here the normalizer quotient contributes four. Central ideles act trivially after multiplying by a rational scalar and choosing its sign to give residue one modulo three.

For an integral model apply [2, Theorem 2.1, Corollary 2.3] first at $V_1(5)$. This level has full determinant image, is maximal at the division primes, and uses $5 \geq 4$ prime to the discriminant six. Add full five-marking and full $\mathcal{O}$-linear three-marking, then forget five-marking by a free

finite quotient. The three-marking sheaf is nonempty because an $\mathcal{O}_3$-stable lattice in the rank-one B_3-module is a principal fractional left ideal. Freeness follows because a polarized automorphism has finite order, while a finite-order element congruent to one modulo three on T_3 is the identity. The universal abelian surface descends. The auxiliary level contributes no factor to the final degree.

Every component has genus at least two, by its negative orbifold Euler characteristic, so the automorphisms extend to its smooth proper model. At $p > 324$ the action is tame. A stabilizer is cyclic and its tangent character is faithful; a nontrivial fixed-point equation has invertible linear term. Unique lifting prevents branch collisions. The three ordered branch sections fix the integral j-coordinate, so no further denominator prime is discarded.

Pull back by $J(u) = -57600F(u)^2/G(u)^6$. If

$$H(u) = (u^2 - 60u - 1800)(13u^2 + 480u + 3600)(11u^2 + 600u + 7200),$$

direct calculation gives $G^6 - 97200F^2 = H^2$. The pullback indices are $2, 6, 2$. Tame normalization replaces e by $e/\gcd(e, a)$ and leaves $2, 1, 1$. The discriminants and resultants have prime support at most 13, so this persists at $p > 324$. The normalized cover of $\mathbf{P}^1_u$ has degree at most 324, not $12 \cdot 324$, and is branched only at the six contacts. Locally it has equation $u - c = z^2$ after unramified extension. Positive even contact valuation is unramified; negative valuation specializes to the unbranched point at infinity. This is exactly the support specified by $M_S(u)$.

The splitting field of a fiber has degree at most F_0. The lattice of ordered homomorphisms among its conjugate abelian surfaces has rank at most $16F_0^2$. A finite subgroup of $\mathrm{GL}_r(\mathbb{Z})$ injects modulo three. Adjoining its field of definition gives a normal field L of degree at most D, ramified only in $S \cup \operatorname{supp} M_S(u)$. It defines all endomorphisms and all ordered conjugacy homomorphisms.

The geometric endomorphism algebra is either B or $M_2(k)$ for an imaginary quadratic field k, by taking the right centralizer of the complex structure. In the first case Skolem–Noether corrects conjugacy isomorphisms to B-linear isogenies over L. These are in particular compatible $\mathbb{Q}(\sqrt{2})$-linear isogenies. Thus the normal-field, endomorphism and compatible-isogeny hypotheses, including condition $(*)$, of [4, Eq. (7.1), Section 7.1.4, Proposition 7.13] hold. With $P_S = \prod_{p \in S} p$ and $M = M_S(u)$, this gives

$$h_F(A) \le (8D)^{288D}(P_S M)^{24}$$

in the non-CM case.

In the CM case retain the degree bound $d_L = [L : \mathbb{Q}] \le D$. The maximal-order reduction of [4, Section 7.3.3, Eq. (7.17)] constructs, over the same field, an isogenous product $A' = \prod A_i^{e_i}$ with maximal CM endomorphism rings. All geometric endomorphisms were already defined over L, so the reduction needs no further field extension. Its intermediate CM-type estimate gives $h_F(A_i) \le 6^{96}(P_S M)^{10}$, since the CM fields are unramified outside the discriminant support of L. As $\sum e_i \le 2$, put $H_0 = 2 \cdot 6^{96}(P_S M)^{10}$; then $h_F(A') \le H_0$. The isogeny estimate [4, Eq. (7.13)], in dimension two, is

$$h_F(A) \le h_F(A') + \tfrac{1}{2}\log \kappa(A'), \qquad \kappa(A') = \big(14^{32} d_L \max(h_F(A'), \log d_L, 1)\big)^{16}.$$

Since $\log 14 < 3$, $D \ge 2$, and $H_0 \ge 1$, it follows that

$$h_F(A) \le 9H_0 + 16D + 768 \le (8D)^{288D}(P_S M)^{24}.$$

For the last inequality use $18 \cdot 6^{96} < 16^{64}$ and $16D + 768 \le (8D)^4$, then absorb their sum in $(8D)^{65}$. This argument bounds the original stable height while retaining the field degree. It does not use the degree-free final conclusion of Proposition 7.14, nor the historical alternative CM bound in S08. The prescribed integer C_{cm} remains unchanged. Its former degree-free interpretation is unnecessary:

the corrected estimate is absorbed by Q_R^{17}. Later comparisons between those integers are numerical comparisons, not uses of the discarded height claim.

The compatible theta normalizations in [3, 5] give

$$j = 3779136\frac{(\prod z_m)^4}{(\sum z_m^8)^5}, \qquad h(j) \le 40h([z_m]) + 22,$$

with ten even theta nulls; the other six vanish. The comparison in [6, Corollary 1.3(2)] yields $h([z_m]) \le 10000 \cdot 2^{24} \max(1, h_F(A))$. The change to the Deligne metric adds $\log \pi$ in dimension two and has been included in this constant. A finite extension for a theta structure changes neither absolute height. The coefficient norms of $-57600F^2, G^6$ are below 4000^6. The rational-root bound gives $h(u) < h(j) + 74$, after removing any initial power of X or dropped leading term. Since $40 \cdot 10000 \cdot 2^{24} + 96 < 10^{13}$, this proves the bound with $C(S)$.

Finally $P_S \le R^R$, $8D \le R^2$, and $6^{100} \le R^{576D}$. Thus

$$C(S) \le 10^{13} R^{576D+24R} \le 10^{13} R^{600R} \le 10^{13} Q_R^{17} \le H_6(S;R)/2.$$

This also proves the assertion for every permitted larger R. □

The final value argument uses $S_{\mathrm{ht}} = \{p : p \le 324\}$. Multiplication uses $S_{\mathrm{mul}} = S_{\mathrm{ht}} \cup \mathcal{H} \cup \mathrm{supp}(c_*)$, where $\mathcal{H}$ is the fixed catalogue. A common numerical R may bound both sets. It does not identify them.

5 The pole predicate and its converse input

Let $S_0 = \{2, 3, 5, 7, 13\}$. With input B and eleven witnesses $b_0, \dots, b_5, e, z, r_1, r_2, w$, define

$$\begin{aligned} m &= (1 - 2b_0)(1 + b_1)(1 + 2b_2)(1 + 4b_3)(1 + 6b_4)(1 + 12b_5), \\ A &= mB, \qquad H = A(A - B^2), \\ T_1 &= A(A - e)(A + 3e), \qquad T_2 = A(Az^2 - eH)(Az^2 + 3eH), \\ Q_{\mathrm{pole}} &= B\Big(\sum_{i=0}^{5}(b_i^2 - b_i)^2 + (eHzw - 1)^2 \\ &\qquad + (r_1^2 - T_1)^2 + (r_2^2 - T_2)^2\Big). \end{aligned}$$

Write $\Phi(B)$ for rational solubility. For $B \ne 0$, m is a signed divisor of 2730, and the equations give points $(A, r_1) \in E_e$ and $(Az^2, zr_2) \in E_{eH}$ with $eHz \ne 0$, where $E_\ell : y^2 = x(x - \ell)(x + 3\ell)$. The polynomial has total degree 75.

Lemma 5.1 (One-sided obstruction)**.** *If $B \ne 0$ and $\Phi(B)$, then $v_p(B)$ is not negative and odd for any $p \notin S_0$.*

Proof. If $k = v_p(B) < 0$ were odd, then $v_p(A) = k$, $v_p(H) = 3k$. Exactly one of e, eH has even valuation. Square scaling makes its parameter a unit, preserving the parity of the abscissa's valuation. For a unit parameter the cubic has valuation $v_p(x)$ if $v_p(x) > 0$ and $3v_p(x)$ if $v_p(x) < 0$. Its being a square forces even valuation. But both points have abscissa in the squareclass A, a contradiction. □

Theorem 5.2 (Pole completeness, PC)**.** *For every rational B with square reduced denominator, $0 < B < 1/2$ and $v_p(B) \ge 4$ for every $p \in S_0$ imply $\Phi(B)$.*

Theorem 5.3 (Restricted converse, RC)**.** *For every positive squarefree* $\ell \equiv 11 \pmod{12}$*,*

$$\operatorname{corank}_{\mathbb{Z}_2} \operatorname{Sel}_{2^\infty}(E_\ell/\mathbb{Q}) = 1 \quad \Longrightarrow \quad \operatorname{ord}_{s=1} L(E_\ell, s) = 1.$$

Here $c_4 = 208\ell^2$, $\Delta = 2304\ell^6$, and $j = 35152/9$. The nonintegral rational j shows these curves are non-CM. They are different from the CM generator curve. Theorem 5.3 is proved in Sections 7–9, with its coefficient and normalization arguments in Appendices E–G. We now prove Theorem 5.2 from this restricted converse.

5.1 The exact descent reduction

The signed divisors of 2730 realize all real, 2-adic and 3-adic squareclass signatures. After $-1, 2, 3$ fix sign and valuation parities, the remaining unit vectors of $5, 7, 13$ are $(0,1,1), (1,0,0), (0,1,0)$ and form a basis. Choose positive m making $A = mB$ square at two and a nonsquare unit squareclass at three. Let β, θ be the positive squarefree representatives of A, H. Then $\beta \equiv 17 \pmod{24}$, $\theta \equiv 1 \pmod{24}$, with disjoint supports, $\beta > 1$, and $\theta = 1$ permitted. Each is a local square at the support of the other. At a prime of β, the relevant odd valuation of B is positive, so $1 - B/m$ is a square unit. At a prime of θ, $v(B)$ must be zero, since otherwise $v(H)$ is even; there $B/m \equiv 1$, so A is a square.

The simultaneous construction in S11–S12 gives $n, n\theta$ positive squarefree and congruent to eleven modulo twelve, both with two-Selmer dimension three and with extra Kummer class $(\beta, 1)$. Its matrix conditions can be stated explicitly. For support primes q_i, set $s_i = [-1/q_i]$, $\lambda_i = [3/q_i]$, and

$$(Kv)_i = \sum_{j \neq i} [q_j/q_i](v_j + v_i).$$

One arranges $\ker K = \langle \mathbf{1}, \mathbf{1}_{\operatorname{supp}\beta} \rangle$ and $\lambda \notin \operatorname{im} K$ on both supports. Add $d-1$ prime slots of type $(s, \lambda) = (0, 0)$ and one slot q_0 of type $(1, 1)$, where $d = \omega(\beta)$. The cross-block has columns $e_i + e_d$. After the rank-one correction $K^\sharp = K + \lambda a^t$, deleting q_0 leaves

$$\begin{pmatrix} D_1 & C_1 \\ C_1^t & 0 \end{pmatrix}, \qquad \mathbf{1}^t D_1 \mathbf{1} = 1.$$

This is nonsingular: $C_1^t x = 0$ makes x constant, and the first equation forces that constant and then y to vanish. Adding the support of θ preserves this block. Free q_0-symbols toggle the new diagonal entries; the determinant's full-product coefficient is one. Thus a single Boolean assignment works on both supports. Reciprocity, CRT and Dirichlet realize it by distinct auxiliary primes.

In Kummer coordinates $(x + 3\ell, x - \ell)$, a positive-second-coordinate Selmer representative is $(3^\delta \prod q_i^{u_i}, \prod q_i^{v_i})$. The necessary local equations are

$$Ku + \operatorname{diag}(\lambda)v = \delta\lambda, \quad (K + \operatorname{diag}(s))v = 0, \quad (\lambda + s)^t v = 0.$$

The final equation is the condition at three. The matrix properties force $v = 0$, $\delta = 0$, $u \in \ker K$, so there are at most eight classes in total. The extra class is actually local at every place, including two, and independent of the two torsion classes, giving dimension three. This sufficiency check for the particular class is essential; arbitrary parity conditions at two would not suffice.

Lemma 5.4 (Dimension three gives corank one)**.** *For an elliptic curve with full rational two-torsion,* $\dim_{\mathbb{F}_2} \operatorname{Sel}_2 = 3$ *implies* $\operatorname{corank}_{\mathbb{Z}_2} \operatorname{Sel}_{2^\infty} = 1$.

Proof. The finite Kummer sequence gives $r + \dim Ш[2] = 1$. Write $Ш[2^\infty] = (\mathbb{Q}_2/\mathbb{Z}_2)^t \oplus F$ with F finite. The Cassels–Tate pairing on the quotient by the maximal divisible subgroup is nondegenerate and alternating for an elliptic curve [7]. Its finite primary cyclic factors are paired, so $\dim F[2]$ is even. Thus $F = 0$ and $r + t = 1$. The restriction of the pairing to $F[2]$ need not be nondegenerate and is not used. □

Applying Theorem 5.3 to both n and $n\theta$, the forward rank-one theorem [17, 18] gives rank one and finite Sha, hence $Ш[2] = 0$. The Kummer class is then represented by actual rational points. Their abscissas have squareclass β, by the cubic equation. Write $H = \theta c^2$ and rescale the second point to E_{nH}. If the first abscissa is Aq^2, divide both parameters by q^2 and both coordinates by q^2, q^3. The abscissas become A, Az^2, giving precisely the eleven witnesses, with $w = (eHz)^{-1}$. If $\theta = 1$, one can reuse a point on the same curve. This proves Theorem 5.2, with both restricted-converse applications in their stated domain.

6 Multiplication and the finite false-triple guard

The actual multiplication template first forces all three input codes to be members of $2E(K_0)$. These membership calls are conjuncts inside the height certificate; they are not assumed caller preconditions. Thus even untyped rational inputs, if accepted, have unique indices. The template has 35,134 rational witnesses. Its construction is S03 and its mathematical specification is S13.

For indices n_1, n_2, n_3, each test group chooses $H = [h]P_0$, $U_i = [z_i]P_0$, and sets $V_0 = H - [c_*]P_0$, $V_i = U_i - [c_* n_i]P_0$, where $c_* = 4M_0$ is fixed. Use

$$\begin{aligned} L(z) = {} & 16575z - 6630d^2z^5 + 11050d^4z^9 - 25500d^6z^{13} \\ & + 68250d^8z^{17} - 198900d^{10}z^{21} + 612612d^{12}z^{25}. \end{aligned}$$

This is 16575 times the degree-25 truncation of the normalized formal logarithm. The twelve tests are four inverse abscissas at H, U_i, their four conjugate counterparts at V_0, V_i, three differences of the corresponding L-ratios, and the product of the first two unprimed ratios minus the third. Identity inverse abscissas are zero; all denominators are guarded.

For a slope $t \notin \mathcal{C} \cup \{1\}$, put

$$R_0 = \eta F(t)/(t-1)^8, \quad \eta^2 = 1, \quad a = 1 - R_0 q^2 \neq 0, \quad R = \ell_0^2 a^2 R_0.$$

Denote the twelve unscaled test values in group g by D_{gj}. Their normalized values are

$$C_{gj} = \frac{D_{gj}}{R_0^{k_j}(t-1)^{k_j+1}}, \qquad k_j = \begin{cases} 1, & 1 \le j \le 8, \\ 25, & 9 \le j \le 12. \end{cases}$$

For six independent groups set $W_g = \prod_j (1 + RC_{gj}^2)$ and $\Theta = \operatorname{Tr} \sum_g W_g^{-1}$, require them nonzero, and impose $\Phi(R/\Theta^2)$. The pole input is nonzero, so the prefactor B in Q_{pole} cannot make these constraints vacuous.

At a positive odd contact prime outside S_{mul}, each factor valuation is the minimum of zero and an odd integer. No cancellation occurs. The trace and Lemma 5.1 force one common group and place where the first eight tests have valuation at least one and the last four at least 25. The first tests put the points in the formal group. At least one of $h, h - c_*$ is a unit at the prime, so its full-log ratio is integral. The truncated logarithm is the full logarithm times a unit congruent to one in valuation at least 28. Its ratio is therefore integral. The difference test transfers integrality to the

other truncated ratio, then to the other full-log ratio. Only now do the relative errors give absolute congruences modulo p^{28}. Multiplying the two ratio congruences by their integer denominators gives $n_i \equiv L(z(U_i))/L(z(H)) \pmod{p^{25}}$. The product test then implies $p^{25} \mid n_1 n_2 - n_3$.

To specify the height certificate, write the abscissa of each input as $x_{i0} + x_{i1} s$, using $x_{i0} = x_{i1} = 0$ at the identity. Choose $a_{i0}, a_{i1}, b_i \in \mathbb{Q}$ with $b_i \neq 0$ and $b_i x_{ij} = a_{ij}$. Express each of these nine rational quantities as a sum of three slopes. Impose the preceding slope predicate on all 27 slopes, with private witnesses, and impose membership on the three input codes. This conjunction is the height certificate. The multiplication predicate conjoins it with the fixed scalar graphs and the inverse of the finite guard below.

Proposition 6.1 (Finite-box deduction)**.** *Assuming these local test identities and the height estimate for S_{mul} with constant A_{ht}, every false triple accepted by the unguarded certificate has $\max_i |n_i| < T = (48A_{\mathrm{ht}})^{13}$.*

Proof. Put $f = n_1 n_2 - n_3 \neq 0$. For each of the 27 slopes, $M_{S_{\mathrm{mul}}}(t)^{25} \mid f$, so $h(t) \leq A_{\mathrm{ht}} |f|^{24/25}$. The nine rational homogeneous quantities for the three abscissas are sums of three slopes. The sum and quotient height bounds give $h(x_j) \leq 6A_{\mathrm{ht}} |f|^{24/25} + 2\log 3$ for each rational component. The fixed coordinate change and the duplication bounds yield

$$2n_i^2 \widehat{h}(P_0) \leq 12A_{\mathrm{ht}} |f|^{24/25} + B_0,$$

where $2\widehat{h}(P_0) > 1$ and $B_0 = 4\log 3 + \frac{3}{2}\log 2 + 2\log(1+\sqrt{2}) + \frac{1}{3}\log 676 < 10$. For $N = \max |n_i|$, $|f| \leq 2N^2$, so $N^2 < 24A_{\mathrm{ht}} N^{48/25} + 10$. If $N \geq 5$, this implies $N^{2/25} < 48A_{\mathrm{ht}}$, hence $N < (48A_{\mathrm{ht}})^{25/2} \leq T$. The case $N \leq 4$ is immediate. □

For each false triple $(a, b, c) \in [-T, T]^3$, write its fifteen code coordinates as A_j/B_j with $B_j > 0$, put $D_{abc} = \prod B_j$, and form

$$g_{abc}(X) = \sum_{j=1}^{15} (D_{abc} X_j - (D_{abc}/B_j) A_j)^2.$$

The guard requires the product of all these factors to have an inverse. A false accepted triple makes its own factor zero. At a true triple every factor is positive by injectivity of codes, and the inverse exists. Thus the sound direction uses the generator and corrected all-S height theorem but does not use pole completeness.

For completeness, S14 constructs, for every integer A and bound B_0, integers u_1, u_2, v with $u_3 = Av - u_1 - u_2$ such that all 22 forms $v, u_j - bv$, $b \in \mathcal{C} \cup \{1\}$, are signed catalogue multiples of primes above B_0. At each such prime the conjugate reduction orders have gcd four and are prime to that prime. The finite-complexity prime asymptotic [8, 9, 10] is used after fixing the catalogue and initial CRT data, then the sieve cutoff L. Twenty-two catalogue primes per unit class allow distinct choices for the twenty-two occurrences. At one catalogue prime the distinguished zero plane avoids at most $42p < p^2$ points; a lift modulo p^2 makes its divided value different from zero and minus one. The fixed affine system is therefore locally admissible. The asymptotic gives a lower count $c_A (\log L)^{-44} X^3/(\log X)^{22}$ for fixed sieve cutoff L. Each pair of forms has a two-by-two coefficient minor equal to ± 1 or a nonzero contact difference of absolute value at most 61. Thus the local conditions are nondegenerate uniformly in A; other minors may involve A. A Gaussian-pair sieve and the 21-form sieve on each nonzero fiber bound the loss by $C_A X^3/(L(\log X)^{22}) + o_A(X^3/(\log X)^{22})$. Choose L first and then X with A, L fixed. A uniform prime-asymptotic error as L varies is neither asserted nor needed. The corrected all-orders inverse theorem is among the external inputs.

Apply this construction to the nine integral homogeneous quantities of a true triple. Each slope has at most six odd-contact primes. At one such prime with reduction orders d_1, d_2, solve

$$h \equiv 0 \pmod{\operatorname{lcm}(M_0, d_1)}, \qquad h \equiv c_* \pmod{d_2}.$$

Compatibility follows from

$$\gcd(\operatorname{lcm}(M_0, d_1), d_2) \mid M_0 \gcd(d_1, d_2) = 4M_0 = c_*.$$

Choose a solution larger than c_*. Each contact prime uses its own private group within its own slope. No common shift across different primes is imposed, and selected prime values need not be distinct. The saved library allocates six private groups per slope and twenty-seven private slope calls. Set $U_i = [n_i]H$ and assign one group to each prime. Fixed local clearing at the old primes can be checked as follows. Put $S_b = S_{\mathrm{ht}} \cup \mathcal{H}$, $B_p = 4$ on S_{ht} and $B_p = 1$ on the remaining catalogue primes, $C_p = 14B_p$, and $D_p = 51C_p$. The selected integral forms give $|v_p(R_0)|, |v_p(t-1)| \le C_p$, while the small formal subgroup in the definition of M_0 gives $v_p(D_{gj}) \ge 0$ at S_b. Consequently $v_p(C_{gj}) \ge -D_p$. The fixed normalization has $v_p(\ell_0) = E_p = 52C_p + v_p(12) + 4$.

Weak approximation chooses q small at S_b, integral at the selected primes, and arbitrarily close to a real root of $1 - R_0 q^2$, without equality; choose η to make $R_0 > 0$. Then a is a unit at the specified finite primes. At $p \in S_b$,

$$v_p(RC_{gj}^2) \ge 2E_p - C_p - 2D_p > v_p(12), \qquad v_p(\Theta - 12) > v_p(12),$$

so $v_p(R/\Theta^2) \ge 103C_p + 8 \ge 4$. At a selected odd-contact prime, $v_p(R) = 1$. Each of the twelve summands of the trace of the six W_g^{-1} has residue zero or one; the assigned group and place contributes one. Since $p > 324$, their sum is a unit and the pole input has valuation one. At every other prime outside S_b, its valuation is even. Thus its reduced denominator is square. In both real embeddings each W_g is positive; as $a \to 0$, $W_g \to 1$, $\Theta \to 12$, and $R/\Theta^2 \to 0$ through positive values. This verifies exactly PC's hypotheses, including all nonzero guards. Negative indices, zero indices and coincident points are handled by the group-law branches. Therefore Theorem 5.2 supplies the required pole witnesses. This proves the full multiplication statement M, including completeness for every integer triple $(n_1, n_2, n_1 n_2)$ and soundness on arbitrary rational input codes.

7 The restricted converse and its dependency order

For a positive squarefree $\ell \equiv 11 \pmod{12}$ put

$$E_\ell : y^2 = x(x - \ell)(x + 3\ell).$$

Its invariants are $c_4 = 208\ell^2$, $\Delta = 2304\ell^6$ and $j = 35152/9$. Thus it has full rational two-torsion and is not a CM curve, since a rational CM j-invariant is an integer. The curve in this section is different from the fixed CM curve used to represent integer indices.

The assertion needed in Theorem 5.3 is

$$\operatorname{corank}_{\mathbb{Z}_2} \operatorname{Sel}_{2^\infty}(E_\ell/\mathbb{Q}) = 1 \quad \Longrightarrow \quad \operatorname{ord}_{s=1} L(E_\ell, s) = 1. \tag{RC}$$

We prove this assertion in this section and the next two sections. Appendices E–G give the fixed-form, theta, nonvanishing and normalization constructions used in this deduction. Fix $E = E_\ell$ and a modular parametrization $\pi : X_0(N) \to E$ with $\pi(\infty) = 0$. Fix the integral Tate lattice and the local squareclasses allowed at the primes of $2N$. Constants depending on this data are chosen before

any conductor, companion field or binary dimension varies. Quotienting by a rational point of order two gives a rational two-isogeny. The Selmer parity theorem [16, Theorem 1.8] therefore gives root number -1 under the left side of (RC). This application does not replace Selmer corank by Mordell–Weil rank and does not assume finite Sha.

Here is the dependency order of the proof. A fixed-form Euler-system calculation gives the even coefficient bound. A varying-companion arithmetic estimate, proved by the two limits below, then gives the genus-point lower bound. Only after that lower bound do we normalize the odd coefficients. Their finite-stage forms and the even forms give one common binary family. Finally a fixed-field determinant argument excludes a zero derivative at the base. The even estimate does not use that final determinant argument on Heegner classes.

7.1 The even coefficient calculation

Use normalized central values $L_*(h)$ with the real and imaginary periods fixed by π and the integral lattice. Only their valuations are used; every subsequent comparison uses this same normalization. For every twist whose two-primary Tate–Shafarevich group is finite, let $s(h) = \operatorname{length}_{\mathbb{Z}_2} \operatorname{Sha}(E^{(h)}/\mathbb{Q})[2^\infty]$.

The fixed-form construction uses one integral symbol combination with nonzero projections to both real signs. Kato's integral maps, with j_* cohomology taken modulo each power of two before passage to the limit, give classes whose inertia restrictions vanish integrally. Indeed the low-degree Leray edge map gives this vanishing before rationalization. Restriction of transfer is the Mackey sum of the corresponding upper inertia restrictions. Hence unnormalized transfer preserves the vanishing even when the intervening extension is ramified. No character projector divided by a growing group order is used.

For a finite binary character group, let F be the least common multiple of the odd parts of all its character conductors, including the base character. Write Z_F for the compatible Iwasawa class obtained by pushing the retained-smoothed Kato classes at levels $F2^n$ through the integral coefficient map. At a character χ its unused-prime factors are

$$E_q(\chi) = 1 - \frac{a_q \chi(q)}{q}\sigma_q^{-1} + \frac{1}{q}\sigma_q^{-2}. \tag{3.1}$$

They are the actual determinants of the unused good local terms. Active ramified good primes have rational inertia -1, so have no such rational local term. The fixed bad and dyadic corrections form a factor F_E with nonzero central value.

The rational divisibility input is Kato's Theorem 13.4(2), together with the nonzero-class input in 13.5(2) [19]. We use the height-one primes not containing two. Rational irreducibility is preserved on a quadratic subgroup of the non-CM open image. That subgroup contains a nontrivial unipotent, which supplies the required one-dimensional fixed space. Oddness supplies the two real eigenspaces, and purity supplies the common Frobenius weight. This application does not use residual irreducibility at two or the stronger hypotheses of Theorem 13.4(3).

The smoothing constants c, d_s can be chosen to satisfy

$$c, d_s \equiv 1 \pmod{3N_{\mathrm{odd}}F}, \qquad c, d_s \equiv 1 + 2^e \pmod{2^{e+1}},$$

where $e \geq \max\{3, v_2(N)\}$ is fixed by the original dyadic data. In weight two and cohomological degree one the two smoothing exponents are one. Kato's explicit reciprocity formula therefore contributes valuation $2e$, independent of F. Quadratic Gauss and square-root generators at odd conductor differ by dyadic units. The numerical sizes of c, d_s may grow; their relevant valuation does not.

For completeness, the integral calculation at two uses the positive complex. Finite continuous cochains are filtered unions of finite free modules and hence flat. Splitting the residual complex into cohomology and disks, then using the nilpotent perturbation calculation of Section 9, gives finite free models with their maps. In the fixed real cyclotomic tower the ordinary two-primary class group vanishes. One proves this inductively: only the prime above two ramifies in each quadratic step; the abelianization of a possible Hilbert two-class tower is generated by that inertia of order two. A finite two-group with cyclic abelianization is cyclic, and inertia excludes a larger cyclic extension. The unit-signature calculation is separate. A norm-minus-one unit in positive layers, and -1 at the base, has signature vector a unit in the characteristic-two cyclic group algebra. For a scalar coefficient the resulting residual dimensions are

$$\dim H^1 = 2^n + s_n - 1, \qquad \dim H^2 = s_n - 1.$$

For each fixed support s_n is bounded. Universal coefficients give generic residual rank one in degree one; the two trivial elliptic factors give rank two. Unit cancellations thus identify the positive complex over $\mathcal{O}[G]$ with $\mathcal{O}[G]^2[-1]$. This last rank is independent of support, although preliminary models need not have a uniform size.

A generator of the fixed real invariant lattice gives a connecting class a. Its central image need not be saturated. Twice an ordinary class lifts because the relevant real H^1 is killed by two. Wedge a with a lift of $2F_E Z_F$ in the original determinant line. This produces a common scalar u^{ev}. The first assertion below holds for the full family. The central valuation identity is asserted only at a constituent of Selmer corank zero with $L_*(h) \neq 0$:

$$2^{C_{\mathrm{ev}}} u^{\mathrm{ev}} \in \Lambda[G], \qquad v_2 u_h^{\mathrm{ev}}(0) = v_2 L_*(h) - 2\omega(h) - s(h) + O_E(1). \tag{3.2}$$

Here $\Lambda = \mathbb{Z}_2[[t]]$ and $\mathcal{O} = \widehat{\Lambda_{(2)}}$. The error and C_{ev} are fixed before the binary group and support. To justify membership in $\Lambda[G]$, first use rational characterwise divisibility and Fourier inversion for membership in $\Lambda[1/2][G]$, without estimating the Fourier denominator. Intersect coefficientwise with the integral $\mathcal{O}[G]$ result. The explicit coefficient construction also covers field discriminant $4h$, with its dyadic character retained before the cofinal norm limit, as in Appendix E. At a corank-zero constituent the same coordinate vanishes centrally if and only if the central value vanishes. At such a nonzero corank-zero constituent, keeping the real saturation and torsion terms gives the local formula

$$v_2 u(0) = v_2 \exp^* z - s - \sum_v g_v + 2g + 1 - r_1 - e_2.$$

At an active good odd prime $g_v = 2$, since $A^{I_v} = E[2]$. Unused terms cancel (3.1); all other terms belong to the fixed local list. This proves (3.2) with the indicated active-prime coefficient.

7.2 Nonvanishing with all support restrictions

A signed unit filter prescribes a real sign and unit squareclasses at a finite set S containing $2, 3$ and the bad primes. Let B be any additional finite set of forbidden primes. If its root number is even, there is a squarefree parameter in this filter, coprime to B, with at most five prime factors and nonzero central value. In an odd filter there is a simple-zero parameter, with no asserted prime-count bound.

Here is the reduction to the precise classical statements. Choose a signed prime q_0 in the prescribed unit classes by Dirichlet's theorem, avoiding $S \cup B$. Apply Hoffstein–Luo [21] in the even case, or Bump–Friedberg–Hoffstein [20] in the odd case, to the actual curve $E^{(q_0)}$. Require their positive discriminant d to split at $S \cup B \cup \{|q_0|\}$ and at the conductor primes. Then $q_0 d$ has the

prescribed original classes. Hoffstein–Luo gives at most four prime factors of d, so at most five in total. The BFH clause used in the odd case is the simple-zero clause.

The root-number check includes ramified dyadic quadratic characters. At the primes in S equal squareclasses give isomorphic local curves. At a good odd prime newly ramified in the parameter, the root factor is $(-1/p)$ [22, Proposition 3.3]. Its product outside S is fixed by the real sign and the dyadic unit class. Thus unit classes 3 or 7 modulo eight are allowed; the field discriminant may be four times the parameter. We do not invoke an odd-discriminant root formula on those parameters.

The extra upper-bound argument. Nonvanishing alone does not bound the valuation of L_*, and (3.2) by itself gives a lower bound. The needed upper bound uses a separate even-symbol selection and transfer argument, which we now give.

For the even coefficient system described below, a unit symbol at a parameter d gives

$$L_*(d) \neq 0, \qquad v_2 L_*(d) \leq 2\omega(d) + C_0. \tag{3.3a}$$

This follows from its same-filter coefficient-square formula and its fixed normalization. That even normalization uses the lower bound from (3.2), not the as-yet undefined odd minimum. Call a prime list partitionable if a permitted contraction forest takes its symbol to a unit seed. The indicated iterated finite difference is nonzero, so its terminal polynomial is nonzero.

For a partitionable parameter d already known to have Selmer corank zero, keep all its prime factors stationary. The one-symbol address construction gives arbitrarily large binary cubes based at d, with no old prime removed, whose other addresses satisfy (3.3a). The fixed-form common scalar in (3.2) has one uniformly cleared denominator on every such cube. Choose a two-adic precision larger than the nonzero-address scalar valuation bound and its fixed clearing exponent. Its sign-evaluation coefficient is a Boolean function with the bounded bit degrees proved in Section 9. A cube dimension larger than their sum therefore supplies a nonzero address congruent to the base at this precision. If the base scalar were zero or had larger valuation than the bound, so would that nonzero address, a contradiction. The corank-zero central comparison in (3.2) consequently gives

$$L_*(d) \neq 0, \qquad v_2 L_*(d) - 2\omega(d) - s(d) \leq C_1. \tag{3.3b}$$

The constants are fixed by the even system. The dimension needed may depend on d; the valuation bound does not. This is the cyclotomic missing-address argument on the fixed-form Kato scalar, not the later Heegner transfer.

Here is why the actual rough companion is partitionable. Prepare an even symbol for every possible total type at the finite fixed support. There are finitely many. In our full rational two-torsion case the alphabet is all split. A least nonempty unit seed has no dependence on internal edge bits: an edge derivative would be a smaller nonempty unit configuration. Its individual labels determine it. The negative-discriminant parity at two makes the total label nonzero, so this seed is nonempty. A list with the same total label and enough copies of the seed labels contracts to it: keep the first $r-1$ seed anchors and merge all remaining vertices into one nonempty cluster. Its label is forced to be the last seed label. Thus the list is partitionable, regardless of the number or labels of the remaining vertices.

Before choosing the rough factor, insert bounded dummy supplies for every prepared total type, including the required anchors and reserves. Realize their labels by distinct primes. For a varying h, place in B the entire fixed support, the support of h, every dummy prime and every other already fixed forbidden prime. If a reduction prime has already been fixed, include it too. Prescribe the total k' to be a negative odd fundamental discriminant, square at every place of $2Nh$ and outside the extra-unit cases. Absorb h and the dummy product into the curve and use the preceding

nonvanishing theorem in its even filter. It adds at most five new primes. For the actual resulting total type, the already chosen dummy supply makes hk' partitionable. Its analytic nonvanishing gives Selmer corank zero by the forward rank-zero theorem. Apply (3.3b), obtaining

$$\omega(k') \le A, \qquad v_2 L_*(hk') \le 2\omega(h) + 2\omega(k') + s(hk') + C_E. \tag{3.3}$$

Here A is the fixed dummy count plus five. This is the argument of the partitionable-even and rough-companion lemmas, with their stationary primes and order of choices retained. Neither prime sizes nor a uniform analytic threshold are asserted. Excluding a later coefficient prime does not enlarge the old Selmer support.

7.3 The genus lower bound precedes normalization

Fix a negative genus partner h_* with $L_*(h_*) \neq 0$. For a known simple-zero h, let P_h^{gen} be the unaveraged conductor-one genus sum over $\mathbb{Q}(\sqrt{hh_*})$. Let $P_{h,k'}^{\mathrm{rc}}$ be the unaveraged conductor-$|h|$ ring-class sum over $\mathbb{Q}(\sqrt{k'})$, weighted by χ_h. They are different points in different native lattices. Write $j_{\mathrm{gen}}(h)$ and $j_{\mathrm{rc}}(h,k')$ for their index valuations.

The explicit height formula [23, Theorem 1.1] applies twice. All level primes split, conductors are prime to N, extra-unit fields are excluded, and the parametrization is the same. Its conductor factor is $c\sqrt{|\operatorname{disc} K|}$ and its sum is unaveraged. The two Rankin products have the common factor $L(E^{(h)}, s)$. Each rational free line has index at most two in its native rank-one lattice: conjugation fixes the free class, and the norm makes twice the class rational. Thus

$$2j_{\mathrm{rc}}(h,k') - 2j_{\mathrm{gen}}(h) = v_2 L_*(hk') - v_2 L_*(h_*) + \epsilon, \qquad |\epsilon| \le 2, \tag{3.4}$$

up to the one fixed initial period convention.

The arithmetic estimate of Section 9, for each fixed A, is

$$2j_{\mathrm{rc}}(h,k') \ge 4\omega(h) + s(h) + s(hk') - C_R(E,A), \qquad \omega(k') \le A. \tag{3.5}$$

Combining (3.3)–(3.5) cancels $s(hk')$ and gives

$$2j_{\mathrm{gen}}(h) \ge 2\omega(h) + s(h) - C. \tag{3.6}$$

This calculation uses no odd normalization.

7.4 A fixed nonzero witness and actual odd forms

Choose one positive squarefree D_w in the odd filter containing one, after adjoining the primes of h_* and their compatible unit tests. Section 7.2 supplies $L'(E^{(D_w)}, 1) \neq 0$. The explicit height formula makes $P_{D_w}^{\mathrm{gen}}$ nontorsion. Its unwanted h_* component has rank zero. One fixed multiplier L kills the bounded-degree torsion and the projection denominator, so $Q = LP_{D_w}^{\mathrm{gen}}$ is a rational nontorsion point on $E^{(D_w)}$.

Let C be complex conjugation on its Tate module and choose a nonzero integral row b with $bC = b$. The Kummer translation image of Q is a full lattice. Indeed, if it vanished, its rational cocycle would descend to the Tate image. A nonidentity central scalar z gives

$$(z-1)c(g) = (g-1)c(z),$$

so that descended cocycle is a coboundary, contrary to Kummer injectivity. Irreducibility makes a nonzero translation span the Tate space, and intersection with any fixed finite kernel preserves full rank. Twisting preserves open image by restriction to an index-at-most-two subgroup.

Choose one such translation v with $2b(v) \neq 0$. Multiplying conjugation by this translation gives square translation $(1+C)v$, detected by $2b(v)$ with one fixed valuation c_w. The element is a single actual Galois element on the compositum. It is trivial on $\sqrt{D_w}$ and nontrivial on $\sqrt{h_*}$ because $D_w > 0 > h_*$. Its restrictions therefore already agree on intersections. Finite Chebotarev, with a place chosen above the resulting prime, gives arbitrarily advanced stages (M_θ, ℓ_0) such that

$$\rho(\mathrm{Fr}_{\ell_0}) \equiv C \pmod{2^{M_\theta}}, \quad (D_w/\ell_0) = 1, \quad (h_*/\ell_0) = -1, \quad v_2\lambda_{M_\theta}(\overline{Q}) = c_w < M_\theta. \tag{3.7}$$

Here $\lambda_{M_\theta} = b\kappa_{M_\theta}$ is the actual finite-field Kummer map on $E(\mathbb{F}_{\ell_0^2})$. It has Frobenius sign $+1$. All reductions use that same place. This is a fixed-witness statement, not a bound for arbitrary points or arbitrary reduction primes.

The weighted ternary theta construction gives actual holomorphic forms at these stages. At original level primes the root weight uses the summand selected by the cyclic level structure. At two, the integral spectral projectors of $(1+x)/2$ specify that summand; the operator itself is not in general idempotent. The nilpotent weight at h_* and the division-algebra residue-line weight at ℓ_0 have the prescribed conjugation and uniformizer signs. Their product is even. The root-character field is fixed before ℓ_0 and M_θ.

Appendix F gives these functions, their exact transformation rules, and the direct extraction using the form Q/m. Poisson summation for the resulting compactly supported local functions gives the scalar transformation law and holomorphy at every cusp. A fixed stabilizer multiple, for example one divisible by $|\mathrm{GL}_4(\mathbb{F}_3)|$, clears the finite automorphism groups. For the Hecke recurrence, a split or irreducible nonzero reduction has $1 + (-n/p)$ invariant lines, a nonzero nilpotent has one, and zero has $p+1$. The unique integral primitive nilpotent neighbor supplies the remaining square term. Tracking the scaling characters gives the recurrence with the same character convention as the coefficient comparison.

For an admissible D with $(D/\ell_0) = 1$, the oriented embedding count [25, Theorem 3.1], with changed set $\{\ell_0\}$, gives

$$a_{M_\theta}(D) = c_0\, 2^{\omega(N)}\zeta_{M_\theta}(D)\lambda_{M_\theta}(\overline{LP_D^{\mathrm{gen}}}) \pmod{2^{M_\theta}}, \tag{3.8}$$

where $c_0 \neq 0$ is fixed and ζ is a permitted root-of-unity phase. For admissible D with $(D/\ell_0) \neq 1$, including $\ell_0 \mid D$, the local Schwartz support instead gives $a_{M_\theta}(D) = 0$. Formula (3.8) is asserted only on the first support. There is no division by a class number. Formula (3.7) makes D_w visible with fixed upper valuation in (3.8). It remains on every stage's support, including the extra test $(D_w/\ell_0) = 1$.

After phase removal, the coefficients used for valuation are integers. Equation (3.6) gives a uniform lower bound for $v_2 a_{M_\theta}(D) - \omega(D)$ on visible bounded-weight sequences. Take the finite integer values constant on a set belonging to the chosen nonprincipal ultrafilter. This set of values is nonempty by D_w, is bounded below, and hence has an attained minimum δ. If the resulting lower inequality failed for a fixed weight bound on an ultrafilter-large set, choosing a violating index and then one of finitely many smaller integer values would contradict minimality. For any fixed finite operator budget, the coarse lower bound handles the large weights and stabilization handles the bounded weights. Consequently every required division is valid at one sufficiently advanced actual stage. No infinite-precision modular form or arbitrary-ring base-change assertion is needed.

7.5 One common family and the final order of choices

The even source is a fixed algebraic vector in the full good-Hecke eigenspace. Waldspurger's Corollary 2, including its same-dyadic-squareclass removal of the auxiliary condition, compares the

actual squarefree Fourier functionals [26]. Local upper-unipotent projections retain a nonzero initial functional; smoothness supplies lower-unipotent invariance. The algebraic linear conditions choose a vector over a number field. Galois conjugation followed by the matching finite character twist shows that each normalized coefficient ratio is a rational number times a two-power root of unity. This supplies the valuation and nonvanishing comparison on the whole filter. Only flat field extension of the algebraic model is used. The comparison is proved at the original support S_0 before any extra coprimality sieve B. The sieve is applied last, and every later good recurrence prime avoids the full support $S_0 \cup B$.

For both coefficient systems the integral operators have the required omission and contraction rules. At a new prime V_p raises the level exponent only to one, with ramified quadratic character. The recurrence makes the relevant differences divisible by the integer two before residue reduction. Multiplying by ordinary theta, then taking a fresh trace, removes the integral unary corrections. Two fresh traces vanish. The matrix trace identity gives $t(gu^2) = t(g)$, hence elementary quadratic labels. At a new odd level prime the weight-two local representation theorem [28] has one unramified and one ramified line; its Euler factor identifies the U_p eigenvalue. The diagonal-matrix calculation gives the two inertia differences. These identities prove omission and primary isolation for the same original form and depth, without choosing another minimum.

If the intrinsic odd limit symbol is a unit at the empty configuration, choose an actual stage carrying that value with $M_\theta > \delta_o$. The index $D = 1$ lies in every moving support and then satisfies $v_2 a_{M_\theta}(1) = \delta_o < M_\theta$. The visibility implication (O.8) gives $\operatorname{ord}_{s=1} L(E, s) = 1$, which is exactly (RC). No final companion or determinant bound is needed in this case. For the rest of the final-family and missing-address argument, assume that the empty configuration is not a unit. The bounded-weight witness then supplies a least nonempty odd seed, as in Appendix G.

Fixing the final companion. Use a separate even symbol on negative discriminants square at $2N$, excluding the fixed odd support, the ramification support of $E[8]$, and the exceptional imaginary fields. Choose a unit k of this symbol. Then $L(E^{(k)}, 1) \neq 0$. Fix k and $K = \mathbb{Q}(\sqrt{k})$ before choosing the binary dimension. At least one prime ramified in K is outside all the old fixed label extensions and torsion ramification. Choose every later moving theta prime ℓ_0 away from k. The intrinsic odd symbol continues to omit the primes of k.

At any such stage, the normal closure of the old odd label extension is unramified outside its old support and ℓ_0. The fresh ramified prime of K lies outside this set, so its intersection with K is $\mathbb{Q}$. Thus each old odd label still has a lift split in K. The added stationary tests at the primes of k may have relations; include those relations in the common alphabet, rather than requiring independent arbitrary prescriptions. Stabilize its separate projection maps on the same ultrafilter. This preserves the old odd minimum. Fresh ramification also makes K independent of $\mathbb{Q}(E[8])$ and of the old fields $\mathbb{Q}(i)$ and $\mathbb{Q}(\sqrt{\ell})$.

The forward theorem at this known nonzero central value gives $s_2(E^{(k)}/\mathbb{Q}) = 0$. The sum-and-difference isogeny over $\mathbb{Q}$ gives

$$s_2(E/K) = s_2(E/\mathbb{Q}) + s_2(E^{(k)}/\mathbb{Q}) = 1.$$

Indeed its kernel and cokernel on the relevant restriction and corestriction maps are killed by a fixed power of two. Tensoring the compact Selmer modules, equivalently the Pontryagin duals of the discrete Selmer groups, with $\mathbb{Q}_2$ gives the direct sum and hence this identity of coranks. No integral direct-sum assertion is used.

The K-split even seed is nonempty. Otherwise its p^3 coefficient and a fresh trace would force $a_p(E^{(k)}) \equiv p + 1 \pmod 8$ for every good split p. Choose $p \equiv 1 \pmod 4$ with $(\ell/p) = -1$ and

$(k/p) = 1$. The three rational points of order two cannot be doubled, since their necessary square differences include $-\ell, \ell, -4\ell$. The two-primary reduction group therefore has order exactly four, contradicting that congruence.

We record the finite combinatorial realization used here. Each label space is a finite elementary two-group; the common alphabet projects onto each separately. Product surjectivity is unnecessary. Take sufficient repeated copies of every label, with total label zero. Keeping the seed anchors, contract the unused copies into the last anchor. Minimal positive seed size makes the internal bits irrelevant, so each of the two terminal polynomials is nonzero. Grounded Laplacian minors express the constant and pair-contraction operators, including singular matrices. A cyclic singular network isolates its target radical and a path network isolates its selected pair coefficient. In the product of the two top forms, taken in the ordinary polynomial ring before Boolean reduction, a variable of exponent two uses one singular program in each factor; exponent one uses one varying program. The total degree kills the other contractions and leaves coefficient one. Thus one choice works for both forms. CRT and Dirichlet realize the prescribed forward residue bits by distinct primes; quadratic reciprocity supplies the reverse bits.

In this remaining branch, for every sufficiently large b there is one finite set Q of distinct good odd primes split in K and outside the fixed support, and nonzero linear forms $\lambda_q : \mathbb{F}_2^b \to \mathbb{F}_2$, such that the family

$$h_x = \prod_{q \in Q} (q^*)^{\lambda_q(x)} \quad (x \in \mathbb{F}_2^b), \qquad q^* = (-1)^{(q-1)/2} q,$$

has $h_0 = 1$, all fixed local types unchanged, and, for $x \neq 0$,

$$\operatorname{ord}_{s=1} L(E^{(h_x)}, s) = 1, \quad L_*(h_x k) \neq 0, \quad j_{\mathrm{gen}}(h_x) \leq \omega(h_x) + C_o, \quad v_2 L_*(h_x k) \leq 2\omega(h_x) + C_e. \tag{3.9}$$

The constants are independent of b. By (3.4), now with fixed k, $2j_{\mathrm{rc}}(h_x, k) \leq 4\omega(h_x) + C_J$.

The order is fixed data and D_w, then the genus lower bound and odd minimum, then the final $K = \mathbb{Q}(\sqrt{k})$ and common alphabet, then determinant precision M, then b and its finite operator budget, then a sufficiently advanced actual theta stage M_θ and its prime realization. In particular M_θ is not M. The finite graph construction, its local conditions and the coefficient tests all occur at that same final stage. Appendix G spells out the absolute five-prime completion, normalization constants, and the final use of the one corrected coordinate.

8 Actual classes, local conditions and central comparison

Let $K = \mathbb{Q}(\sqrt{k})$ be imaginary, split at $2N$, and different from $\mathbb{Q}(i)$ and $\mathbb{Q}(\sqrt{-3})$. Active primes are good odd primes split in K. Derivative primes are good inert primes outside all current supports. The actual finite coefficients are

$$A_i = (\mathbb{Z}/2^i\mathbb{Z})[t]/t^i, \quad M_i = (T_2E \otimes A_i)(\chi_h \Psi_i).$$

For a fixed binary group G, use $A_i[G]$ and the universal quadratic character. The character Ψ_i is finite, trivial at $2N$, ramified at a moving conductor prime r_i, and has prescribed odd scalar exponents at the finite active list. The prime r_i varies with precision. There is no assertion that one fixed odd prime supports an infinitely ramified $\mathbb{Z}_2$ character.

8.1 One multiplier before any conductor varies

Choose $n \geq 2$ with $1 + 2^n M_2(\mathbb{Z}_2)$ contained in the Tate image, and put $u = 1 + 2^n$. A scalar lift g has Tate action uI. For complex conjugation τ set

$$z = g^2 \tau g^2 \tau^{-1}.$$

The element g^2 fixes every quadratic field. Conjugation acts by inversion on every ring-class quotient of an imaginary quadratic field, so z fixes each such ring-class field and every rational quadratic character. Its Tate action is $u^4 I$, with $v_2(u^4-1) = n+2$. Thus the relevant invariant torsion has one bounded exponent, independent of K and conductor.

The required matrix-span bound is equally uniform. Squares of preimages in the open subgroup give, inside every quadratic subgroup, the matrices

$$I, \quad I + (2^{n+1} + 2^{2n})E_{11}, \quad I + 2^{n+1}E_{12}, \quad I + 2^{n+1}E_{21}.$$

Their span has fixed finite index in $M_2(\mathbb{Z}_2)$. These facts are the non-CM input. They are never applied to the CM generator curve.

Put $e = n+2$. In inflation–restriction for the complete conductor extension, both the obstruction and ambiguity groups with invariant coefficients are killed by 2^e. Lift 2^e times an invariant Kummer class, then multiply its lift by 2^e. The resulting descended class is independent of the lift, is additive, and restricts to 2^{2e} times the original class. Apply this once to the complete set of derivative primes, not once per prime.

At an original bad place, these conductor extensions are unramified. The first cohomology of the identity component vanishes under finite unramified cyclic descent: Lang's theorem handles the connected special fiber and the normal-basis calculation handles each additive formal quotient. Completeness passes through the formal filtration. The component-group order therefore kills the remaining obstruction. Only finitely many local quadratic types occur at $2N$. Together with the fixed cuspidal correction of π, their component orders and 2^{2e} give one integer L. It is fixed before K, h, r_i, derivative labels and binary dimension vary.

8.2 Norms, derivatives and retained boundaries

Let H_c be the ring-class field and y_c the consistently oriented CM point mapped by π. For a new prime not dividing $6cND_K$ the unaveraged norm identities are

$$\operatorname{Tr} y_{c\ell} = a_\ell y_c \quad (\ell \text{ inert}), \qquad \operatorname{Tr} y_{cq} = (a_q - \sigma_q - \sigma_q^{-1})y_c \quad (q \text{ split}). \tag{4.1}$$

The arbitrary-old-conductor version is [24, Theorem 2.15]; no squarefree-inert restriction on c is inserted. The unit factor is one by the exclusions on K.

At an inert prime the reduction identity is $\overline{y}_{c\ell} = \operatorname{Fr}_\ell \overline{y}_c$ for a compatible place. Locally $\mathcal{O}_c \otimes \mathbb{Z}_\ell$ is the maximal unramified quadratic order. Its $\ell+1$ raising isogenies reduce to the supersingular Frobenius quotient, and prime-to-ℓ level structure is transported by Frobenius. The parametrization commutes with reduction. This proves the same identity after all old class-group translates.

For $D_\ell = \sum_{j=1}^{\ell} j\sigma^j$,

$$(\sigma - 1)D_\ell = (\ell+1) - \operatorname{Tr}.$$

When the chosen two-power divides a_ℓ and $\ell+1$, (4.1) gives the invariant class to which the preceding descent applies. The finite and singular coordinates satisfy, at retained precision,

$$f = (a_\ell F - (\ell+1))\overline{Q}, \qquad s = (F - a_\ell)f + \varepsilon, \qquad 2^e\varepsilon = 0.$$

Here $F^2 - a_\ell F + \ell = 0$. Taking the initial precision at least e larger, and F congruent to the chosen conjugation matrix J with one additional bit, gives

$$f(P_{I+\ell}) = 0, \qquad s(P_{I+\ell}) = Jf(P_I). \tag{4.2}$$

The vanishing follows also from $\overline{D_\ell y_{c\ell}} = \ell(\ell+1)F\overline{y}_c/2$. The extra bit pays for this division by two.

Weighted transfer is the Shapiro map followed by $v \otimes [\delta] \mapsto \theta(\delta)v$. It is a finite sum without division by a class number. Local Mackey decomposition transports the places and Frobenius, so (4.1)–(4.2) hold term by term before weighting, also over the universal binary group ring. Whenever equality in the Selmer complex is asserted, retain an actual boundary witness at the same finite stage.

8.3 The derived Kummer condition

For the actual local twist $B = E^{(h)}$ over $F = K_v$, put

$$U_F = B(F)^{\wedge}_2[-1] \longrightarrow R\Gamma(F, T_2B).$$

The completion includes finite two-primary torsion. Derived reduction modulo 2^a has $B(F)[2^a]$ in degree zero and $B(F)/2^aB(F)$ in degree one. At $v \mid 2N$, Ψ_i is trivial, so derived coefficient extension defines the scalar A_i condition for this actual twist. In the universal binary family every h_x has the same local squareclasses at $2N$ as $h_0 = 1$. Hence every sign character evaluates the universal character ε trivially on G_{K_v}, and therefore $\varepsilon|_{G_{K_v}} = 1$ in G, before reduction modulo two. The $A_i[G]$ Kummer condition and its null-homotopy are consequently derived scalar extensions of the fixed local data. Omitting the scalar twist, the universal local-squareclass condition, or degree-zero torsion changes this argument.

The local finite dualities [12, I.2.3, I.3.2–I.3.5] make the Kummer map adjoint to $H^1(F, B[2^a]) \to H^1(F, B)[2^a]$. The latter has Kummer image as kernel, so the Kummer image is its exact annihilator. To obtain the integral chain statement, write $M = B(F)^{\wedge}_2$. The restricted cup map $M[-1] \otimes^L M[-1] \to \mathbb{Z}_2[-2]$ has homotopy class determined by $\mathrm{Hom}(M \otimes M, \mathbb{Z}_2)$. Finite isotropy makes every value divisible by every power of two, hence zero. A projective resolution therefore supplies a null-homotopy. The possible ambiguity group in degree minus one vanishes. Reduction modulo two, including the torsion in degree zero, gives the perfect quotient-to-dual comparison; derived Nakayama lifts it. The global Selmer duality theorem [13, Sections 5.1, 5.2.4, 5.4.3 and 6.3.4] is applied to these actual local maps and homotopies.

For clarity the Selmer complex is the cone

$$C_U = \mathrm{Cone}\left(R\Gamma(G_{K,S}, M_i) \oplus \bigoplus_{v \in S} U_v \longrightarrow \bigoplus_{v \in S} R\Gamma(K_v, M_i) \right)[-1]. \tag{4.3}$$

The compatible trace in (4.3) is coefficient-valued. A top-coefficient functional used to test a finite Frobenius ring's perfection is not used as a coefficient-reduction-compatible arithmetic trace.

8.4 Actual support comparisons

At a tame split prime the local free model has

$$d^0(v) = (Av, Cv), \quad d^1(v, w) = -Cv + Dw, \quad A = \Phi - 1,\ C = g - 1,\ D = \Phi - N_q(g),$$

$$N_q(g) = \frac{q+1}{2} + \frac{q-1}{2}g, \qquad DC = CA. \tag{4.4}$$

At augmentation $g = 1$, this is the direct sum of the unramified complex $[\Phi - 1]$ in degrees zero and one and the singular quotient $[\Phi - q]$ in degrees one and two.

The support-change theorem [13, Proposition 7.8.8(ii)] retains arbitrary old local conditions and puts the unramified condition at new places. Its finite-module proof applies at two. Use it only when the actual coefficient representation is unramified at the removed place. There are five

uses here: universal augmentation at Q; an unused scalar prime $q \nmid h$; characteristic-two residual removal after $\chi_h = 1$; the moving prime after $t = 0$; and a new good inert derivative prime outside h, r_i and the old support. In the last case the coefficient module is unramified and the chosen finite local plane is the unramified condition. It is not applied to an active ramified scalar prime or to r_i before central specialization. Replacing the unramified condition by the full condition then gives the singular quotient by a cone of Selmer cones. In particular the old complex is an actual subcomplex of an upper triangular model, not just a complex with the same Euler characteristic.

The simultaneous character construction solves the finite ideal relation lattice before choosing r_i. Prescribed odd valuations at the active primes are compatible with the Kummer radicals; the cyclotomic tower is unramified there. On the Tate side, commutators of a fixed open subgroup give a fixed open subgroup in the allowed coset. The functions $\operatorname{tr} \pm 2$ are not identically zero on any such open coset. There are only finitely many cosets at the fixed level, giving a bound c_E independent of the active list. At each coefficient stage Chebotarev realizes the single compatible target. Thus the same Ψ_i has odd deformation exponent at every active prime and bounded central moving-prime determinants.

8.5 The same functional at every localization

After the inner limit, put $\mathcal{O} = \widehat{\mathbb{Z}_2[[t]]_{(2)}}$. In (4.4), the residual active Frobenius is $(1+t)^a I$ with a odd. Thus A, D are invertible over $\mathcal{O}[G]$. The formulas

$$h^1(v, w) = A^{-1}v, \qquad h^2(u) = (0, D^{-1}u)$$

contract the complex; the identities follow from $DC = CA$. Combining these local comparisons with the retained cup map gives a closed functional λ, with $2^a\lambda$ integral for one fixed exponent a. This is a statement after the inner limit and base change, not an inverse of t in A_i.

For every height-one prime $\mathfrak{p} \neq (2)$ of Λ, construct a representative separately. At an active scalar prime $C = -2$ is invertible. At r_i, $C = t$ times a unit is invertible away from (t). There use

$$h^1(v, w) = C^{-1}w, \qquad h^2(u) = (-C^{-1}u, 0).$$

At (t) the bounded central Frobenius determinants permit the first contraction by adjugates. This includes primes such as $(t + 2)$: after inverting two, t is a unit there. All representatives invert the same generic comparison and compose with the same cup map. Their equality in the generic derived category, not $\mathcal{O}$-integrality alone, identifies the generic functional.

For a constituent whose generic cohomology has dimensions $(1, 1)$ in degrees one and two and vanishes otherwise, use the determinant line

$$\mathcal{D} = \det^{-1} C \simeq H^1(C) \otimes H^2(C)^*.$$

The class-functional coordinate u is the coordinate of $[z] \otimes [\lambda] \in \mathcal{D}$ in the retained integral determinant basis. Here z is the actual class and λ is the closed functional just constructed. For a generically acyclic constituent set $u = 0$. The primitive and relaxed subscripts specify the corresponding local conditions on C.

8.6 Central identity, torsion and the primitive coordinate

The split norm relation gives an actual finite-stage boundary

$$z_i(0) - L m_i \, j_i(b_h) = d v_i, \qquad m_i = a_{r_i} - 2\chi_h(r_i). \tag{4.5}$$

The equality in global cohomology lifts to the Selmer cone because the local degree-zero maps are onto and the degree-one maps inject. Retain v_i. The uniform trace bound bounds $v_2(m_i)$ and the local determinants by c_E; their adjugate identities multiplied by a fixed power of two survive the limit.

For the ordinary compact Kummer complex C_K, universal coefficients and the local degree-zero identifications give

$$H^1(C_K)_{\mathrm{tors}} = E^{(h)}(K)[2^\infty], \qquad 0 \to H^1(C_K) \otimes \mathbb{Q}_2/\mathbb{Z}_2 \to \mathrm{Sel}_{2^\infty} \to H^2(C_K)_{\mathrm{tors}} \to 0.$$

The last finite group is Selmer modulo its maximal divisible subgroup. Self-duality identifies the degree-three torsion with the degree-one torsion and gives a unimodular pairing of the free degree-one and degree-two parts. Thus their rational dimensions equal full Selmer corank, including a possible divisible part of Sha.

At known simple zeros, forward Gross–Zagier and Kolyvagin [17, 18] give rank one and finite Sha. For restriction of scalars and the product of the two twists, the sum/difference isogenies are dual and compose to [2]. Their free-lattice indices satisfy $IJ = 2$, with regulator ratio $2/I^2$. At odd primes the differential lattices agree; at two the pullback determinants have bounded total valuation. Milne's isogeny ratio [12, proof of I.7.3, (7.3.1)] follows from duality without assuming the BSD leading-term equality. It applies here with finite Sha, canonical heights and pullback measures.

At every active $p \mid h$, K splits, so the isogeny local factors cancel exactly. At either split place the full/Kummer quotient has length $\dim_{\mathbb{F}_2} E[2]^{\mathrm{Fr}_p} = 2$, since all of $E[2]$ is rational. The two places therefore contribute exactly -4. At a companion prime $p \mid k$ the ramified twist has component factor four, while the restriction of scalars has connected special fiber. Hence its error t_p is at most two. The result is

$$v_2 u_{\mathrm{prim}}(0) = 2j_{\mathrm{rc}}(h,k) - 4\omega(h) - s(h) - s(hk) - \sum_{p \mid k} t_p + O_{E,\pi}(1). \tag{4.6}$$

Only the companion sum can be absorbed when $\omega(k)$ is bounded. No class number or unbounded degree-two torsion length is absorbed.

At an unused prime the exact point factor is $e_q = a_q - z - z^{-1}$. The second factor comes from the same closed functional, as follows. Let $i : C_P \to C_R$ be the primitive-to-relaxed map, and let $j : C_S \to C_{P^\perp}$ be the paired strict-to-primitive map. All old orthogonal conditions and their homotopies are retained. At q these are the maps from the unramified condition to the full condition and from zero to the unramified condition. Naturality of the cup pairing [13, Sections 6.4.1–6.4.3] gives

$$\beta_R(v, iw) = \beta_P(jv, w).$$

The actual norm identity is $z_R = e_q i(z_P)$, with its retained boundary. The generic paired strict lift satisfies $jz_S = e_q^c z_{P^\perp}$. Conjugation sends z to z^{-1}, so $e_q^c = e_q$. Consequently

$$\lambda_R \circ i = e_q \lambda_P.$$

Thus the class and its paired functional each contribute one e_q; no integral self-duality of the primitive condition is assumed. The two quotient determinants satisfy

$$\delta_q = \det(z\,\mathrm{Fr}_q - q)\det(z^{-1}\,\mathrm{Fr}_q - q) = q^4 l_q.$$

In the original integral determinant bases, the support triangle therefore gives

$$l_q u_{\mathrm{rel}} = e_q^2 \varepsilon_q u_{\mathrm{prim}}, \qquad \varepsilon_q \in \Lambda^\times. \tag{4.7}$$

To see the unit assertion, retain the finite mapping-cone bases. Their comparison is a sign; replacing them by the retained homotopy equivalent free models contributes only determinants of unit disks. These are A_i-units. Their limits and inverse limits are Λ-units, and q^{-4} is one also. An arbitrary $\mathcal{O}$-unit would not suffice.

The universal correction η is integral over $\mathcal{O}[G]$: $l_q/e_q^2 = 1 + 2v_q$, and $1 + v_q(1 + g_q)$ selects that factor precisely at an inactive character. Put $\widetilde{u} = \eta u_{\mathrm{rel}}$. It already is one common family; no interpolation of the individual ε_x is required. Formula (4.7) makes $\widetilde{u}_x = \varepsilon_x u_{\mathrm{prim},x}$. The central zero and valuations therefore transfer unchanged. The multiplier η_x may itself have a pole at $t = 0$. Only the product $\widetilde{u}_x$, identified with the regular primitive coordinate times a Λ-unit, is specialized there.

Finally, suppose the unknown base derivative vanishes. The conductor-one height formula makes its point torsion. Use a torsion annihilator and (4.5) to retain its central boundary. The central rational dimensions in degrees one and two are $(1, 1)$ by full Selmer corank. This does not assert that the central H^1 vanishes: the actual point class is zero because the point is torsion. The generic complex is acyclic or has dimensions $(1, 1)$. In the acyclic case the specified class-functional coordinate is zero. In the latter case equality of central and generic dimensions rules out a nonunit disk over $\Lambda_{(t)}$: each such disk would add a central class in both adjacent degrees, including its Tor contribution. After cancellation, determinant base change therefore identifies the same coordinate with the central class paired with the central functional. The retained boundary makes this zero. Thus $u_{\mathrm{prim},0}(0) = \widetilde{u}_0(0) = 0$. No finite-Sha assertion at the unknown base enters this argument.

9 The two limits and determinant transfer

This section uses the actual diagrams of Section 8. Their cohomology ranks alone would not suffice: maps to original cochains, local comparisons, cup products, classes and boundary witnesses are retained together. The same retained data include the integral duality maps, their homotopy inverses, and the homotopies expressing compatibility of duality with localization.

9.1 Finite free models and the inner limit

Let A be an Artin local ring with nilpotent maximal ideal $\mathfrak{m}$. A flat A-module is free, including in infinite rank: lift a basis modulo $\mathfrak{m}$, use nilpotence for surjectivity, and flatness and nilpotence for vanishing of the kernel. A continuous finite-coefficient cochain module is a filtered union of modules of functions on finite clopen partitions, hence is flat.

Suppose the residual cohomology of a complex of such modules is finite and bounded. Split the residual complex into cohomology and disks, and lift graded bases. Let d_0 be the lifted disk differential and $d = d_0 + \epsilon$, where ϵ has entries in $\mathfrak{m}$. Use the graded decomposition into cohomology, disk sources and disk targets. Let i_0, p_0 include and project the cohomology summand, and let h_0 invert d_0 from disk targets to disk sources, zero on the other summands. Then

$$p_0 i_0 = 1, \quad h_0^2 = h_0 i_0 = p_0 h_0 = 0, \quad 1 - i_0 p_0 = d_0 h_0 + h_0 d_0.$$

With this normalized contraction, the finite geometric series give

$$\begin{aligned} d_C &= p_0 \epsilon (1 + h_0 \epsilon)^{-1} i_0, \quad & i &= (1 + h_0 \epsilon)^{-1} i_0, \\ p &= p_0 (1 + \epsilon h_0)^{-1}, & h &= h_0 (1 + \epsilon h_0)^{-1}. \end{aligned}$$

Substitution using $d^2 = 0$ gives $pi = 1$ and $1 - ip = dh + hd$. The resulting bounded finite free model carries actual maps and homotopies to the original cochains. Transport any fixed finite collection

of arithmetic maps by $f \mapsto p'fi$. The identity $1 - ip = dh + hd$ transports their compatibility homotopies as well, including the second homotopies in the cup-product diagrams.

Fix one nonprincipal ultrafilter on coefficient stages. Bounded ranks can be fixed on a set in that ultrafilter. Each finite tuple of matrix entries then has a compatible limit at every coefficient precision, giving matrices over Λ. Every polynomial identity between the matrices survives. Retained homotopy inverses remain homotopy inverses. Mere stagewise quasi-isomorphisms without those maps would not justify a nonflat coefficient change.

The residual rank estimate needs a precise order of operations. Over $B_i = \mathbb{F}_2[t]/t^i$, the quadratic character χ_h is literally trivial. Thus an actual smaller-support representation exists, with old support, two places over r_i, and the fixed number j of derivative labels. Cancel only units over B_i. The index-two pro-two generator bound controls its residual H^1. It does not by itself control H^2. For that use the totally imaginary global Euler characteristic $\chi = -2$ and vanishing above degree two. Let $v = \omega(2N)$ and let s count rational primes in the smaller ramification support, including primes dividing $\operatorname{disc} K$, the moving prime and j derivative labels. Since $K = \mathbb{Q}(\sqrt{k})$ and two is already counted in $2N$, the condition $\omega(k) \le A$ gives $s \le v + A + j + 1$. This adds the quadratic field's ramification to the rational generator count, not new Selmer local conditions. The local contributions give the explicit bound

$$N(E, A, j) = 8(s + 1) + 6 + 48(v + 1 + j) \le 56v + 8A + 56j + 70. \tag{5.1}$$

For each fixed active support and family, retain the actual comparison cones, maps and homotopies in a padded tuple whose ranks are bounded over coefficient stages, and pass to $\mathbb{F}_2[[t]]$. The ranks of this whole retained tuple may grow with the number of active cones. The bound $N(E, A, j)$ concerns the reduced generic model, not that whole tuple uniformly in the active support. Only now invert t. The active local determinants are nonzero because their scalar exponents are odd; their cones become acyclic. This proves the generic residual bound for the original complex without inventing a fixed-r infinite character. The bound is independent of the number and sizes of active primes and of $v_2(r_i - 1)$.

9.2 Evaluation and the outer valuation ring

For fixed K let $\Gamma_K = \prod_i G_K$ and include one simultaneous complex conjugation. Evaluation of retained cochain maps at arbitrary sequences $g = (g_i), h = (h_i)$ gives

$$e^1_g d = (\rho(g) - 1)e^0, \qquad e^2_{g,h} d = \rho(g)e^1_h - e^1_{gh} + e^1_g. \tag{5.2}$$

One fixed ultrafilter supplies these identities for all sequences. A cycle therefore defines an abstract crossed homomorphism of the product group. We do not assert that its restriction to one original Galois group is a continuous pointwise limit.

Evaluation remains injective after field extension. Residually, take representatives V for H^1 and coefficient space W. Evaluate (c, w) by $b_c(g) - (\rho(g) - 1)w$. The intersection of all kernels is $(0, W^\Gamma)$. A finite list of at most $\dim V + \dim W$ evaluations already has that intersection. The comparison to $[W \to W^d]$ is an isomorphism on H^0 and injective on H^1. Its cone has no residual cohomology in degrees at most zero. Cancel unit disks there so that the free cone starts in degree one; the same is true after every coefficient extension. The long exact sequence gives the claimed injection. For Selmer complexes this applies because the local maps are isomorphisms on H^0 and injections on H^1.

Now allow K_j, h_j to vary with $\omega(k_j) \le A$. For copies $\mathcal{O}_j$ of $\mathcal{O}$, form

$$\mathcal{O}_\infty = \left(\prod_{\mathcal{U}} \mathcal{O}_j \right) / \{(x_j) : v_2(x_j) \to_{\mathcal{U}} \infty\}. \tag{5.3}$$

A nonzero element has valuation one fixed nonnegative integer on a set in $\mathcal{U}$ and is that power of two times a unit. Every nonzero ideal has an element of least valuation and is principal. Thus (5.3) is a DVR with uniformizer two and characteristic-zero fraction field. Completeness is unnecessary.

Integral minimal-model matrices and (5.2) pass to (5.3). So do the retained duality maps, inverse homotopies and duality-to-localization compatibility homotopies. After minimal cancellation an integral homotopy equivalence is degreewise invertible: its residual matrices are invertible, and Nakayama applies. These inverse matrices and all compatibility identities pass to the same outer quotient. Thus perfect duality and exact orthogonality are preserved there; they are not inferred merely from isotropy. Approximate cycles suffice: if $v_2(dw_j) \to \infty$, the degree-two term in (5.2) tends to zero, so w_∞ is an actual crossed homomorphism. The finite detecting lists have bounded length by (5.1); retain their cone contractions to preserve injection. One must not replace approximate cycles by unexplained exact cycles.

9.3 One new prime detects two classes

The exact kernel consists of sequences trivial on the full Tate module, all relevant ring-class fields and quadratic characters. The scalar element of Section 8 is central modulo this kernel, though not in the kernel itself. For a cocycle zero on the kernel,

$$(u^4 - 1)b(g) = (g - 1)b(z)$$

makes it a coboundary over characteristic zero. The same fixed scalar survives the outer quotient. To eliminate residual invariants, use moving-prime inertia: its scalar difference is t, invertible in $\mathbb{F}_2((t))$. These are two separate arguments.

On the characteristic-zero tests used for rank reduction, the fixed finite matrix-span witness gives absolute irreducibility of each rank-two summand. When the scalar character is not quadratic, moving inertia has distinct characters on the two summands and simultaneous conjugation exchanges them. The induced rank-four representation is therefore absolutely irreducible with scalar endomorphisms. The outer fraction field is one such characteristic-zero test; the quadratic character exception at an inner test is treated separately below.

For two independent cohomology classes on these tests, the span of their joint values on the exact kernel projects onto both copies of this absolutely irreducible representation. A proper submodule with both projections onto would be the graph of a scalar isomorphism, contradicting independence. Thus the span is the full sum. If the determinant of their two finite localizations vanished on every actual kernel element, additivity and polarization would make it vanish on the entire span, impossible. Hence one actual element supplies a rank-two localization on these characteristic-zero tests.

The polarization identity itself is also valid in characteristic two, but no residual irreducibility or residual rank-two localization is asserted. Here $E[2]$ is trivial of rank two, so its scalar twist is isotypic and the induced residual representation has multiplicity two. At the residual test we use only scalar-inertia vanishing of H^0, the bounded residual cohomology dimensions, and finite detecting lists from intersections of linear evaluation kernels. Those lists need not consist of one rank-two localization. The rank-two selection for the uniform valuation bound occurs only after passage to the characteristic-zero fraction field of the outer DVR.

Chebotarev is applied at finite stages, not to the product group. Each of the finitely many original basis cochains in degrees zero through two factors through a finite quotient. Include their quotient, the required Tate precision, ring-class and quadratic fields and old diagrams in one compositum. The element just found, multiplied by conjugation, is a single actual automorphism of

that compositum. Its restrictions agree on intersections. Choose the new prime and a place above it to realize that representative.

This also resolves the precision issue caused by localization. Choose the new prime to match all original basis-cochain evaluations at a common retained precision $n(i) \to \infty$. Taking the same inner ultrafilter gives an exact equality of Λ-matrices

$$L_j = E_{\sigma_j^2}. \tag{5.4}$$

Only afterward compose with the inclusion I_j from an $\mathcal{O}$-minimal model: $L_j I_j = E_{\sigma_j^2} I_j$. The denominators in I_j may depend on j; equality (5.4) makes this irrelevant. Long t-jets are never treated as approximations in the two-adic valuation of $\mathcal{O}$, where t is a unit.

9.4 Switches and the varying-companion estimate

At a new inert prime the finite and singular planes are free of rank two, with cross pairing $\beta(x, y) = \varepsilon\langle x, Jy\rangle$. Suppose the old actual class A spans generic H^1 and has $f(A) = v \neq 0$. Its derivative B satisfies $f(B) = 0$, $s(B) = Jv$. The strict H^1 vanishes. The transported duality and localization identities identify the relaxed localization image with the kernel of its dual localization map, hence with its exact annihilator [13, Sections 6.3.3–6.3.4]. This image contains $(v, 0)$ and $(0, Jv)$, so these span it. Hence the new generic H^1 is spanned by B.

The determinant comparison is exact even if v is divisible. In the fraction field choose x with $v \wedge x$ of unit volume, and similarly y with $Jv \wedge y$ of unit volume. The finite and singular determinant exact sequences have the same strict factor, so their class-functional coordinates have equal valuations. All strict torsion remains in that common factor. Requiring x integral would be wrong when v is divisible.

The starting generic kernel has rank one. Indeed, at a known simple zero the ordinary class is nonzero. Equation (4.5), with its bounded nonzero multiplier, makes the retained central class nonzero. Over $\Lambda_{(t)}$ cancel the degree-zero and degree-three disks. If the generic kernel of the resulting two-term free complex vanished, the actual lifted cycle would vanish, contradicting its central value. Semicontinuity bounds that kernel by the central dimension one.

Suppose no uniform lower bound exists. Choose instances with $v_2 u_j \to -\infty$, $\omega(k_j) \leq A$, and write their minimal completed models as $[\mathcal{O}_j^n \xrightarrow{D_j} \mathcal{O}_j^n]$, fixing the bounded n on an outer-large set. In each generic one-dimensional kernel choose a primitive vector v_j. Its outer image is nonzero, so the outer kernel dimension s is at least one. If $s > 1$, one new derivative prime gives rank-two localization including that vector. The relaxed localization image is self-annihilating and contains the whole finite plane, hence equals that plane. The strict kernel has dimension $s - 2$ and the singular switch adds no localized vector, so the new outer dimension is $s - 2$.

The actual inner class is a nonzero scalar multiple of v_j. Even if this scalar tends to zero, its generic localization is nonzero on an outer-large set. Exact switch preservation therefore keeps each inner determinant valuation. The new inner kernels again have primitive vectors. Thus their outer dimension remains at least one; a proposed drop to zero is already a contradiction. There can be only finitely many drops. For each fixed number of them (5.1) bounds the models; no bound uniform in an unbounded number of switches is needed.

At terminal outer dimension one a maximal boundary minor Δ is nonzero. Its inner minors have one finite valuation b on an outer-large set. Determinant-one Schur operations in a square two-term rank-one-kernel complex give $u = \pm\alpha\beta/\Delta$, where α is an integral surviving cycle coordinate and β has valuation at least $-a$. Consequently $v_2 u_j \geq -a - b$, the required contradiction. We obtain

$$v_2 u_{\mathrm{prim}} \geq -C(E, A). \tag{5.5}$$

Here (5.5) is asserted for the known simple-zero ring instances used to obtain the varying-companion estimate. Their nonzero actual central class supplied the rank-one start. It is not a uniform valuation assertion at the unknown corank-one base. Bounded matrix size alone would not prove it: $[2^j]$ is a counterexample to that inference.

At a height-one $\mathfrak{p} \neq (2)$ the residue field has characteristic zero. If the specialized scalar character is not quadratic, the same argument applies. If it is quadratic, the induced representation splits into two nonisomorphic rank-two signs. When both occur, one evaluation detects one class from each. When only one occurs with dimension greater than one, a first switch preserves total dimension but introduces the other sign: the local symmetric hyperbolic plane has exactly the finite and singular axes as isotropic lines. A second switch then drops dimension by two. This quadratic exception is necessary, for example at a specialization where the moving character becomes -1.

Use the separate integral functional representatives constructed in Section 8. The terminal minimal complex has one term in each of degrees one and two and zero differential. Its class-functional coordinate is integral over $\Lambda_{\mathfrak{p}}$. Exact switch preservation transports integrality back. Intersecting these height-one localizations gives $u_{\mathrm{prim}} \in \Lambda[1/2]$. This regularity statement also applies at the unknown full-corank-one base: if its generic class is zero the coordinate is zero; otherwise its nonzero generic cycle and the central dimension bound give the rank-one start for the height-one argument. It does not require a uniform outer valuation bound at that base. Combining (5.5), on the known simple-zero instances, with (4.6) and $\sum_{p|k} t_p \leq 2A$ proves (3.5). No fixed-field class number is absorbed into this varying-field bound.

9.5 Actual augmentation and the fixed-field count

We now use the final K in the remaining branch of Section 7, where the intrinsic odd empty configuration is not a unit. There are no derivative labels in this last binary family. At augmentation every graph prime is unramified, and the actual support triangle gives

$$C_{\mathrm{old}} \longrightarrow (C_b)_0 \longrightarrow \bigoplus_{q \in Q} Q_q \longrightarrow C_{\mathrm{old}}[1], \tag{5.6}$$

with C_{old} an actual subcomplex of an upper triangular model. Each two-place local determinant modulo two has t-order at most four by the same odd scalar exponents.

Let $d_K = \dim_{\mathbb{F}_2} \mathrm{Sel}_2(E/K)$, $t_K = \mathrm{length}\, E(K)[2^\infty]$ and $h_K = \mathrm{length}\, H^2(C_K)_{\mathrm{tors}}$. The last number is finite without finite Sha at the unknown base. After central specialization the two moving-prime singular blocks have total determinant valuation at most $2c_E$. The long exact sequence and duality give

$$\mathrm{rank}_{\mathrm{total}}\, C_{\mathrm{old}} \leq 2d_K + 12, \quad m \leq d_K + 5, \quad h_1 \leq h_K + 2c_E + 2t_K. \tag{5.7}$$

These bounds depend on fixed K, not on b or the graph size. The j-dependent bound (5.1) is not used with $j = b$.

A homotopy equivalence of minimal free complexes over a local ring is degreewise invertible, by reduction and Nakayama. Cancel unit disks and add them back as needed to lift (5.6) upstairs. The augmentation kernel is in the Jacobson radical, so invertible augmented matrices lift to invertible matrices. The block containing the local quotients has

$$B_0 = A = \mathrm{diag}(A_q), \qquad T_0 = 0$$

in a differential $\left(\begin{smallmatrix} P & Q \\ T & B \end{smallmatrix}\right)$. Over $\mathcal{O}[G]$, cancel B. The surviving Schur differential $P - QB^{-1}T$ augments to P_0.

Write $\det A_q = t^{d_q} v_q + 2w_q$, $d_q \le 4$, with v_q a power-series unit, and put $C_q = \operatorname{adj}(A_q)/(t^{d_q} v_q)$. For a character x, $B_x C = 1 + 2F$ with both C and F having no Laurent terms below degree -4. Therefore

$$B_x^{-1} = C \sum_{j=0}^{M-1} (-2F)^j \pmod{2^M} \tag{5.8}$$

has pole at most $4M$, independent of the number of blocks. The adjacent surviving differentials are unchanged by $d^2 = 0$. Every entry of the remaining old-rank complex has the same pole bound.

9.6 Determinant clearing and the missing-address contradiction

In integral central bases decompose the old complex into m Smith disks and one free generator in each of degrees one and two. In each differential group all disk sources into E_i and all their targets into B_{i+1}. Cancel the entire block $A_i : E_i \to B_{i+1}$, once per degree, and set

$$\Delta = \prod_i \det A_i.$$

A Schur complement can change the complementary part of that same differential. It does not change the adjacent surviving differentials, so it does not change another grouped A_j. Thus Δ has degree m in the original differential entries and $v_2 \Delta_0(0) \le h_1$.

The projected class, projected functional and determinant comparison each use at most one denominator Δ. For the integral unused-prime multiplier η, there is $Q \in \mathcal{O}[G]$ with

$$\widetilde{u}_x = 2^{-a} \eta_x Q_x / \Delta_x^3$$

when $\Delta_x \neq 0$. The integral unit from the cancelled local-block determinant is absorbed in Q. Put

$$f = 2^a \Delta^4, \qquad d = \eta \Delta Q.$$

Then $f_x \widetilde{u}_x = d_x$ at every character, including $\Delta_x = 0$, where both sides vanish. Here

$$c = v_2 f_0(0) \le c_{\det} := a + 4(h_K + 2c_E + 2t_K), \qquad L_T(M) = 16(d_K + 5)M \tag{5.9}$$

bounds the negative Laurent support of $f \bmod 2^M$. There is no required pole bound for the complicated numerator d. The common functional bound alone would not prove (5.9); the actual augmentation and grouped-disk calculation are essential.

For a coefficient of any element of $\mathcal{O}[G]$, its sign-value function on $\mathbb{F}_2^b$ expands as

$$A(x) = a_\varnothing + \sum_{I \neq \varnothing} 2^{|I|} a_I \prod_{i \in I} x_i.$$

Its low bit is constant and its bit of index $j \ge 1$ has Boolean degree at most 2^{j-1}. One can see this by extracting the Y^{2^j} coefficient of $(1+Y)^{A(x)}$ modulo two: the first possible contribution of x_I has degree at least $2^{|I|}$ in Y, and $|I| \le 2^{|I|-1}$. Thus q coefficient equalities modulo 2^M have sum of Boolean degrees at most $q(2^{M-1} - 1)$.

If that sum is less than b, multiply the complementary indicators of the equations. Its degree is less than b, so its sum over the Boolean cube is zero: every monomial misses a variable. The support of this indicator is the simultaneous solution set. It has even cardinality and contains zero, hence another address.

Apply this to the coefficients of f from $-L_T(M)$ through zero and the single constant coefficient of d. It suffices to choose

$$M > \max(c_{\det}, B + 2c_{\det}), \qquad b > (L_T(M) + 2)(2^M - 1), \tag{5.10}$$

where B bounds the nonzero-address central valuations. These choices precede the graph and its theta precision. At the resulting nonzero address, negative coefficients of f_x vanish modulo 2^M, and its constant coefficient agrees with $f_0(0)$ and has valuation c. Positive coefficients may have smaller valuation. Nevertheless $v_{\mathcal{O}}(f_x) \le c$, so integrality of $d_x = f_x \widetilde{u}_x$ gives $v_{\mathcal{O}}(\widetilde{u}_x) \ge -c$. This estimate comes from the displayed product at the selected nonzero address; it does not apply (5.5) to the unknown base. Both $\widetilde{u}_x$ and $\widetilde{u}_0$ lie in $\Lambda[1/2]$ by the height-one argument above, so neither has negative powers of t. In particular, since $f_0 \in \Lambda$, $[t^0]d_0 = f_0(0)\widetilde{u}_0(0)$. The convergent constant-term convolution yields

$$v_2\big(\widetilde{u}_x(0) - \widetilde{u}_0(0)\big) \ge M - 2c > B. \tag{5.11}$$

By (3.9), (3.4), (4.6) and the unit comparison (4.7), every nonzero address has $v_2\widetilde{u}_x(0) \le B$ for one B fixed before b; the finite Sha terms have negative sign and may be discarded for this upper bound. If the base derivative vanished, Section 8 would give $\widetilde{u}_0(0) = 0$, contradicting (5.11). Hence $L'(E/K, 1) \neq 0$. The complementary value $L(E^{(k)}, 1)$ is nonzero, and $L(E/K, s) = L(E, s)L(E^{(k)}, s)$, so E has analytic rank one. This proves (RC), and hence Theorem 5.3, using the classical applications and finite constructions specified above.

10 Exact precision computation and polynomial compilation

Encode a signed integer by

$$\operatorname{enc}(n) = \begin{cases} 2n, & n \ge 0, \\ -2n - 1, & n < 0. \end{cases}$$

The five natural inputs decode in the fixed order (n, κ, a, l, j). The decision predicate rejects unless $n \notin \mathcal{C}$, $\kappa \neq 0$, $a, l \ge 1$, and $j = na$. For nonzero multiples of P_0, $z = -x/y$ is well-defined. Set

$$z([l]P_0) = u_l + v_l s, \qquad \epsilon_l = |u_l| + |v_l|, \qquad r(a, j) = \tfrac{1}{2} \operatorname{Tr}_{K_0/\mathbb{Q}} \frac{z([j]P_0)}{z([a]P_0)}.$$

With the fixed constant $A_{\mathrm{ht}}^{(0)} = H_6(S_{\mathrm{ht}})$, its final tests are

$$|r(a, j) - n| < \epsilon_l, \qquad 4\epsilon_l < 4^{-A_{\mathrm{ht}}^{(0)} |\kappa F(n)|^{24}}.$$

These are strict tests on exactly computable rationals and integers.

The reference program S21 uses exact quadratic-field arithmetic and capped integer operations. The ideal register intermediate representation S22 computes the same predicate on every five-natural tuple. Its independent all-input argument covers normalized signed arithmetic, origin and zero-ordinate cases, equal-abscissa addition, and every early rejection. Nine exact polynomial identities verify the nonexceptional projective formulas and preservation of $Y^2 = X^3 + AXZ^4$. Scalar loops decrease by halving; the precision loop divides by four; factorial iteration is bounded. This is a proof about the ideal intermediate representation, rather than an assertion that a budgeted simulator always finishes.

For example, if $\epsilon = U/V > 0$ with natural U, V and integer $B \ge 0$,

$$4\epsilon < 4^{-B} \quad \Longleftrightarrow \quad 4^B \le \left\lfloor \frac{V - 1}{4U} \right\rfloor.$$

This permits a finite cap. For positive L, $\sigma(x) = \min(x, L+1)$ commutes with saturated addition and multiplication. Powers and factorials can be computed with this saturation. Capping an exponent is safe for base at least two because an exponent above L already makes the value exceed L. Base zero, base one and exponent zero are separate cases. This proves equality with the fixed-height expression without requiring its full evaluation when the available precision is too small.

10.1 The natural polynomial and the power interface

The saved precision polynomial has 1 305 811 natural variables, the first five being inputs. Its SHA256 is

`f3e163f5c60b972c98b48647274f956dd7d78428e27ddc0543574a5f18120b8c.`

Source S23 specializes a fixed universal natural halting polynomial at the code of a wrapper. The wrapper inverts four Cantor pairs and then runs the unchanged precision machine. Its pairing equations are

$$2t_i = (c_i + t_{i+1})(c_i + t_{i+1} + 1) + 2t_{i+1} \quad (0 \le i < 4), \qquad t_4 = c_4.$$

Their triangular intervals prove unique natural unpairing. The new polynomial sums the square of the old halting polynomial and the four squared differences above.

The universal compiler packs a finite run of a fixed m-state increment/decrement/branch program into base-B integers. Acceptance means reaching the unique final halt with all registers zero; the fixed interpreter includes a finite cleanup stage. Choose

$$c > 0, \quad K = 2^c, \quad B = 4K > m, \quad T = B^q, \quad F_q = 1 + B + \cdots + B^q, \quad D = (K-1)F_q,$$

and require every input to be below K. For natural integers, say X is masked by Y when every set binary bit of X occurs in Y. Require each register history R_j to be masked by D and satisfy $R_j < T$. Each state history S_i is masked by F_q, and

$$\sum_{i=0}^{m-1} S_i = F_q, \qquad S_{\text{halt}} = T.$$

Thus register digits are below K, terminal register digits are zero, and state digits are zero or one. Since $B > m$, the state sum has no carries and selects exactly one state at each time.

The nonzero mask N_j is determined by

$$KN_j = (R_j + D) \text{ AND } (KF_q).$$

Indeed each digit $r + K - 1 < 2K < B$ has its K-bit set exactly when $r > 0$. For a decrement/branch instruction at state i testing register j, impose

$$V_i = S_i \text{ AND } N_j, \qquad S_i = V_i + W_i.$$

These are the nonzero and zero branch histories. Form the packed increment indicator A_j from states incrementing j, the decrement indicator C_j from the appropriate V_i, and each incoming-state indicator I_i from the states or branches directed to i. They have zero-one digits, since there is just one active instruction and branch at each time. The transition equations are

$$R_j + BC_j = \text{initial}_j + BR_j + BA_j, \qquad S_i = \mathbf{1}_{i=0} + BI_i.$$

Every register coefficient on either side is at most $K < B$; uniqueness of base-B digits gives the initial conditions and each exact transition. The possible extra final slot vanishes because halt has no outgoing instruction and the terminal registers are zero. For $q = 0$ the same equations require the initial state already to be halt and the inputs to vanish. Conversely, any finite accepting run fits after choosing c sufficiently large, and supplies all these masks and equations.

10.2 The literal alpha and power formulas

Define $U_n = \alpha_b(n)$ by

$$U_0 = 0, \qquad U_1 = 1, \qquad U_{n+2} = bU_{n+1} - U_n \quad (n \geq 0).$$

For $b \geq 3$ this recurrence gives $0 \leq U_n < U_{n+1}$ and $U_n \geq n$. For $n \geq 1$, the matrix

$$M = \begin{pmatrix} b & -1 \\ 1 & 0 \end{pmatrix}, \qquad M^n = \begin{pmatrix} U_{n+1} & -U_n \\ U_n & -U_{n-1} \end{pmatrix}$$

has determinant one. Thus adjacent terms are coprime and $U_n^2 - bU_nU_{n-1} + U_{n-1}^2 = 1$. Parameter congruence follows directly from the recurrence: $b \equiv b' \pmod{d}$ implies $\alpha_b(n) \equiv \alpha_{b'}(n) \pmod{d}$. In particular $\alpha_2(n) = n$. This is parameter congruence, not merely congruence between indices.

Every natural solution of $x^2 + y^2 = 1 + bxy$, $b > 2$, is an unordered adjacent pair. To prove this, order it as $X > Y$. If $Y = 0$, $X = 1$. Otherwise $Z = bY - X$ satisfies $XZ = Y^2 - 1$, hence $0 \leq Z < Y$, and (Y, Z) satisfies the same equation. Descent reaches $(1, 0)$; reversing the steps generates exactly the adjacent pairs. No ordering assumption on the original pair is needed.

For $k > 0$, reduction of M^k modulo U_k is the invertible scalar $U_{k+1}I$. Dividing an index by k and using strict growth proves $U_k \mid U_m$ if and only if $k \mid m$. More precisely, for $m > 0$,

$$U_k^2 \mid U_m \quad \Longleftrightarrow \quad kU_k \mid m. \tag{C.1}$$

For $k = 0$ both sides are false. For $k > 0$ write $m = kq$ and expand $M^{kq} = (U_kM - U_{k-1}I)^q$ modulo U_k^2. Its lower-left entry is $qU_k(-U_{k-1})^{q-1}$, so coprimality proves (C.1), including $U_k = 1$.

For $m > 0$ put $v = U_{m+1} - U_{m-1} = \operatorname{tr}(M^m)$. Cayley–Hamilton gives $M^{2m} \equiv -I \pmod{v}$. Extend the recurrence by $U_{-j} = -U_j$. Every $n \geq 0$ has $n = 2lm + \varepsilon j$ with $l \geq 0$, $0 \leq j \leq m$ and $\varepsilon \in \{1, -1\}$; then $U_n \equiv (-1)^l \varepsilon U_j \pmod{v}$. The value sign need not equal the index sign. If $b \geq 4$, $v = bU_m - 2U_{m-1} > 2U_m$.

The actual alpha formula has natural witnesses r, s, t, u, v, w, x, y and the following fifteen conditions:

$$\begin{gathered} b > 3, \quad u^2 + t^2 = 1 + but, \quad s^2 + r^2 = 1 + bsr, \quad r < s, \quad u^2 \mid s, \quad v + 2r = bs, \\ w \equiv b \pmod{v}, \quad w \equiv 2 \pmod{u}, \quad w > 2, \quad x^2 + y^2 = 1 + wxy, \\ 2X < u, \quad 2X < v, \quad x \equiv X \pmod{v}, \quad 2N < u, \quad x \equiv N \pmod{u}. \end{gathered} \tag{C.2}$$

A strict inequality $a < b$ is $b = a+1+d$; divisibility is $b = ad$; congruence is $A + Mu = B + Mv$, with fresh natural witnesses. At zero modulus the last equation says equality, so there is no exception hidden in this replacement.

For soundness, the unordered Pell result gives $u = U_k$, $k > 0$, and $x = \alpha_w(n)$. The ordered pair gives $s = U_m, r = U_{m-1}$, $m > 0$. By (C.1), $u \mid m$. Parameter congruence gives $x \equiv U_n \pmod{v}$ and $x \equiv n \pmod{u}$. Reflect n to $j \in [0, m]$. Since $2X < v$ and $2U_j < v$, the congruence $X \equiv \pm U_j \pmod{v}$ forces $X = U_j$, with the negative case forcing both zero. Then $j \leq X$, $2j < u$ and $2N < u$ force $N = j$ from $N \equiv \pm j \pmod{u}$. Thus $X = U_N$.

Conversely choose an odd $u = U_k > 2U_N$. Such terms are unbounded, since adjacent terms are coprime and the sequence grows. Set $t = U_{k+1}$, $m = ku$, $s = U_m$, $r = U_{m-1}$ and $v = bs - 2r$. Equation (C.1) gives $u^2 \mid s$, and the required strict bounds hold. The moduli u, v are coprime: a common divisor is odd, divides s and $2r$, hence divides both adjacent terms. CRT gives $w \equiv 2 \pmod{u}$, $w \equiv b \pmod{v}$ with $w > 2$. Take $x = \alpha_w(N)$, $y = \alpha_w(N + 1)$. Parameter congruence verifies every remaining condition. This includes $N = X = 0$.

The literal power formula separates $Q = 0, R = 0, P = 1$ and $Q = 0, R > 0, P = 0$. For $Q \geq 1$ it imposes

$$a_0 = \alpha_{Q+4}(R+1), \quad b = a_0 + Q^2 + 2, \quad m + Q^2 + 1 = bQ, \quad P < m,$$
$$a_1 = \alpha_b(R), \quad a_2 = \alpha_b(R+1), \quad P + ba_1 \equiv Qa_1 + a_2 \pmod{m}. \tag{C.3}$$

Induction gives $a_0 \geq (Q+3)^R$, so $m > Q^R$ and $m > 0$. Let $E_R = QU_R - bU_R + U_{R+1}$. The recurrence and $bQ \equiv Q^2 + 1 \pmod{m}$ give $E_0 = 1$ and $E_{R+1} \equiv QE_R \pmod{m}$. Thus (C.3) says $P \equiv Q^R \pmod{m}$. Both values lie in $[0, m)$, forcing equality. Conversely equality supplies every witness. This includes $R = 0$, $Q = 1$ and $0^0 = 1$.

These formulas make the binomial and bit operations polynomial. For $B = 2^{n+1}$, the equation

$$(1+B)^n = B^k(Bq + C) + r, \quad C < B, \quad r < B^k$$

selects exactly $\binom{n}{k}$, since all its base-B coefficients are below B, including zero for $k > n$. The identity $(1+t)^n = \prod_{i:\,\mathrm{bit}_i(n)=1}(1+t^{2^i})$ over $\mathbb{F}_2$ makes odd binomial coefficients equivalent to bit inclusion. Bit disjointness of a, b is equivalent to inclusion of a in $a + b$: at the least shared bit there is no incoming carry and that bit is cleared. Finally $x = z + a, y = z + b$ with z, a, b pairwise bit-disjoint says exactly $z = x$ AND y.

10.3 Interpreter, lowering and five-input specialization

The universal program numbering is $e + 1 = 2^w(2D + 1)$. At $w = 0$ it halts. Otherwise the four base-2^w digits at positions $4p, \ldots, 4p+3$ give the opcode, register and two targets at state p. Opcode one increments, opcode two decrements or takes its zero branch; every other opcode halts. Missing digits are zero, so decoding is total.

The interpreter stores a finite tape as 0 or $\mathrm{cons}(h, t) = 2^h(2t + 1)$. Repeated halving decodes a nonempty head and tail; the tail is smaller than the code. To update a register, save the finite reversed prefix, replace the entry and restore the prefix. Absent entries are zero. The reserved jump register remains zero. Copies restore their source; aliased addition snapshots its operand before overwriting it. Division maintains $x = qy + r$ and subtracts only when $r \geq y$; the interpreter's divisors are at least two. These invariants prove that one interpreter loop executes exactly one decoded instruction, with finite work. A simulated halt enters cleanup and clears all registers. An infinite simulation never enters cleanup. Thus its all-zero halt is equivalent to virtual halting, without requiring the virtual final tape itself to be zero.

The precision IR uses normalized signed pairs for integer arithmetic and pairs over the basis $1, \sqrt{2}$ for field arithmetic. Norm zero implies the field element is zero: after clearing denominators, a primitive solution of $a^2 = 2b^2$ would have both entries even. Every division is guarded by that exact test. The Jacobian point formulas satisfy the curve equation and the affine chord or tangent formulas after multiplying denominators. Equal abscissas split into doubling and opposite-point branches; zero ordinate doubles to the identity. The fixed projective point is the same affine P_0. These are polynomial identities, independent of the generator theorem.

The scalar loop halves its remaining count. The precision loop divides its remainder by four. The capped arithmetic computes exactly $\min(x, L + 1)$; zero bases, unit bases and zero exponents are separate. For bases at least two an exponent exceeding L already exceeds the cap. Thus the ideal IR terminates and agrees with the stated rational predicate on all five natural inputs, including every early rejection.

Each primitive lowering uses fresh auxiliary registers, snapshots overlapping sources, and preserves the live-register map at the end of each finite block. Saturating decrement gives natural subtraction, and repeated addition gives multiplication. All variable divisions are guarded. Induction

on completed loop iterations proves the same claim for loops. Acceptance enters cleanup; rejection enters a permanent self-loop. Hence the primitive machine halts exactly on acceptance, although the ideal decision program is total.

For $\mathrm{pair}(a,b) = (a+b)(a+b+1)/2 + b$, unpairing maintains remainder = input $- s(s+1)/2$ and subtracts $s+1$ while $s <$ remainder. At exit $0 \le$ remainder $\le s$, so $b =$ remainder, $a = s - b$. Four repetitions put the five codes into registers zero through four. Fresh scratch lies above the old register range and is cleared; old work registers remain zero. Relocating every target by the prefix length therefore preserves the precision machine.

The polynomial specialization fixes the universal program input to this wrapper code, maps the old packed input to variable five, and maps every old variable $i \ge 2$ and private-block start by $i \mapsto i + 7$. The four pairing variables occupy five through eight, disjoint from the five inputs. The appended four squared equations force their unique Cantor values. Thus, for the exact saved polynomial C,

$$\forall \mathbf{c} \in \mathbb{N}^5, \qquad \exists \mathbf{z} \in \mathbb{N}^{1305806}\ C(\mathbf{c}, \mathbf{z}) = 0 \quad \Longleftrightarrow \quad \text{accepts_naturals}(\mathbf{c}) = \text{True}. \tag{C.4}$$

The natural compiler proof has no height, restricted-converse or rational-model premise. Its use of the height expression concerns only computation of that fixed integer expression.

10.4 Translation to point codes

Apply (C.4) using the exact signed index model. Induction on the actual source DAG gives each node the code of its signed integer value. Natural guards constrain source variables, not intermediate sums or products. Raw addition receives canonical codes; globally typed multiplication justifies the natural guards. Fixed generator equations and scalar templates supply constants.

The power-value recipe has three signed code inputs, seventy-nine private natural witnesses, and an output code for the value of the power-template polynomial. That value is not independently required to vanish. Its saved root equality is $[17751579, 15]$. The precision parent has 54056 explicit variables and 15845 disjoint private blocks of seventy-nine variables. No parent variable node accesses a private coordinate. Replacing each block by its private power-value call therefore preserves the full existential assignment. Only the final parent root is equated to the identity.

At each primitive formula, equality is a difference, conjunction is a sum of squares, and disjunction is a product. A topological induction, including the globally conjoined fixed equations, proves equality of polynomial values under substitution, not merely equisatisfiability of an unrelated formula. This is the required link to the saved AST.

11 A variable-shift relation for ordinary values

We prove the ordinary-value deduction from the exact index model, the compiled precision relation, the height theorem and PC. These are supplied by Sections 3, 10, 4 and Theorem 5.2, respectively. Capital letters below denote five-coordinate point codes; their lower-case indices are ordinary integers. The external variable t is any rational number.

11.1 Approximation for every fixed shift

Define $\mathrm{Close}(N, \mathcal{K}, t)$ by choosing A, L, J, linking all five signed codes to precision inputs in the order (n, κ, a, l, j), and imposing $|t - r(a,j)| < \epsilon_l$ on those same coordinates. Put

$$B(n,\kappa) = H_6(S_{\mathrm{ht}})|\kappa F(n)|^{24}.$$

The precision contract gives

$$\mathrm{Close}(\iota(n),\iota(\kappa),t) \Longrightarrow |t-n| < e^{-B(n,\kappa)}, \tag{1}$$

$$\mathrm{Close}(\iota(n),\iota(\kappa),n) \quad \text{for every } n \notin \mathcal{C},\ \kappa \neq 0. \tag{2}$$

The first follows from $|t-n| < 2\epsilon_l < 4^{-B}/2 < e^{-B}$. For the second, positive multiples of P_0 approach the identity simultaneously in the compact group $E(\mathbf{R}) \times E(\mathbf{R})$. This follows, for instance, by partitioning the compact closure of the cyclic subgroup into arbitrarily small neighborhoods and subtracting two sufficiently close multiples. If the resulting positive differences were bounded, a fixed nonzero multiple would be the identity; infinite order excludes this. Choose l first to make the positive ϵ_l small enough. Since the derivative of $[n]$ at the identity is n, $r(a, na) \to n$ along positive multiples a approaching it. Choose such an a and put $j = na$. This works for every already fixed nonzero shift κ, regardless of its size.

11.2 The six local tests

For $N = \iota(n)$, $\mathcal{K} = \iota(\kappa)$, form the code for κn, choose $H = \iota(h)$, $U = \iota(s_0)$, and put $V_0 = H - \mathcal{K}$, $V_1 = U - \iota(\kappa n)$. Require $\kappa \neq 0$, $H, V_0 \neq O$. Define $w(Q) = 1/x(Q)$ for $Q \neq O$, $w(O) = 0$. For $t \notin \mathcal{C}$ use

$$D_1 = w(H), \qquad D_2 = w(U), \qquad D_3 = \sigma w(V_0), \qquad D_4 = \sigma w(V_1),$$
$$D_5 = z(U)/z(H) - t, \qquad D_6 = \sigma(z(V_1)/z(V_0)) - t.$$

Here σ is conjugation in K_0. Choose $\eta, q \in \mathbb{Q}$ and impose

$$\eta^2 = 1, \qquad R_0 = \eta 2730^6 F(t), \qquad a_0 = 1 - R_0 q^2 \neq 0,$$
$$R = R_0 a_0^2, \qquad C_j = D_j/F(t), \qquad W = \prod_{j=1}^{6}(1 + RC_j^2) \neq 0,$$
$$\Theta = \mathrm{Tr}_{K_0/\mathbb{Q}}(W^{-1}) \neq 0, \qquad B_{\mathrm{pole}} = R/\Theta^2, \qquad \Phi(B_{\mathrm{pole}}).$$

Call this relation $\mathrm{Test}(N, \mathcal{K}, t)$. All quotients are guarded polynomially. There is one common denominator $F(t)$, one product W, one trace and one pole instance.

Lemma 11.1 (Divisibility)**.** *If* $\mathrm{Test}(\iota(n),\iota(\kappa),t)$ *holds, then* $M_{S_{\mathrm{ht}}}(t) \mid \kappa F(n)$.

Proof. Let $p \mid M_{S_{\mathrm{ht}}}(t)$. Exactly one $c \in \mathcal{C}$ has $\delta = v_p(t-c) > 0$ odd. The contact differences are units, so $v_p(F(t)) = \delta$; the field is unramified at p. Put $b = v_p(a_0)$. If $q = 0$, then $b = 0$. Otherwise $\delta + 2v_p(q)$ is odd and nonzero, so $b = \min(0, \delta + 2v_p(q)) \leq 0$. In either case $\alpha = v_p(R) = \delta + 2b$ is odd. At each place above p,

$$v(1 + RC_j^2) = \min(0, \alpha + 2(v(D_j) - \delta)).$$

Let $\lambda = \max_{v|p} v(W) \leq 0$. Then $v_p(\Theta) \geq -\lambda \geq 0$. The pole input has odd valuation $\alpha - 2v_p(\Theta)$, which is positive by Lemma 5.1. Therefore $\alpha > 0$ and $\lambda > -\alpha/2$.

At a place attaining λ, each nonpositive factor valuation is at least λ. Consequently

$$v(D_j) \geq \delta - 3\alpha/4 = \delta/4 - 3b/2 > 0 \quad (1 \leq j \leq 6).$$

Integral valuations are at least one, at the same place for all tests. The first four put H, U in one formal group and V_0, V_1 in the conjugate formal group. At the identity we set both $z(O) = 0$

and $\log(O) = 0$; thus the cases $U = O$ and $V_1 = O$ obey the same ratio identities below. Their denominators are nonzero by the guards on H and V_0. For a nonzero point in either formal group,

$$\log(Q) = z(Q)(1 + e_Q), \qquad v(e_Q) \geq 1.$$

The invariant differential is integral, and integration loses at most $v_p(r)$ in degree r; for $r \geq 2$, $(r-1)v(z) - v_p(r) \geq 1$. The relation $[s_0]H = [h]U$ gives $\log(U)/\log(H) = s_0/h$. The fifth test makes this integral and congruent to t modulo p. Likewise the sixth gives $(s_0 - \kappa n)/(h - \kappa) \equiv t$. Multiplying by the integer denominators and subtracting yields $v_p(\kappa(n-t)) \geq 1$. Since $t \equiv c$, we get $p \mid \kappa(n-c)$, hence $p \mid \kappa F(n)$. The squarefree product divides as asserted. No unit assumption on κ was used. □

Lemma 11.2 (One suitable shift)**.** *For every $n \in \mathbb{Z} \setminus \mathcal{C}$ there is $\kappa \neq 0$ with* $\mathrm{Test}(\iota(n), \iota(\kappa), n)$.

Proof. Choose the sign so $R_0 = \eta 2730^6 F(n) > 0$ and let S' be its finite prime support. Each sufficiently small local formal neighborhood has finite index in the compact local group. A positive common multiple of the quotient orders gives κ with $[\kappa]P_0$ in every chosen neighborhood. Take

$$H = [2\kappa]P_0, \quad U = [2\kappa n]P_0, \quad V_0 = [\kappa]P_0, \quad V_1 = [\kappa n]P_0.$$

The formal parameter and the derivative of $[n]$ make each C_j integral above S' after sufficiently shrinking the neighborhoods. Infinite order preserves the nonzero guards.

Choose rational q integral at S' and arbitrarily close to $1/\sqrt{R_0}$ in $\mathbf{R}$, without equality. Denominators $R_0 m + 1$ with nearest integer numerators give such choices, adjusting by one if equality occurs. At S', a_0 is a unit and $W \equiv 1$, $\Theta \equiv 2$ modulo $p^{v_p(R_0)}$. Integral trace justifies this also at the ramified place two. The pole input has valuation at least four at two and at least six at the other primes of S_0. Outside S', its valuation is $2v_p(a_0) - 2v_p(\Theta)$ and is even, so its reduced denominator is a square. In both real embeddings, R tends to zero positively, $W \to 1$ and $\Theta \to 2$. All guards hold and $0 < B_{\mathrm{pole}} < 1/2$ for a sufficiently close choice. Theorem 5.2 supplies the pole witnesses. □

11.3 Exact ordinary value

Define $\mathrm{Val}(N, t)$ by membership of N and the disjunction

$$\bigvee_{c \in \mathcal{C}} (N = \iota(c) \wedge t = c) \quad \vee \quad \exists \mathcal{K}\, (\mathrm{Close}(N, \mathcal{K}, t) \wedge \mathrm{Test}(N, \mathcal{K}, t)).$$

The same five coordinates $\mathcal{K}$ are used in both calls. In an accepting main branch $n \notin \mathcal{C}$, $\kappa \neq 0$, and Lemma 11.1 and the height estimate imply

$$h(t) \leq H_6(S_{\mathrm{ht}})|\kappa F(n)|^{24} = B(n, \kappa).$$

For $t = a/b \neq n$ in reduced form, $|a - nb| \geq 1$, hence

$$|t - n| \geq 1/b \geq e^{-h(t)} \geq e^{-B(n,\kappa)},$$

contradicting (1). Thus $t = n$. Conversely choose κ by Lemma 11.2, then use (2) for that same shift. Private witness blocks are disjoint, so the two calls can be satisfied simultaneously. The six remaining integers have exactly their displayed branches.

Therefore $\mathrm{Val}(\iota(n), t)$ is exactly the graph $t = n$. Existentially quantifying N gives ordinary $\mathbb{Z}$ in $\mathbb{Q}$. S01 and S16 multiply the exception and main-branch polynomials and conjoin membership by a sum of squares. The literal substitutions in Section 2 and Appendix C identify this polynomial with the saved root P_*. This proves the zero-set part of Lemma 1.2.

12 The cutoff and polynomial size

The literal constants in S03 define the catalogue product H_{cat}, its maximum $H_{\max}$, $N_s = P_s H_{\mathrm{cat}}$ with $P_s = \prod_{p \le 324} p$, the least formal multiple M_0, and $c_* = 4M_0$. They take

$$R = \max(D, H_{\max}, c_*, 10^{12}), \quad A_{\mathrm{ht}} = 2 \cdot 10^{13} \max(1, C_{\mathrm{cm}}, Q_R^{17}), \quad T = (48 A_{\mathrm{ht}})^{13}.$$

The pole clearing constant is $\ell_0 = 12 P_s^{2916} H_{\mathrm{cat}}^{732}$. The symbol $F_0 = 324!$ is distinct from the six-contact polynomial $F(u)$.

Proposition 12.1 (Literal cutoff)**.** *The fixed prescription satisfies*

$$R = D, \qquad T = (96 \cdot 10^{13})^{13} D^{2210 G_0^2 D^3} < 2^{2^{2^{9013}}}.$$

This is a statement about the prescribed integer, independently of its semantic role in excluding false multiplication triples.

Proof. We give the bounds of S25. Exact small calculations give $10^5 < W < 2^{512}$ and $2^{2000} < F_0 < 2^{2250}$. For $q \ge 10^5$, the unconditional progression estimate [11, Theorem 1.2, (1.10)–(1.11), Corollary 6.17 with $A = 1$] applies for $x \ge \exp(0.03\sqrt{q}(\log q)^3)$, including the possible exceptional-zero case. At $X = 2^{W^3}$ its error is at most $X/(160 \log X)$ and the hypothesis follows from $0.03\sqrt{W}(\log W)^3 < (\log 2) W^3$. Hence $\vartheta(X; W, a) > X/(2W)$, giving more than $X/(2W^4) > 22$ primes in each unit class. No prime at most 324 belongs to such a class. Thus

$$H_{\max} < X, \quad H_{\mathrm{cat}} \le X^{22W}, \quad \log_2 N_s < 2^{2054}.$$

For every odd prime dividing N_s, take the actual reduction order of P_0 at each place above it, and let their product be m_0. All preceding positive multiples stay in the dyadic formal subgroup, since P_0 starts there and the subgroup is closed under addition. A nonidentity point there has negative abscissa valuation, so its reciprocal is defined; the constructor accepts the identity directly. Thus every earlier search step is defined without a nontorsion premise. The point has nonsingular reduction also at the bad places over seven. Each order is at most p^3 and there are at most two places, giving $m_0 \le N_s^6$. Multiplication by N_s^4 raises the formal depth enough that the rational coefficients of the reciprocal abscissa are divisible by N_s^4, with denominator prime to N_s. At two use the valuation with uniformizer $\sqrt{2}$, so $v(2) = 2$. The initial valuations are $v(x) = -6$, $v(y) = -9$, $v(z) = 3$, and $v([2]z) \ge v(z) + 2$ at this depth. Therefore the literal search succeeds by $N_s^4 m_0$ and $M_0 \le N_s^{10}$. This uses actual finite orders, not numerical upper bounds incorrectly treated as multiples of those orders.

Consequently $c_* < 2^{2^{2059}}$, whereas $2^{2^{8008}} < D < 2^{2^{9010}}$, so $R = D$. Next $C_{\mathrm{cm}} < 2^{50000 D^2}$ and

$$\log_2 Q_D^{17} = 170 G_0^2 D^3 \log_2 D > 50000 D^2.$$

The maximum is therefore Q_D^{17}, giving the identity for T. Finally $G_0 < 2^{37}$, $E_0 < D^3$, $\log_2 Q_D < D^5$, $\log_2 A_{\mathrm{ht}} < D^7$, and $\log_2 T < D^8 < 2^{2^{9013}}$. □

12.1 Quantitative syntax bounds

Use coefficient ℓ^1 norm for integer polynomials and the weighted pair norm $\|a + bs\| = \|a\|_1 + 2\|b\|_1$. They are submultiplicative. The division-polynomial recurrences give

$$\deg f_n \le (n+1)^3, \qquad \|f_n\| \le 2^{100(n+1)^3}.$$

For the even recurrence the exponent slack is $4m^3 - 24m - 27 \geq 9$ for $m \geq 3$; for the odd recurrence it is at least ten for $m \geq 2$. This absorbs the fixed factors and addition.

The following facts have been checked for the actual saved expression: the absolute values of all literal fixed constants and all occurring division-polynomial indices are at most T; the specified syntax-height traversal has root height at most 256; each model recipe has at most 2^{20} constraints and each composition recipe has at most 2^{20} terms. The check accounts for all 23 templates, seven recipes, private witness blocks, 173 scalar constant nodes and 50 finite constructors. Its exact artifact identity and independent replay are recorded in the source map. Together with the constructor bounds below, this discharges the quantitative part of B. The all-input arithmetic interpretation of the same constituents is the separate substitution induction of Section 10.

Put $B_* = T + 1$. Coordinates of $\iota(n)$ for $|n| \leq T$ have absolute numerator and positive denominator at most 2^{K_*}, with $K_* = 2^{30}(T+1)^3$. Evaluate the division polynomials at P_0 and use the fixed coordinate formulas; at most eight quadratic-field operations are needed, each taking an available height bound H to at most $3H^4$. There are at most $8(T+1)^3$ guard factors. The resulting bounds for each finite constructor are

$$\deg \leq B_*^4, \qquad \log_2 \max(1, \|\cdot\|_1) \leq B_*^8.$$

Give variables syntax height zero, constants height one, and add one for each arithmetic operation or finite constructor. A template call adds one plus the callee's intrinsic height and the maximum argument height. The substitution inequalities

$$\deg P(Q) \leq \deg P \max(1, \max_i \deg Q_i), \qquad \|P(Q)\|_1 \leq \|P\|_1 \max(1, \max_i \|Q_i\|_1)^{\deg P}$$

give degree and logarithmic norm at most B_*^{8h} at positive height h, with the bounded sums in the actual recipes treated as in S26. This general call rule applies to polynomial arguments. In each outer composition recipe the arguments are coordinate variables, so the call is only a renaming, even when coordinates repeat. For any such map σ,

$$\deg P(Y_{\sigma(i)}) \leq \deg P, \qquad \|P(Y_{\sigma(i)})\|_1 = \sum_\beta \left| \sum_{\alpha \mapsto \beta} c_\alpha \right| \leq \sum_\alpha |c_\alpha| = \|P\|_1.$$

Thus these calls retain the callee's majorant height before charging the sum-of-squares or product connective. The seven recipe heights are $228, 233, 235, 251, 252, 254, 256$. A rule charging an extra unit for each of these mere renamings would describe a looser syntax-depth bound, not this proved degree-and-norm majorant. For the model wrappers use the safe height bound four for the generator equations and root differences; the callee contribution dominates this bound. The literal height-two claim for those equations in S26 is not used. This height is a majorant for the compact constructor syntax, not for an expanded monomial tree. The height-256 bound yields

$$\deg P_* \leq (T+1)^{2048}, \qquad |[X^\alpha]P_*| \leq 2^{(T+1)^{2048}}.$$

Since T is an integer, $T + 1 \leq U = 2^{2^{2^{9013}}}$. Hence

$$(T+1)^{2048} \leq U^{2048} = 2^{2^{2^{9013}+11}} < 2^{2^{2^{9014}}}.$$

As $9014 < 65536 = 2 \uparrow\uparrow 4$, the degree is less than $2 \uparrow\uparrow 7$ and the coefficient magnitude less than $2 \uparrow\uparrow 8$. The root recipe declares 292 890 231 904 witnesses. Together with the preceding estimates, these complete the bounds for the same P_* in Theorem 1.1.

13 Consequences of the integer definition

The proof of Theorem 1.1 combines the following results.

Result	Mathematical inputs	Proof location
Index group	Two-isogeny descent, reductions, finite height search	Section 3
Six-contact bound	Quaternionic model, tame pullback, compatible-isogeny height theorem, theta comparison	Section 4
Pole completeness	Simultaneous descent, restricted converse, forward rank-one theorem	Section 5
Multiplication	Index group, height bound, selected primes, pole completeness, finite guard	Section 6
Precision relation	Decision program, lowering, power and packed histories	Section 10
Ordinary value	Shared-shift Close and Test, height and rational separation	Section 11
Bounded P_*	Literal substitutions and quantitative syntax	Sections 2, 12; Appendix C

13.1 Three corollaries

We apply the exact equality between the rational zero projection of P_* and ordinary $\mathbb{Z}$ proved in Theorem 1.1.

Corollary 13.1 (Negation of Mazur's finite-component conjecture)**.** *Mazur's finite-connected-components conjecture for the real-topological closure of rational points is false.*

Proof. Mazur's Conjecture 3 in [32, p. 165] asserts that the closure of the rational points of any variety over $\mathbb{Q}$ in its real locus has finitely many connected components. In that source a variety means a reduced scheme of finite type; irreducibility is not required. Put $r = 292\,890\,231\,904$ and

$$V = \left(\{P_*(t, w_1, \dots, w_r) = 0\} \subset \mathbf{A}^{r+1}_{\mathbb{Q}}\right)_{\mathrm{red}}, \qquad C = \overline{V(\mathbb{Q})} \subset V(\mathbf{R}).$$

Reduction changes neither the rational nor the real points. Let $\pi : V(\mathbf{R}) \to \mathbf{R}$ be projection to t. Theorem 1.1 gives $\pi(V(\mathbb{Q})) = \mathbb{Z}$. Since $\mathbb{Z}$ is closed in $\mathbf{R}$ and π is continuous, $C \subset \pi^{-1}(\mathbb{Z})$, so $\pi(C) \subset \mathbb{Z}$. Every integer already occurs on $V(\mathbb{Q})$, hence $\pi(C) = \mathbb{Z}$. The image of any connected component of C is a connected subset of $\mathbb{Z}$, therefore a singleton. Points projecting to distinct integers belong to distinct components. Thus C has infinitely many connected components, contradicting the stated conjecture. □

Corollary 13.2 (A universal rational Diophantine polynomial)**.** *There exist $m \geq 1$ and a fixed polynomial $U \in \mathbb{Z}[e, n, x_1, \dots, x_m]$ such that, for every $e, n \in \mathbb{N}$,*

$$\exists \mathbf{x} \in \mathbb{Q}^m \; U(e, n, \mathbf{x}) = 0 \quad \Longleftrightarrow \quad \textit{machine } e \textit{ halts on input } n.$$

Here e and n are ordinary integer parameters, with $\mathbb{N}$ embedded in $\mathbb{Q}$, and are not elliptic-curve coordinates. In particular, every computably enumerable subset of $\mathbb{N}$ is represented by a suitable fixed value of e in this one polynomial.

Proof. Fix an effective enumeration of machines. The Davis–Putnam–Robinson–Matiyasevich theorem, [33, Basic Theorem, p. 3], applied to the universal halting relation gives a fixed $D \in \mathbb{Z}[e, n, z_1, \dots, z_s]$, with $s \geq 1$, such that, for $e, n \in \mathbb{N}$,

$$\text{machine } e \text{ halts on } n \quad \Longleftrightarrow \quad \exists \mathbf{z} \in \mathbb{Z}^s \; D(e, n, \mathbf{z}) = 0.$$

To use integer witnesses in this statement, replace each nonnegative witness by a sum of four integer squares. In the positive-integer convention of the cited source, apply the theorem to the predicate that machine $a-1$ halts on $b-1$, substitute $a=e+1$, $b=n+1$, and use one plus that sum for each witness; see also [33, p. 2]. A dummy witness ensures $s \geq 1$ if needed. For each z_i take a separate block $\mathbf{w}_i$ of r rational variables, where r is the witness count of P_*. Define

$$U(e, n, \mathbf{z}, \mathbf{w}_1, \ldots, \mathbf{w}_s) = D(e, n, \mathbf{z})^2 + \sum_{i=1}^{s} P_*(z_i, \mathbf{w}_i)^2.$$

A rational zero makes every summand zero. Theorem 1.1 then forces every z_i to be an integer and yields the displayed halting condition. Conversely, integer witnesses for D have separate rational witness blocks for P_*, giving a zero of U. All blocks are private and the polynomial is fixed independently of e, n. No variable, degree or coefficient bound proved for P_* is asserted for this new polynomial U. □

Corollary 13.3 (The many-one degree of $\mathrm{HTP}(\mathbb{Q})$)**.**

$$\mathrm{HTP}(\mathbb{Q}) \equiv_m 0', \qquad 0' = \{e \in \mathbb{N} : \textit{machine } e \textit{ halts on input } e\}.$$

Here $\mathrm{HTP}(\mathbb{Q})$ *is the set of codes of integer-coefficient polynomials, of arbitrary finite arity, having a rational zero, under a fixed effective coding.*

Proof. Given e, specialize the fixed polynomial of Corollary 13.2 at its ordinary integer parameters to obtain $U(e, e, \mathbf{x})$. Computing its coefficient list and code is effective, and

$$e \in 0' \quad \Longleftrightarrow \quad \ulcorner U(e, e, \mathbf{x}) \urcorner \in \mathrm{HTP}(\mathbb{Q}).$$

This proves $0' \leq_m \mathrm{HTP}(\mathbb{Q})$.

Conversely, from the code of a polynomial f construct a machine that ignores its input and enumerates all rational tuples of the appropriate arity. It evaluates f using exact rational arithmetic and halts as soon as it finds a zero. Its index $j(f)$ is computable from the finite code of f, and

$$f \in \mathrm{HTP}(\mathbb{Q}) \quad \Longleftrightarrow \quad j(f) \in 0',$$

where the left side uses the code of f. Thus $\mathrm{HTP}(\mathbb{Q}) \leq_m 0'$. Invalid polynomial codes, if present in the coding convention, are sent to a fixed nonhalting index. Using rational-coefficient polynomial codes gives the same result, since clearing their finitely many denominators is effective and preserves rational zeros. The two reductions prove the assertion. □

More generally, an integer Diophantine problem translates to a rational one by adjoining P_* for every integer witness, with private witness blocks. Natural witnesses can first be expressed as sums of four integer squares. Bounds for the resulting polynomial require their own composition calculation; the bounds for P_* do not automatically transfer.

For example, a machine that enumerates ZFC proofs and halts upon finding a contradiction has a rational Diophantine halting condition by Corollary 13.2. This equivalence does not assert that a contradiction exists, and no additional explicit polynomial is specified here.

A Dependency order in the arithmetic argument

The even coefficient bound uses stationary-base selection as well as integrality. The odd minimum uses a fixed nonzero witness and the prior genus-index lower bound. Support removal is applied only to the actual unramified specialization. The functional bound and determinant clearing are proved separately, and the same corrected coordinate is used at every address. These constructions are given in Sections 7–9 and Appendices E–G.

The conversion to ordinary integers uses the canonical model and total multiplication, followed by the exact computation relation. Only the shared-shift separation argument in Section 11 identifies the external rational variable with the ordinary value of its integer index. The literal substitution and size calculation then apply to the same expression P_*.

B Source identities and scope

The companion `SOURCE_MAP.json` records exact paths, hashes and locators, distinguishing mathematical and construction sources, artifacts, and version records. The base collection contains 83 sources; parity, construction and uniformity supplements are in separate addenda. The principal local sources are listed below; full paths are in the source map.

ID	Source basename	Use
S01	`ordinary-integer-polynomial.js.txt`	Outer construction
S02	`rational-circuit-translator.js.txt`	Substitution semantics
S03	`finite-polynomial-expressions.js.txt`	Fixed constructors
S04	`variable-shift-polynomial-library.js.txt`	Close and Test
S05	`elliptic-generator-proof.txt`	Generator certificate
S06	`explicit-elliptic-data.txt`	Rank, torsion, codes
S07	`verify-elliptic-generator.js.txt`	Exact finite algorithm
S08	`six-contact-height-proof.txt`	Contacts and constants
S09	`z-in-q-height-independent-audit.txt`	Height proof
S10	`explicit-pole-predicate.txt`	Pole polynomial
S11	`z-in-q-pole-completeness-audit.txt`	Simultaneous descent
S12	`parity-frozen-4eaee875.tex`	Raw local matrices
S13	`multiplication-certificate-correctness-proof.txt`	Multiplication proof
S14	`twenty-two-prime-pattern-proof.txt`	Selected primes
S15	`z-in-q-variable-shift-bridge.txt`	Ordinary-value graph
S16	`z-in-q-ordinary-integer-assembly-proof.txt`	Outer composition
S17	`z-in-q-restricted-converse-assembly.txt`	RC assembly inputs
S18	`ring-limits-frozen-4eaee875.tex`	Finite models and limits
S19	`paired-functional-and-uniformity-proof.txt`	Paired functional
S20	`determinant-transfer-audit.txt`	Clearing and transfer
S21	`variable-shift-precision.py.txt`	Exact decision program
S22	`precision-register.js.txt`	Ideal IR and lowering
S23	`specialize-precision.js.txt`	Wrapper specialization
S24	`integer-halting-correctness-proof.txt`	Natural compiler
S25	`z-in-q-explicit-cutoff-bound.txt`	Literal cutoff
S26	`z-in-q-polynomial-size-bound-proof.txt`	Syntax/norm argument

ID	Source basename	Use
S27	`z-in-q-numerical-polynomial-bounds.txt`	Tower comparison
S28	`determinant-switch-audit.txt`	Switches and outer rank
S29	`ring-determinants-frozen-4eaee875.tex`	Ring data and derivatives
S30	`preliminaries-frozen-4eaee875.tex`	Selmer coranks and lengths
S31	`fixed-certificate-constants.js.txt`	Exact least-multiple search
S32	`pointwise-frozen-4eaee875.tex`	Distinct point indices

The supplement manifest also records the revised fixed-degree CM proof, the complete natural compiler proof, actual arithmetic and analytic producer arguments, finite-Artin and determinant-count clarifications, the fixed BFH witness application, exact substitution checks, and the new fixed-form, weighted-theta, operator and normalization derivations. Primary theorem applications remain explicit in the mathematical text. The version records identify the source files used in the construction.

C Literal finite-constructor specification

The supplementary file `ordinary-integer-definition.json` is part of this specification. It is a finite directed acyclic description, not an expanded monomial list. The following rules determine one polynomial from its bytes. Diagnostic labels and intended-theorem strings add no equations.

The root is `integers`. Variable zero is the external input and variables one through 292890231904 are its rational witnesses. A nonnegative point reference r means $X_r, \dots, X_{r+4}$. References $-1, -2, -3$ respectively mean

$$(1,0,0,0,0), \quad (0, g_0, -145, g_1, g_2), \quad (0, g_0, -145, -g_1, -g_2).$$

Every model recipe conjoins $8g_0 - 1641 = 0$, $4g_1 + 15465 = 0$, $32g_2 - 87483 = 0$.

C.1 Scalar nodes and fixed expressions

Read nodes in their stored topological order. A `const` is its decimal integer; `var` is its local variable; `add` and `mul` are ordinary polynomial operations. For a `call` to a template with p parameters, substitute the specified node polynomials for variables $0, \dots, p-1$, and X_{w+j} for variable $p+j$, where w is the witness start. The value is the callee's entire root polynomial.

A `fixed_integer` evaluates a finite expression using literals, references to preceding definitions, addition, multiplication, nonnegative powers, maximum, factorial, the catalogue product or maximum, and the least formal multiple. The catalogue uses the first 22 primes in increasing order in each unit class. The searches are deterministic and terminate by Section 12. They produce constants, not quantified witnesses.

A `psi_component` uses the division-polynomial recurrence in Section 2, pair multiplication $(a,b)(c,d) = (ac + 2bd, ad + bc)$ and $A = (-11243, 7950)$. It returns component zero or one; every index is fixed. The positive fixed point codes use

$$x_n = x(P_0) - \frac{\psi_{n+1}(P_0)\psi_{n-1}(P_0)}{\psi_n(P_0)^2}, \qquad y_n = \frac{\psi_{2n}(P_0)}{2\psi_n(P_0)^4}.$$

Nontorsion makes these fixed denominators nonzero. Negative indices negate both ordinate components.

A `finite_guard_product` ranges over $(a, b, c) \in [-T, T]^3 \cap \mathbb{Z}^3$ with $ab \neq c$, with fixed T. Write the fifteen coordinates of the three codes as reduced A_j/B_j, $B_j > 0$, and put $D = \prod_j B_j$. Its factor is

$$\sum_{j=1}^{15} (DX_j - (D/B_j)A_j)^2.$$

Multiply these factors, with empty product one. This is an integer polynomial with precisely the finite false-triple zero set used above.

C.2 Recipe constraints and witness scope

A model-evaluation recipe sums squares of the generator equations, listed natural calls, scalar-constant calls, add/multiply/power-value calls, and the five coordinate differences of `rootEquality`. A source-variable row with a non-null natural witness start contributes exactly that natural call. Operation kinds zero, one and two select addition, multiplication and power value. Source-node and provenance fields add no equations.

A sum-of-squares composition sums squares of its callee polynomials; a product composition multiplies them. Parameter `[var,i]` means X_i. Each private block uses its recorded start and length. The checked ranges are disjoint from parameters and other private blocks. These rules include globally conjoined fixed equations in active callees.

The following parameter and witness counts are extracted from the exact specification, without expanding its polynomial.

Template	Parameters	Witnesses	Root node
`member`	5	7	149
`raw_add`	15	6	308
`raw_z`	7	1	68
`raw_inverse_x`	7	0	54
`log25`	4	0	229
`pole`	1	11	106
`test_group`	44	162	515
`slope`	22	1,298	2681
`height_certificate`	36	35,106	260
`multiply.scalar`	10	1	136
`multiply.guard`	15	1	22
`multiply`	15	35,134	83
`natural`	5	140,604	79
`value.signed_encode`	10	140,654	138
`value.abs`	2	4	23
`value.close_coordinates`	16	53	271
`value.test`	11	35,282	775
`value.exceptions`	6	406	487
`scalar.integer_constant_0`	10	1	136
`scalar.integer_constant_1`	10	1	136
`scalar.integer_constant_2`	10	1	136
`scalar.integer_constant_3`	10	1	136

Template	Parameters	Witnesses	Root node
`scalar.integer_constant_4`	10	1	136

Recipe	Parameters	Witnesses	Total variables
`power_value`	20	17,786,698	17,786,718
`precision`	25	292,889,492,836	292,889,492,861
`close`	11	292,890,196,199	292,890,196,210
`main`	6	292,890,231,486	292,890,231,492
`value_union`	6	292,890,231,892	292,890,231,898
`value`	6	292,890,231,899	292,890,231,905
`integers`	1	292,890,231,904	292,890,231,905

The five scalar-constant templates use the values in `library.fixedConstants`. Large program literals are retained byte-for-byte in the supplementary specification. Replacing them by unspecified sufficiently large constants would change this definition.

D Primary theorem applications

Input	Exact application boundary
Selmer parity [16]	Rational two-isogeny, $p = 2$, full Selmer corank; no finite-Sha hypothesis.
Kato [19]	Fixed weight-two form and symbol, integral transfer and fixed-valuation smoothing; rational divisibility is Theorem 13.4(2), with oddness, rational irreducibility and the unipotent condition.
BFH and Hoffstein–Luo [20, 21]	Actual signed-prime twist with trivial character and arbitrary finite splitting set. Only the nonvanishing clause has the four-prime bound before the extra prime.
Waldspurger and Ramsey [26, 27]	Full good-Hecke eigenspace, same dyadic squareclass, and algebraic vectors by flat field extension. No arbitrary-ring base change in weight 3/2.
CM norms and heights [24, 23, 25]	Split level, conductors coprime to level and the new prime, no extra units, unaveraged sums and one fixed parametrization.
Duality [12, 13]	Actual derived Kummer conditions and their homotopies. Support change only after the coefficient representation is unramified at the removed place.
Forward finiteness [17, 18]	Only at known analytic nonvanishing/simple zeros or a known nontorsion Heegner trace.
Isogeny ratio [12, proof of I.7.3]	Finite Sha, dual sum/difference maps, canonical heights and pullback measures; no BSD leading-term assumption.
Height [4, 6]	Compatible isogenies or same-field maximal-order replacement with bounded degree.
Primes [8, 9, 10, 11]	Fixed finite-complexity system after fixing sieve cutoff, and the unconditional explicit progression estimate.

The source package includes the specification, natural precision polynomial and construction sources in `SUPPLEMENT_MANIFEST.json`. The larger proof corpus is identified by hashes and stable locators in `SOURCE_MAP.json`. A source identity is not a proof verdict.

E Fixed-form arithmetic and the even coefficient source

This appendix supplies the constructions used in (3.1)–(3.3). Throughout, $E = E_\ell$ is fixed, with the lattice, parametrization and period convention of Section 7. Constants may depend on these data and on the prescribed finite list of local types. They do not depend on the number or sizes of the later good conductor primes.

E.1 One retained class, including discriminant $4h$

Fix the weight-two newform of E and an integral modular-symbol combination ξ whose projections on both period lines are nonzero. Such a combination exists because the symbol span is not the union of the kernels of these two nonzero rational functionals [19, Sections 13.6–13.7]. A fixed lattice comparison sends its cohomological lattice into T_2E after multiplication by 2^{c_E}. This is a comparison for the original form, before any character or trace level is chosen.

Kato's classes are formed in j_* cohomology at finite coefficients. The Leray edge map identifies this group with the classes whose restrictions to inertia away from two vanish. Restricting a corestriction is a Mackey sum of transfers of upper inertia restrictions. Each summand is zero for the retained class. Consequently even a ramified transfer preserves the integral condition. Tensoring by a character line is performed over a field on which that character is trivial. There is no division by an inertia index.

For a finite binary family with group G, use the common odd modulus F and retained-smoothed class Z_F defined in Section 7. The coefficient map is the integral map on group bases

$$[a] \longmapsto \chi_0(a)[a|_\Gamma][\psi(a)].$$

Inverse-action Shapiro gives the quadratic twist and inverse cyclotomic action in this convention. In particular this is not an averaged character projector. At a constituent χ with odd conductor $d \mid F$, the actual classes

$$y_m = \mathrm{Cor}_{\mathbb{Q}(\mu_{dm})/\mathbb{Q}(\mu_m)}(z_{dm} \otimes e_\chi)$$

satisfy the twisted norm relation. Transitivity of the same transfers gives

$$(Z_F)_\chi = \prod_{q|F/d} \left(1 - \frac{a_q \chi(q)}{q}\sigma_q^{-1} + \frac{1}{q}\sigma_q^{-2}\right) y_{2^\infty}. \tag{E.1}$$

The factors are present in the classes before any divisibility theorem is applied.

For an odd signed squarefree parameter h, write its fundamental discriminant as Δ_h and

$$|\Delta_h| = 2^a d, \qquad \chi_h = \chi_2 \chi_{\mathrm{odd}}.$$

Here $a = 0$ or 2; allowing the finite list $a \le 3$ does not change the argument. At cyclotomic levels $n \ge \max(a, 2)$ use the actual map

$$[a_0] \longmapsto \chi_2(a_0)\chi_{\mathrm{odd}}(a_0)[a_0|_{\Gamma_n}]. \tag{E.2}$$

Its coefficients are signs. A real-sign quotient is integral. If the dyadic character restricts nontrivially to Γ, the resulting change is the integral automorphism

$$1 + t \longmapsto -(1 + t), \qquad t \longmapsto -2 - t.$$

It must be retained in the module action. To compare with the Euler system, define y_m using $|\Delta_h|m$ in place of dm. Kato's norm factor is a product over newly introduced primes [19, Proposition 8.12]. Increasing an already present power of two therefore contributes no factor. Factoring the transfer at $m = 2^n$ through $\mathbb{Q}(\mu_{d2^n})$ proves that this y_{2^n} is exactly the evaluation of (E.2) on a cofinal tail. This proves the discriminant-$4h$ extension on actual classes.

Choose smoothing integers greater than one with

$$c, d_s \equiv 1 \pmod{3N_{\mathrm{odd}}F}, \qquad c, d_s \equiv 1 + 2^e \pmod{2^{e+1}},$$

where $e \geq \max\{3, v_2(N)\}$ is fixed by the old dyadic data. Their sizes may depend on F, but the central weight-two, degree-one smoothing factor has valuation exactly $2e$ [19, Sections 4.2.4, 6.6 and 9.7]. For the signed fundamental discriminant, the quadratic Gauss identity is

$$\tau(\chi_h)^2 = \chi_h(-1)|\Delta_h| = \Delta_h.$$

Its dyadic valuation is $a/2$; the ratio $\sqrt{|\Delta_h|}/\sqrt{|h|}$ also belongs to this finite dyadic list. Neither contributes a term growing with $\omega(h)$.

E.2 Rational divisibility and the integral determinant

Apply Kato's Theorem 13.4(2), not its stronger part (3). The coefficient prime two is allowed in part (2), which concerns height-one primes away from two. Oddness gives two one-dimensional real eigenspaces. Purity gives the required good Frobenius weight. Non-CM open image, intersected with a quadratic kernel, gives rational irreducibility and a nontrivial unipotent with one-dimensional fixed space. The fixed ξ has nonzero period projections. Explicit reciprocity and Theorem 13.5(2) give infinitely many nonzero cyclotomic specializations in each sign, hence generic nonvanishing. No residual irreducibility at two is used.

This theorem initially controls H^2_0, the kernel of localization at two. Choose a fixed integral factor F_E to pay for local Iwasawa H^2 at two and the finitely many bad-prime determinants. Its central value is nonzero: local control and $H^2(\mathbb{Q}_v, V_h) = 0$, by local duality and finiteness of local elliptic torsion, exclude a factor t. Take the product over the finite allowed local and dyadic types before varying h.

At an active good prime rational inertia is -1, so its rational j_*-to-full cone vanishes. At an unused good prime this cone is

$$[\, H^1(I_q, V_h) \otimes A \xrightarrow{1-\sigma_q^{-1}\,\mathrm{Fr}_q} H^1(I_q, V_h) \otimes A\,], \qquad A = \Lambda[1/2],$$

in degrees one and two. The map is injective, including after localization at a prime dividing its determinant. The added degree-two length is the valuation of the Euler factor already in (E.1). Thus divisibility for the retained $F_E Z_F$ is the required divisibility for full global cohomology. At a bad prime one uses $H^1(I_q, V_h)$ itself, not an unjustified replacement by $V_h^{I_q}(-1)$.

Let C^+ be the fiber of ordinary global cochains to ordinary real cochains. Real H^0 is included. Over a finite Artin quotient, continuous cochains are filtered unions of finite free function modules, hence flat and therefore free. A residual splitting into finite cohomology and disks lifts by the nilpotent perturbation series to a finite free model with actual homotopies. Minimal models at

successive quotients are related by invertible matrices: a homotopy equivalence reduces to an isomorphism of minimal residual complexes, and Nakayama applies. Lifting these matrices and comparison homotopies makes the models compatible. Their acyclic, termwise surjective comparison cones have acyclic inverse limit. This proves the needed perfection and coefficient change.

For clarity, the residual dimension calculation has two separate arithmetic inputs. In the real cyclotomic tower $K_n = \mathbb{Q}(\zeta_{2^{n+2}})^+$ the ordinary two-class group vanishes. In a quadratic step only the prime above two ramifies. A possible Hilbert two-class extension would give a finite two-group whose abelianization is generated by inertia of order two. Its Frattini quotient has one generator, so the group is cyclic and equals that inertia. This excludes the extension. Separately the unit $1 + \zeta + \zeta^{-1}$ has norm -1; its signature has augmentation one in $\mathbb{F}_2[C_{2^n}]$, hence is a unit. Its conjugates span all signatures. The base uses -1. Kummer theory and the Brauer invariant relation then give, for a scalar coefficient,

$$\dim H^1(C_n^+) = 2^n + s_n - 1, \quad \dim H^2(C_n^+) = s_n - 1, \quad H^i = 0 \ (i \neq 1, 2).$$

For each fixed support s_n is bounded, since a fixed odd prime has open decomposition group in the cyclotomic tower. Universal coefficients over $\mathbb{F}_2[[t]]$ leave generic rank one only in degree one. The two trivial composition factors of $E[2]$ leave rank two. Minimal cancellation over $B = \mathcal{O}[G]$ therefore gives

$$C^+ \otimes B \simeq B^2[-1]. \tag{E.3}$$

This uses the local ring B, not semisimplicity of $\mathbb{F}_2[G]$.

Choose a generator of the fixed real invariant lattice and its connecting class a. Since real H^1 is killed by two, $2^{c_E+1} F_E Z_F$ has an integral positive lift b. For a basis θ of the original inverse determinant line, define

$$u^{\mathrm{ev}}\theta = 2^{-c_E} a \wedge b. \tag{E.4}$$

Equation (E.3) proves $2^{c_E} u^{\mathrm{ev}} \in \mathcal{O}[G]$. Rational divisibility gives $u^{\mathrm{ev}} \in A[G]$. Fourier inversion is used only for this latter membership, with no bound asserted for its denominator. In the original group basis,

$$A \cap \mathcal{O} = \Lambda, \qquad 2^{c_E} u^{\mathrm{ev}} \in \Lambda[G]. \tag{E.5}$$

Indeed an element of A has bounded powers of two as denominators in its power-series coefficients; membership in $\mathcal{O}$ makes their minimum valuation nonnegative. The coefficientwise intersection retains c_E, without a term $v_2|G|$.

E.3 Central saturation and the numerical lower bound

At a corank-zero constituent put

$$g = \ell_{\mathbb{Z}_2} E^h(\mathbb{Q})[2^\infty], \quad g_v = \ell_{\mathbb{Z}_2} E^h(\mathbb{Q}_v)[2^\infty], \quad r_i = \ell_{\mathbb{Z}_2} H^i(\mathbf{R}, T_h) \quad (i = 1, 2).$$

Let i be the index length of the image of global H^1 in the free rank-one quotient $L_2 = H^1(\mathbb{Q}_2, T_h)/H^1_f(\mathbb{Q}_2, T_h)$. Orthogonal Kummer duality, with its real terms, gives

$$\ell H^2_{\mathrm{glob}} = s(h) - i + \sum_v g_v + r_2 - g. \tag{E.6}$$

Write $K = \ker(H^1_{\mathrm{glob}} \to H^1_{\mathbf{R}})$, let k be its torsion length and m its free index. If $P = H^1(C_h^+)$, p_0 is its torsion length and a_s the saturation index of the real image in P/P_{tors}, exactness gives

$$\ell \operatorname{coker}(H^1_{\mathrm{glob}} \to H^1_{\mathbf{R}}) = r_1 - g + k - m, \qquad k = p_0 + a_s.$$

For the numerical calculation suppose also $L_*(h) \neq 0$. The wedge in (E.4) adds $a_s + 1 + v_{\mathrm{free}}(z) - m$. Conversion to the inverse determinant adds $p_0 - \ell H^2(C_h^+)$. Substitution of these equalities cancels m and a_s exactly:

$$v_2 u_h^{\mathrm{ev}}(0) = v_2 \exp^* z - s(h) - \sum_v g_v + 2g + 1 - r_1 - e_2, \tag{E.7}$$

where e_2 is the valuation of $\exp^*$ on a generator of L_2. Primitivity of the real image after inverting t has not been specialized to $t = 0$.

Explicit reciprocity for the single unnormalized character sum gives

$$v_2 \exp^* z = v_2 L_*(h) + \sum_{q \text{ unused}} v_2 \# E^h(\mathbb{F}_q) + \epsilon_{\mathrm{rec}}.$$

At unused good primes $g_q = v_2 \# E^h(\mathbb{F}_q)$, so these terms cancel. At an active good prime inertia invariants in $E^h[2^\infty]$ are $E[2]$, of length two for our curve. All remaining local terms have fixed bounds, including global torsion by its injection at two. Thus

$$v_2 u_h^{\mathrm{ev}}(0) = v_2 L_*(h) - 2\omega(h) - s(h) + \epsilon_h, \qquad |\epsilon_h| \leq C_{\mathrm{loc}}. \tag{E.8}$$

The central rational positive complex has only rank-two H^1, so this is specialization of (E.4). Corank-zero duality also shows $u_h^{\mathrm{ev}}(0) = 0$ if and only if $L_*(h) = 0$. At $L_*(h) \neq 0$, the forward rank-zero theorem supplies the corank-zero and finite-Sha hypotheses just used. Equations (E.5) and (E.8), already with $G = 1$, yield

$$v_2 L_*(h) \geq 2\omega(h) + s(h) - c_E - C_{\mathrm{loc}}. \tag{E.9}$$

No binary-family dimension is needed for this lower bound.

E.4 The actual even form and its phase

Fix a sign σ and unit classes at a finite support S_0 containing two and every prime dividing N. Let $D_0 > 0$ be squarefree in these prescribed classes, with $L_*(\sigma D_0) \neq 0$. In (E.10)–(E.11), D ranges over positive squarefree parameters in these same unit classes at every prime of S_0. Fix also a finite extra exclusion set B with $\gcd(D_0, \prod_{p \in B} p) = 1$. Waldspurger's nonzero local functional at D_0 supplies a global holomorphic cuspidal weight-3/2 vector in the appropriate packet [26, Proposition 18]; the holomorphic archimedean member and its unary-orthogonal realization are the ones in [29, Theorem 1 and Proposition 1].

Here is how one obtains the needed scalar character without losing that coefficient. At an old prime choose a local vector v with Whittaker functional $\ell_{D_0}(v) \neq 0$. The additive projection

$$P_R v = \int_{p^{-R}\mathbb{Z}_p} e_p(-D_0 b) \pi(n(b)) v \, db$$

retains this functional. Under the normalized unit action $M(u)$, its value vanishes unless $u^2 \equiv 1 \pmod{p^R}$. Average against a finite two-power character α satisfying $\alpha(-1)$ equal to the central sign. For large R the surviving neighborhoods of 1 and -1 have equal nonzero contributions. At two these are neighborhoods $\pm 1 + 2^{R-1}\mathbb{Z}_2$, not four independently chosen phases. The average is a unit-character vector; smoothness supplies lower unipotent invariance at sufficiently deep level. Together these give a scalar K_0 character. Finite additive projections impose the required squareclass support and preserve the D_0 coefficient. Away from S_0 keep the spherical vector. Its unramified

coefficient c_p^0 is one at squarefree valuations zero and one [26, Section I.3, pp. 377–378]. This includes primes dividing D_0, so this choice preserves the nonzero reference coefficient.

Let the resulting character be ψ and put $\eta = \psi\chi_{-\sigma}$. Both have two-power order. Work first in the full good-Hecke eigenspace at this S_0-supported level, retaining all old multiplicities. The packet identification is

$$\phi_0 = f_E \otimes \eta, \qquad \chi_0 = \psi\chi_{-1}, \qquad \phi_0 \otimes \chi_0^{-1} = f_E \otimes \chi_\sigma.$$

For every vector f in this whole space, Waldspurger's Corollary 2 and its same-$\mathbb{Q}_2$-squareclass removal of the auxiliary condition give

$$a_f(D)^2 = \psi(D/D_0)\frac{L_*(\sigma D)}{L_*(\sigma D_0)}a_f(D_0)^2. \tag{E.10}$$

Polarization makes the two coefficient functionals proportional on the whole space, including its old multiplicities.

The algebraic model is the finite-dimensional space of sections on $X_1(4N_{\mathrm{aux}})$ in [27, Definition 3.1 and Theorem 4.2], where N_{aux} is sufficiently divisible and supported on S_0. This auxiliary level includes the depths just chosen and need not equal the conductor N of E. Good-Hecke and support conditions are algebraic linear conditions. Only flat extension of the coefficient field is used for H^0. No arbitrary-ring base-change assertion at weight $3/2$ is required. Choose an algebraic vector g and normalize $a(D_0) = 1$.

For an automorphism acting on the two-power characters by an odd power e, put $\nu = \eta^{(e-1)/2}$. The vectors g^τ and $g \otimes \nu$ have the same character and good eigenvalues at a common S_0-supported level. The full-space proportionality in (E.10) therefore gives $\tau(a(D)) = \nu(D/D_0)a(D)$. Choose compatible local square roots of D/D_0 and let z_D be their η-value. Then $\tau(z_D) = z_D^e$, so

$$a(D) = z_D r_D, \quad r_D \in \mathbb{Q}, \quad r_D^2 = \frac{L_*(\sigma D)}{L_*(\sigma D_0)}. \tag{E.11}$$

In particular the nonzero coefficient valuations are integers.

Only now impose an extra finite coprimality sieve B. Finite Fourier translations remove coefficients divisible by a prime of B, without changing the desired coefficients or D_0. Comparison (E.11) was proved before this enlargement. The good recurrence is henceforth asserted only for $p \notin S_1 = S_0 \cup B$:

$$a(p^2n) = \{\eta(p)a_p(E) - \psi(p)(-n/p)\}a(n) - p\eta(p)^2a(n/p^2). \tag{E.12}$$

Its square-shell multipliers are integers after removal of the η phase. Combining (E.9) and (E.11) gives a fixed lower bound for $v_2a(D) - \omega(D)$ and hence an attained finite minimum. Restricting later fresh primes by further exclusions does not authorize choosing another minimum.

F Actual odd theta forms and their operators

We now construct the forms in (3.8), with the same h_*, D_w, L and actual Kummer stages as in Section 7. Write $m = |h_*|$. An admissible D is positive, squarefree, prime to the fixed support and in the prescribed unit classes. Then $K_D = \mathbb{Q}(\sqrt{Dh_*})$ is imaginary, split at $2N$, has no extra units, and Dh_* is an odd fundamental discriminant. The conductor-one point is the unaveraged sum

$$P_D = \sum_{[\mathfrak{a}] \in \mathrm{Pic}(\mathcal{O}_{K_D})} \chi_{h_*}(\mathrm{N}\,\mathfrak{a})\,\pi(x_D^{[\mathfrak{a}]}). \tag{O.1}$$

It is distinct from the conductor-$|h|$ ring-class point used in Section 9.

F.1 A fixed witness and one compatible sequence

The rational two-isogeny parity theorem gives root number -1 for E under the full-corank-one assumption. The two clauses of Bump–Friedberg–Hoffstein [20, pp. 543–544], with all old primes in their finite splitting set, supply first a negative h_* with nonzero central value and then a positive D_w with simple zero, after the primes of h_* have been added. The filters include the required squareclasses at $2N$; choose $|h_*| > 4$ to exclude extra units. The explicit unaveraged height formula [23, Theorem 1.1] applies with conductor one. Its Rankin product has factors $L(E^{D_w}, s)$ and $L(E^{h_*}, s)$, so P_{D_w} is nontorsion. The forward rank-zero theorem makes the unwanted h_* component rank zero.

One integer $L \neq 0$, fixed before D varies, kills the finitely many Atkin–Lehner cusp constants and all needed torsion and projection denominators. For the torsion assertion, the points lie over fields of degree at most four. At two fixed distinct good rational primes, prime-to-residue-characteristic torsion injects into one of finitely many groups $E(\mathbb{F}_{p^f})$, $f \leq 4$. Their orders bound all primary parts. A further factor four clears the biquadratic projections. Thus $Q = LP_{D_w}$ is a rational nontorsion point on E^{D_w}.

Here is a direct compatibility proof for its detecting stages. Let H be the kernel of the Tate representation and c_Q the Kummer cocycle. If $c_Q(H) = 0$, its rational class descends to the Tate image. A nonidentity central scalar z gives

$$(z-1)c_Q(g) = (g-1)c_Q(z),$$

so the descended class is a coboundary, contrary to Kummer injectivity. The closed translation image is a nonzero invariant $\mathbb{Z}_2$-submodule. Non-CM open image [31] makes its rational span the full Tate space, so it contains a full lattice. Intersecting H with the kernel of any fixed finite auxiliary extension preserves that property.

Fix complex conjugation τ, its matrix C, and an integral row $b \neq 0$ with $bC = b$. Choose $h \in H$ in these fixed kernels with $v = c_Q(h)$ and $2b(v) \neq 0$. For $\sigma = h\tau$ the cocycle identity gives

$$c_Q(\sigma^2) = (1 + C)v, \qquad bc_Q(\sigma^2) = 2b(v).$$

This is one actual Galois element. Its restrictions already agree on all intersections: it fixes $\sqrt{D_w}$, acts nontrivially on $\sqrt{h_*}$, and has Tate matrix C. At each sufficiently large finite precision, Chebotarev applied to the common torsion, division-point and fixed auxiliary quotient supplies a good ℓ_0, with a place realizing that element. Since $\mathrm{Fr}_{\ell_0}^2 = 1$ on $T/2^{M_\theta}$, finite-field Kummer gives

$$\lambda_{M_\theta} = b\kappa_{M_\theta} : E(\mathbb{F}_{\ell_0^2}) \longrightarrow \mathbb{Z}/2^{M_\theta}, \qquad v_2\lambda_{M_\theta}(\overline{Q}) = c_w < M_\theta \tag{O.2}$$

for the fixed $c_w = v_2(2b(v))$. Its Frobenius sign is $+1$. Later finite exclusions of rational primes do not change c_w. Additional field prescriptions would require a compatible common lift; arbitrary independent prescriptions are not asserted.

F.2 Local functions and the theta sum

Let B be the quaternion algebra ramified at ℓ_0 and infinity, R an Eichler order of level N, $V = B^0$, and $Q_B(x) = -x^2$. The finite set

$$X = B^\times \backslash \widehat{B}^\times / \widehat{R}^\times$$

is the set of supersingular pairs. Put $R_z = B \cap z\widehat{R}z^{-1}$ and $P_M(z) = \lambda_M(L\pi_{\ell_0}(z))$. This is an actual function on those pairs. Its Hecke, Atkin–Lehner and Frobenius identities hold modulo 2^M. Choose

arbitrary integer lifts; they need not satisfy any characteristic-zero eigenrelation. In this appendix $M = M_\theta$.

Define a finite Schwartz function $f = \prod f_p$ as follows. At $p \nmid 2Nh_*\ell_0$ use $\mathbf{1}_{R_p^0}$. At $p^e \| N$ require an integral split quadratic embedding. The cyclic level quotient selects one of its two summands; if r is the eigenvalue of x on this summand, give it weight $\alpha_p(r)$, where the fixed two-power character satisfies

$$\alpha_p(-1) = w_{p^e}\chi_{h_*}(p)^e. \tag{O.3}$$

Conjugation by $R_p^\times$ preserves the selected summand. Scaling x by a unit u multiplies the weight by $\alpha_p(u)$.

At two impose $(1+x)/2 \in R_2$ and x^2 a unit square. When $2 \mid N$, the operator $(1+x)/2$ is not itself asserted to be idempotent. Its eigenvalues $(1+r)/2, (1-r)/2$ differ by the odd unit r. Its integral spectral projectors distinguish the two summands, including after reduction modulo two. For odd u,

$$(1+ux)/2 = u(1+x)/2 + (1-u)/2.$$

The summands are preserved and the selected eigenvalue of x becomes ur. Selection is not by parity of the eigenvalue of $(1+x)/2$. This proves the dyadic root-character rule at every power of two in N.

For $p \mid h_*$ require $x \in M_2(\mathbb{Z}_p)^0$ with $v_p(Q_B(x)) = 1$. Its reduction is nonzero nilpotent. With $X_0 = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ put

$$\epsilon_p(gX_0g^{-1}) = (\det g/p).$$

The centralizer has square determinant, so this is well-defined. Conjugation has multiplier $(\det u/p)$ and scaling by a has multiplier (a/p).

At ℓ_0, the residue of a trace-zero element is in a one-dimensional trace-zero line in $\mathbb{F}_{\ell_0^2}$. Fix $\beta \neq 0$ on this line and give $t\beta$, $t \neq 0$, weight (t/ℓ_0); give zero weight to zero. Units act trivially on the residue field; a uniformizer acts by Frobenius and sends β to $-\beta$. Its multiplier is -1 since $\ell_0 \equiv -1 \pmod 4$. Scaling has character (u/ℓ_0). Also $-\beta^2$ is a square, and $(m/\ell_0) = 1$, so the extracted indices have $(D/\ell_0) = 1$.

Extra fixed unit conditions are characteristic functions and preserve these scaling rules. Altogether,

$$f(u^{-1}xu) = \chi_{h_*}(\mathrm{Nrd}\, u)^{-1} f(x).$$

The product scaling character is even: its value at -1 is

$$\left(\prod_{p^e \| N} w_{p^e}\chi_{h_*}(p)^e\right)\chi_{h_*}(-1)(-1) = 1.$$

The first factor is the positive functional sign of E^{h_*}. Write η_B for the resulting classical character, using the inverse-idele convention; its value at a positive good prime is the product of the finite scaling values.

Every finite $R_z^\times$ embeds by left multiplication in $\mathrm{GL}_4(\mathbb{Z})$. Reduction modulo three is injective on a finite subgroup: a prime-order element $1 + 3^a A$ in its kernel has nonzero first term in its prime-th power, modulo 3^{a+1}, or modulo 3^{a+2} for order three. Thus one may take

$$c_0 = |\,\mathrm{GL}_4(\mathbb{F}_3)| = 24\,261\,120, \qquad v_2(c_0) = 9.$$

It is independent of ℓ_0. Define

$$H_M(n) = c_0 \sum_{z \in X} \frac{P_M(z)\chi_{h_*}(\mathrm{Nrd}\, z)}{|R_z^\times|} \sum_{\substack{x \in V(\mathbb{Q}) \\ Q_B(x) = n}} f(z^{-1}xz). \tag{O.4}$$

The conjugation rule makes this independent of representatives. Its coefficients are integral in the fixed character ring.

F.3 Transformation law and recurrence

Use the Weil representation with self-dual measures and the standard global additive character [30, I.13, II.25, II.28, III, Theorem 6]. Upper unipotents multiply by $e_p(bQ_B(x))$, so integral ones fix the local function. If it is invariant under translations by a lattice L'_p, its Fourier transform is supported in the dual lattice. For sufficiently large A_p, $p^{A_p}Q_B((L'_p)^\vee)$ is integral. Fourier conjugation therefore gives invariance under lower unipotents of that depth. This argument also applies at two.

In a presentation $B = (-a, -b)$, Q_B has diagonal coefficients a, b, ab, with square product. Its diagonal Weil factor relative to the ordinary theta multiplier cubed is consequently $(\det Q_B, u)_p = 1$. Factoring a matrix of sufficiently deep K_0 into lower unipotent, diagonal and upper unipotent proves the scalar character η_B. Conjugation by z is an isometry, so the level works in every summand. The Gaussian at infinity is the holomorphic lowest-weight vector of weight $3/2$. At any cusp its finite Weil transform is again Schwartz, and positive definiteness gives only nonnegative rational exponents. The series converges normally on compact subsets. This proves holomorphy at every cusp, not only the formal transformation law.

Every nonzero index in (O.4) is divisible by m. Replace Q_B directly by Q_B/m on the same Schwartz space. It is integral on the support, so the same proof gives the actual form

$$g_M(\tau) = \sum_n H_M(mn)e^{2\pi in\tau}.$$

This justifies extraction without invoking an unspecified half-integral-weight U_m operation. The determinant squareclass becomes m, hence

$$\psi = \eta_B(m/\cdot), \qquad \eta = \eta_B\chi_{h_*}, \qquad \psi/\eta = \chi_{-1}. \tag{O.5}$$

For a good p let $s = \eta_B(p)$. Lattice neighbors are indexed by lines in $\mathbb{F}_p^2$. A nonzero trace-zero reduction has $1 + (-n/p)$ invariant lines; a nonzero nilpotent has one and zero has $p + 1$. The nonintegral contributions are $x = y/p$ with y primitive of norm p^2n. In a basis reducing y to X_0, the lower-left entry is divisible by p^2, so exactly its kernel-line neighbor makes y/p integral. The two parts of the neighbor sum are

$$\begin{aligned} &\text{primitive part:} && s^{-1}H_M(p^2n) - H_M(n), \\ &\text{integral part:} && (1 + (-n/p))H_M(n) + psH_M(n/p^2). \end{aligned}$$

The orbit-stabilizer weights in (O.4) validate the count even with extra automorphisms. Multiply by the norm character $\chi_{h_*}(p)$ and use the actual relation $T_pP_M = a_p(E)P_M \pmod{2^M}$. After extraction this gives

$$a_M(p^2n) = \{\eta(p)a_p(E) - \psi(p)(-n/p)\}a_M(n) - p\eta(p)^2a_M(n/p^2) \pmod{2^M}. \tag{O.6}$$

No characteristic-zero eigenform assertion has entered.

F.4 The unaveraged point comparison

For this comparison assume also $(D/\ell_0) = 1$ at the current stage. Together with $(h_*/\ell_0) = -1$, this makes ℓ_0 inert in K_D. If $(D/\ell_0) \neq 1$, the local Schwartz support instead gives $a_M(D) = 0$; no supersingular reduction formula is asserted there. At norm mD the vectors in (O.4) are exactly

optimal embeddings of $\mathcal{O}_{K_D}$ into R_z. The condition at two is the integrality of the generator $(1+\sqrt{Dh_*})/2$. There are two orientations at each $p^e \| N$, two at ℓ_0, and one elsewhere. The cyclic level quotient must lie in one idempotent summand, since a nonzero quotient in both summands would not be cyclic.

Apply [25, Theorem 3.1, Corollary 3.2, Sections 4.9–4.10]. The base is $\mathbb{Q}$, the source algebra is $M_2(\mathbb{Q})$, the changed set is $\{\ell_0\}$, and the target is B. The parity condition is $1+0+1$ even; ℓ_0 is inert in K_D. The level is prime to ℓ_0 but may have prime powers. Both fine conductors are one, so the ring-class ratio and the local factor $N_{\ell_0}(0,0,0)$ are one. Equivariance identifies the ideal-class orbit with reduction of (O.1), retaining the level and ℓ_0 orientations.

The stabilizer of an embedding is $\{\pm 1\}$. Its $R_z^\times$-orbit has size $|R_z^\times|/2$, so (O.4) contributes a factor $1/2$. Reversing an old orientation multiplies the weight and the norm-weighted point by the same sign (O.3). Reversing the ℓ_0 orientation gives -1 from each. Thus all $2^{\omega(N)+1}$ choices contribute equally and

$$a_M(D) = c_0\, 2^{\omega(N)} \zeta_M(D) \lambda_M(\overline{LP_D}) \pmod{2^M}, \tag{O.7}$$

with a two-power root of unity $\zeta_M(D)$. There is no class-number average. The character field is generated by the fixed α_p; the moving prime contributes quadratic values only. Formula (O.2) gives fixed finite valuation $c_w + 9 + \omega(N)$ at D_w for every sufficiently advanced stage.

For a known simple-zero D, write $P_D = m_D G + T$ in its native rank-one free lattice. The same L kills T. An integral finite-field Kummer map cannot decrease $v_2(m_D)$, so (O.7) gives the lower bound by the native index. The rational free line has index at most two in that native lattice, by conjugation and norm. Absorb this and the fixed initial convention in C_{det}. The first inequality below holds for a known simple-zero D; the visibility implication holds for every admissible D:

$$\begin{aligned} v_2 a_M(D) &\geq \min\{M, j_{\mathrm{gen}}(D) - C_{\mathrm{det}}\}, \\ v_2 a_M(D) < M \implies \operatorname{ord}_{s=1} L(E^D, s) &= 1, \quad j_{\mathrm{gen}}(D) \leq v_2 a_M(D) + C_{\mathrm{det}}. \end{aligned} \tag{O.8}$$

If an odd-sign D is not a simple zero, the height formula gives a torsion genus point, killed by L, and its coefficient is zero modulo 2^M. Combined with the first inequality on simple-zero parameters, this gives the all-D truncated bound used below, without assigning a rank-one lattice index to a torsion point.

F.5 Integer divisions and actual trace operators

All new primes below avoid the full coefficient support S, including every late sieve and the current ℓ_0. Here S is the actual coefficient support. The coefficients of g vanish at all indices not prime to S. For the odd source this is forced at $2, N, h_*, \ell_0$ by the local Schwartz support and the extraction at $|h_*|$. At every additional coefficient-sieve prime it is imposed by an exact zero-or-retain Fourier sieve, which commutes with the good recurrence outside the enlarged support. This is the vanishing used in the condition $(m, S) = 1$ in the unary series below. Later stationary prime-selection tests may enrich the common label alphabet without resieving the intrinsic odd source; such tests do not enlarge this coefficient-support condition. Use the minimum-depth inequalities proved in Appendix G. The general bookkeeping assigns weight one to a split prime of $\mathbb{Q}(E[2])$ and one-half to a prime with transposition Frobenius, called simple. In our actual full-two-torsion family every prime is split, $w(U) = |U|$, and the depth δ is an integer. Thus only matching parity, meaning $e_U = 0$, is needed here; the displayed general formula also records where the unused simple-prime term would enter. For a finite set U, put

$$n_U = \lceil \delta + w(U) \rceil, \quad e_U = n_U - \delta - w(U), \qquad B_U = 2^{-n_U} \prod_{p \in U} (U_p - \eta(p) V_p) g,$$

where $U_p(\sum a(n)q^n) = \sum a(pn)q^n$ and $V_p g(\tau) = g(p\tau)$. These are operations on the actual form g. The degeneracy transformation for V_p, and $U_p = U_{p^2}V_p$, put each new prime at level exponent at most one, with character $\psi(p/\cdot)$. The double coset for U_{p^2} has representatives $\left(\begin{smallmatrix} 1 & r \\ 0 & p^2 \end{smallmatrix}\right)$. For a matrix in $K_0(p)$ the rule $r' = (b+rd)/(a+rc) \pmod{p^2}$ permutes them, since $a+rc$ is a unit. This verifies the classical bad-prime extraction at that level.

Write an index as Jrm^2 with $J \subset U$ and r squarefree, prime to U. Products of prime sets are understood here. The originating squarefree parameter in the coefficient of g is $D = (U \setminus J)r$, not the squarefree part Jr of the tested index. Remove the phase $\eta(p)^s$ in (O.6) and put

$$b_{-1} = 0, \quad b_0 = 1, \quad b_1 = a_p - \kappa, \quad b_{s+1} = a_p b_s - p b_{s-1}, \quad \kappa = (\psi/\eta)(p)(-D/p).$$

For $p \in U \setminus J$ the factor $b_s - b_{s-1}$ is odd; for $p \in J$ the factor $b_{s+1} - b_s$ is even. Consequently, for the even source,

$$v_2 B_U[Jrm^2] \geq w(r) + \tfrac{1}{2}\#\{p \in J : p \text{ is simple}\} - e_U. \tag{O.9}$$

The right side is an integer on the allowed filter and is nonnegative. In our full-two-torsion application all primes are split and $w = \omega$. For the odd source, fix an integer $R \geq 1$ and a weight bound W first. At one sufficiently advanced actual stage, simultaneously for all U with $w(U) \leq W$ and every Fourier index, (O.9) holds with its right side replaced by its minimum with R. Thus the untruncated bound on a prescribed bounded range of squarefree weights follows by choosing R and then the stage. For the unary-removal and two-trace tests below, choose $R \geq 2$ after fixing the operator budget. For a split prime $a_p \equiv p + 1 \pmod 4$. The recurrence gives the same residue on each divided shell,

$$(b_{s+1} - b_s)/2 \equiv c_p(D) := \{a_p - (\psi/\eta)(p)(-D/p) - 1\}/2 \pmod 2. \tag{O.10}$$

Choose $i \in \{0, 1\}$ with $\eta\chi_{-4}^i$ odd and let

$$\Theta_J = \sum_{\substack{m>0 \\ (m,S)=1}} \eta(m)\chi_{-4}(m)^i m q^{Jm^2}.$$

Poisson summation gives this genuine unary weight-3/2 form. Its scalar character is $\eta\chi_{-4}^i(-J/\cdot)$ and every $p \mid J$ enters the level at exponent one. In matching parity the shell recurrence gives $B_U[Jm^2] - B_U[J]\eta(m) \in 2\mathcal{O}_{\text{coef}}$. Thus

$$H_U = \frac{B_U - \sum_J B_U[J]\Theta_J}{2} \tag{O.11}$$

is integral; in opposite parity take $H_U = B_U$. This is division by the integer two, not reduction modulo the maximal ideal of a ramified coefficient ring. For odd-stage forms, the coarse bound and bounded-weight stabilization proved below pay every fixed denominator before choosing the stage.

Multiply (O.11) and the undivided corrections by $\theta = 1 + 2\sum_{m>0} q^{m^2}$. They are actual weight-two forms. For a fixed stage and U, choose a finite complete coefficient-field extension containing all constituents and retain their old multiplicities and diamond translates in the finite-dimensional characteristic-zero space. Use modular Galois representations on cuspidal constituents and the two character summands on Eisenstein constituents. Holomorphy alone does not exclude the latter. Diamond translates remain integral: their correction coefficients have the form

$$\tfrac{1}{2}B_U[J]\{\psi_U(u) - \eta(u)\chi_{-4}(u)^i(-J/u)\},$$

whose character ratio is quadratic. Good Hecke operations preserve integrality by $(T_q f)[n] = f[qn] + q(\langle q\rangle f)[n/q]$. Finitely many Fourier coordinates inject this space into a finite coordinate

space. The module generated by these translates is therefore finite over the coefficient DVR, and closed. Chebotarev and continuity make scalar trace and determinant operators limits of good Hecke and diamond operations, so they preserve this module. No integral eigenspace projector is used.

Reading an added vertex. For the actual full-two-torsion family, all primes are split, $\delta \in \mathbb{Z}$, and $n_U = \delta + |U|$. Put $F_U = \theta H_U$ and let $q \notin S \cup U$ be a fresh prime, where S includes every late sieve and the current moving theta prime. Write $u = \prod_{p \in U} p$ and let $\mathcal{A}(D)$ denote the normalized valuation symbol on the tested range: it is one exactly when $v_2 a(D) = \delta + \omega(D)$ on the filter. For an odd source, that range has a fixed weight bound chosen before the valid actual stage, as specified before (F.7). No finite-stage value at larger weights is asserted; the compatible limit tables used later are a separate object. Expanding the operators, every term containing a V_p has zero coefficient at q, and no unary index Jm^2 with $J \subset U$ equals q. Therefore

$$B_U[q] = 2^{-n_U} a(uq), \qquad H_U[q] = 2^{-n_U - 1} a(uq).$$

At coefficient one the diamond term in the good Hecke formula vanishes. Since $F_U \equiv H_U \pmod{2\mathcal{O}_{\mathrm{coef}}}$,

$$\overline{(t(\mathrm{Fr}_q) F_U)[1]} = \overline{H_U[q]} = \overline{2^{-\delta - |U| - 1} a(uq)} = \mathcal{A}(uq) \in \mathbb{F}_2.$$

Here residues lie in the coefficient residue field, with $\mathbb{F}_2$ its prime subfield. Every permitted phase is a two-power root of unity and reduces to one. For an odd source, fix the finite operator budget $|U| \le W$ and $R \ge 2$ first; the stage choice in (O.9) may be taken with $M_\theta \ge \max(0, \delta + W) + R$. Thus $M_\theta - n_U - 1 \ge 1$, so the normalized finite-stage error is still divisible by the integer two. The same reading holds with value zero off the fixed or moving coefficient support.

The diamond correction above is a sum of integral multiples of $\theta \Theta_J$; its fresh coefficient q is zero after reduction. It introduces neither an extra phase nor a denominator in this reading. All readings and subsequent omissions use this same g, η, ψ, δ and actual stage; only the set U changes.

A fresh trace kills every unary correction after reduction, by its squareclass support. The exact two-by-two matrix identity then gives, on the resulting trace orbit,

$$d(u)t(v) = t(v), \qquad t(u)t(v) = t(uv) + t(uv^{-1}), \qquad t(1) = 0.$$

Two fresh traces vanish by (O.9), with the unary division paid first. Hence $t(gu^2) = t(g)$. The relevant trace functions factor through the elementary quadratic quotient unramified outside $S \cup U$. This says nothing about abelianity of the characteristic-zero Galois representations.

At $p \in U$ a new constituent has conductor one and primitive quadratic determinant; an old constituent is unramified there. For the former, the conductor formula excludes nonzero monodromy: a special representation either has unramified determinant or conductor at least two. On cuspidal constituents, local compatibility [28, Theorem A] gives one unramified line and one quadratically ramified line. The coefficient prime is two and p is odd. The base $\mathbb{Q}$ has odd degree, so the extra even-base-degree condition in that theorem is not required. For an Eisenstein constituent the same description follows directly from its two character summands. In either case, in a suitable basis,

$$\rho(\tau) = \mathrm{diag}(1, -1), \quad \rho(\phi) = \mathrm{diag}(\lambda, \mu), \quad U_p^{(2)} = \lambda.$$

A direct matrix calculation proves

$$U_p^{(2)} \{ t(g\tau\phi) - t(g\phi) \} = d(\phi) \{ t(g\tau) - t(g) \}. \tag{O.12}$$

Old constituents have zero inertia difference on both sides. Thus the identity holds with every old multiplicity retained.

Apply (O.12) at coefficient one. Put $V = U \setminus \{p\}$, $v = \prod_{r\in V} r$, so $n_U = n_V + 1$. For a fresh prime r set

$$C_p(vr) = \frac{a_p - (\psi/\eta)(p)(-vr/p) - 1}{2} \in \mathbb{Z}, \qquad c_p(vr) = C_p(vr) \bmod 2.$$

The two surviving operator terms at pr and the good recurrence give

$$H_U[pr] \equiv \eta(p) C_p(vr)\, 2^{-n_V-1} a(vr) \pmod{2\mathcal{O}_{\mathrm{coef}}}.$$

For the even source this is an equality before reduction. For the odd source the error is in $2^{M_\theta - n_U - 1}\mathcal{O}_{\mathrm{coef}}$, which is contained in $2\mathcal{O}_{\mathrm{coef}}$ at our chosen stage. Choose fresh q, q_τ, r_0, r_1 representing respectively $g, g\tau, g\phi, g\tau\phi$ in one sufficiently precise joint quotient for the U and V forms and all retained characters. The omitted form uses the same g, η, ψ, δ and stage, and its level omits the entire rational prime p. Thus the two V symbols at vr_0, vr_1 agree. Their Legendre signs at p are opposite, so $c_p(vr_0) + c_p(vr_1) = 1$ in $\mathbb{F}_2$. The coefficient-one reading and the reduced trace identity now give

$$\mathcal{A}_U(Uq_\tau) + \mathcal{A}_U(Uq) = \mathcal{A}_V(Vr_0) \qquad \text{in } \mathbb{F}_2.$$

Here the subscript records the operator set, not a new coefficient source or minimum. The factor $\eta(p)$ reduces to one; no factor two is cancelled in (O.12). Retained quadratic labels multiply under $g \mapsto g\phi$, including all external incidences. This is the omission rule. A split vertex with trivial omitted label has trace $t(1) = 0$; contracting a primary cluster to it proves isolation. These rules concern the actual arithmetic symbols. Their simultaneous prime realization and finite graph budget are treated next.

G Nonvanishing, normalization and the final family

We give the choice order underlying (3.3)–(3.9). This also separates the two valuation arguments: the even upper transfer uses the common Kato scalar, while the final contradiction uses the common Heegner coordinate of Section 9.

G.1 An absolute bound of five new primes

The theorem of Hoffstein–Luo [21, p. 436] supplies fundamental discriminants with nonzero central value and at most four prime factors. Its finite splitting set is arbitrary. The bound four is absolute; no uniform size threshold for those discriminants is needed.

Let $A/\mathbb{Q}$ be any fixed modular elliptic curve for this application, S contain two, three and all its bad primes, and B be any finite exclusion set. Fix a nonempty signed unit filter of even root number. By CRT and Dirichlet choose one signed prime q_0 in its classes, outside $S \cup B$. Apply the theorem to the actual curve A^{q_0}, putting

$$S' = S \cup B \cup \{|q_0|\} \cup \{p : p \mid N_{A^{q_0}}\}$$

in its splitting set. The theorem gives a fundamental d with $\chi_d(p) = 1$ for all $p \in S'$ and at most four prime factors. The functional equation forces $d > 0$, since A^{q_0} has even root number and every conductor prime splits. Splitting at two gives $d \equiv 1 \pmod 8$. Thus $q_0 d$ has all original unit classes, is prime to B, is squarefree, and has at most five prime factors. If $q_0 d \equiv 3$ or $7 \pmod 8$, its field

discriminant is $4q_0d$; only d was required to be fundamental in the theorem. This is compatible with the coefficient construction (E.2).

For the rough companion the curve itself varies: write a for the bounded dummy product and take $A = E^{ha}$. Choose q_0 so $k_0 = aq_0$ is negative, 1 modulo eight and a square at every prime of $2Nh$, with the required exclusions. Then k_0 is odd fundamental, prime to N_{E^h}, and

$$w(E^{hk_0}) = w(E^h)\chi_{k_0}(-N_{E^h}) = +1.$$

This proves the sign directly for the actual application. Apply Hoffstein–Luo to E^{hk_0} with every conductor, old, h, dummy and forbidden prime in its splitting set. The resulting $k' = k_0d$ adds at most five new primes. The constants and sizes in the analytic theorem may depend on this varying curve and the whole finite set; the bound five does not.

G.2 The even upper transfer

For one fixed even source write (E.11) as

$$v_2L_*(\sigma D) = 2v_2a(D) + \kappa_e.$$

The lower bound (E.9) and a nonzero reference coefficient imply that the nonempty set of integers $v_2a(D) - \omega(D)$ has a finite attained minimum δ_e. A unit of the normalized symbol means precisely equality to this minimum. It therefore gives

$$L_*(\sigma D) \neq 0, \qquad v_2L_*(\sigma D) = 2\omega(D) + C_{e,\mathrm{unit}}, \quad C_{e,\mathrm{unit}} = 2\delta_e + \kappa_e. \tag{F.1}$$

This constant belongs to the fixed source, not to a curve later introduced solely to obtain nonvanishing.

For the negative types used in the rough companion, the unsigned index is 7 modulo eight. Its total $(-1/D)$ label is nonzero, so every unit seed is nonempty. A seed of least positive size has no dependence on internal edge bits: a nonzero edge derivative would be a smaller nonempty unit. Its labels therefore suffice. If a list with the same total label contains the first $r-1$ seed anchors and a nonempty remainder, contract that remainder into the last anchor. Its total label is forced to be the remaining seed label. This proves partitionability regardless of how many old h-primes are in the remainder.

For a partitionable d already known to have Selmer corank zero, the one-symbol address construction keeps every prime of d stationary and gives arbitrarily large cubes d_x, $d_0 = d$, with unit symbol at each $x \neq 0$. The scalar in (E.5) is common to that cube. By (E.8) and (F.1),

$$v_2u_x^{\mathrm{ev}}(0) \leq B_e := C_{e,\mathrm{unit}} + C_{\mathrm{loc}} \quad (x \neq 0).$$

Choose $M - c_E > B_e$. The constant coefficient of $2^{c_E}u^{\mathrm{ev}} \in \Lambda[G_b]$ has sign-value bits of total Boolean degree at most $2^{M-1} - 1$. For a larger cube dimension, the Boolean comparison in Section 9 gives $x \neq 0$ with $v_2(u_x^{\mathrm{ev}}(0) - u_0^{\mathrm{ev}}(0)) \geq M - c_E$. If the base were zero or had valuation greater than B_e, this would contradict the upper bound at x. Therefore

$$v_2L_*(d) - 2\omega(d) - s(d) \leq C_{e,\mathrm{unit}} + 2C_{\mathrm{loc}}. \tag{F.2}$$

The required dimension may depend on d; this bound does not. It precedes the odd minimum and uses no Heegner transfer.

Prepare these sources and nonempty seeds for every total type at the fixed support. There are finitely many. A bounded supply of dummy anchors and reserves works for all of them. For each h realize that supply by distinct primes avoiding the entire fixed support and h. Put every dummy

and every other already fixed forbidden prime in the exclusion set of the preceding nonvanishing construction. It gives k' with $\omega(k') \le A$, where A is the fixed dummy count plus five. For the total type that actually occurs, hk' is partitionable. Nonvanishing gives corank zero by the forward theorem, so (F.2) applies. Taking the maximum over the finitely many prepared constants proves

$$v_2 L_*(hk') \le 2\omega(h) + 2\omega(k') + s(hk') + C_E. \tag{F.3}$$

No new even minimum is chosen after h or its exclusion set. Those excluded primes are not added as relaxed places in a Selmer complex.

G.3 The odd minimum is chosen after its lower bound

The arithmetic result (3.5) and the two native-lattice height comparisons (3.4) give, using (F.3),

$$2j_{\mathrm{gen}}(h) \ge 2\omega(h) + s(h) - C_{\mathrm{gen}}, \qquad C_{\mathrm{gen}} = 2A + C_R + C_E - v_2 L_*(h_*) + C_H. \tag{F.4}$$

The variable $s(hk')$ cancels. This estimate precedes the theta prime, precision, odd minimum, final field and binary dimension.

Combining (F.4) with (O.8) gives on all admissible D

$$v_2 a_M(D) \ge \min\{M, \omega(D) - C_{\mathrm{coarse}}\}, \qquad C_{\mathrm{coarse}} = C_{\mathrm{gen}}/2 + C_{\mathrm{det}}. \tag{F.5}$$

At visible coefficients (O.8) first gives a simple zero, allowing (F.4); at the other odd-sign parameters the coefficient is zero modulo 2^M. This avoids a circular use of the future minimum.

Fix one nonprincipal ultrafilter on the actual stages with $M = M_\theta \to \infty$. Consider the integers r for which a bounded-weight sequence of visible D_M has $v_2 a_M(D_M) - \omega(D_M) = r$ on a set in the ultrafilter. The fixed witness D_w and (O.2),(O.7) make this set nonempty. Equation (F.5) bounds it below. Let its attained minimum be δ_o. For every fixed W, on a set in the same ultrafilter,

$$v_2 a_M(D) \ge \omega(D) + \delta_o \quad \text{for every visible } D,\ \omega(D) \le W. \tag{F.6}$$

Otherwise choose a violating D_M at each bad stage. One of the finitely many integers between $-C_{\mathrm{coarse}}$ and $\delta_o - 1$ would occur on an ultrafilter-large set, a contradiction. Invisible coefficients satisfy the required truncated inequality once $M > W + \delta_o$.

For a fixed finite operator budget $|U| \le W_U$, put $N_0 = \max(0, \delta_o + W_U)$ and fix $R \ge 2$. Choose an integer $T \ge W_U + 1$ with $T - C_{\mathrm{coarse}} \ge N_0 + R$, then choose a stage in the ultrafilter-large set where (F.6) holds through weight T, with $M \ge N_0 + R$ and $M > T + \delta_o$. For the index Jrm^2 put $D = (U \setminus J)r$. The integral shell recurrence has an error in $2^M\mathcal{O}$. If $\omega(D) > T$, (F.5) gives

$$v_2\big(2^{n_U} B_U[Jrm^2]\big) \ge N_0 + R.$$

If $\omega(D) \le T$, (F.6) for visible coefficients and $v_2 a_M(D) \ge M$ for invisible ones instead give

$$v_2\big(2^{n_U} B_U[Jrm^2]\big) \ge \min\{M, n_U + \omega(r)\}.$$

Consequently $v_2 B_U[Jrm^2] \ge \min\{R, \omega(r)\}$ for every Fourier index and every U in the budget at this one stage. At $r = 1$ the normalized originating coefficient $2^{|J|-n_U} a_M(U \setminus J)$ is integral, so (O.10) proves the unary congruence modulo the integer two. The additional division in (O.11) leaves error valuation at least $R - 1 \ge 1$. In particular $M - n_U - 1 \ge 1$ for every reading and first-shell omission in the budget. It does not select one stage for infinitely many budgets.

More precisely, fix a squarefree-output cutoff L. On a set of stages in the same ultrafilter, (F.6) holds through L and $M_\theta > \delta_o + L$. Define $\mathcal{A}_{M_\theta,L}(D)$ only for positive squarefree D with $\omega(D) \le L$: it is zero off the full coefficient filter, one when $v_2 a_{M_\theta}(D) = \delta_o + \omega(D)$ on that filter, and zero otherwise. This is exhaustive on the tested range: visible coefficients satisfy (F.6), while invisible ones have valuation at least $M_\theta > \delta_o + \omega(D)$. In particular a unit in this range is visible and satisfies

$$\operatorname{ord}_{s=1} L(E^D, s) = 1, \qquad j_{\mathrm{gen}}(D) \le \omega(D) + C_o, \quad C_o = \delta_o + C_{\mathrm{det}}. \tag{F.7}$$

For the preceding operator budget take $L \ge W_U + 1$ and include its valid-stage requirements; all fresh-vertex and omission outputs then lie in this range. The constant C_o is independent of L and of the stage. No value is assigned by this definition at unbounded weights of one finite stage. No minimum is changed when a vertex is omitted or a stationary condition is later added.

G.4 Nonempty rational seed and a common finite stage

Choose the final negative even unit k as in Section 7, with fresh ramification outside the old odd label and torsion fields. Fix $K = \mathbb{Q}(\sqrt{k})$ before the binary dimension. The forward theorem gives corank zero for E^k; the sum-and-difference isogeny, on compact Selmer modules tensored with $\mathbb{Q}_2$, gives full corank one for E/K. The intrinsic odd symbol omits the primes of k. Later ℓ_0 avoids them. Fresh ramification implies that every old odd label still lifts to a K-split label. Additional stationary tests can have relations; use their actual common alphabet and its separate surjective projections.

The final E^k source is transported from the fixed negative source in which k was a unit. For $(D, k) = 1$, its coefficient comparison gives

$$v_2 a_{\mathrm{new}}(D) = v_2 a_{\mathrm{old}}(kD) - v_2 a_{\mathrm{old}}(k) \ge \omega(D).$$

Thus normalization at index one has this specific inherited lower bound; nonvanishing at index one alone would not prove it. A unit has equality and hence $v_2 L_*(h_x k) \le 2\omega(h_x) + C_{e,\mathrm{final}}$, where the conversion from the fixed E^k period convention changes only a fixed constant. Its K-split seed must be nonempty. Suppose otherwise. The integral coefficient valuations then improve the preceding lower bound to $\omega(D) + 1$ on every nonempty product of allowed K-split primes. For a fresh such p, apply the trace identity to $f = \theta H_{\{p\}}$:

$$U_p^2 t(\phi) = U_p^3 + d(\phi) U_p, \qquad (t(\phi) f)[p^2] = f[p^3] + (d(\phi) f)[p].$$

Choose a fresh K-split p' matching ϕ in the finite Galois quotient and all fixed rational characters. The product pp' is in the prescribed square filter even when p alone is not. The left coefficient is a unit times $(a_p - 1)a(pp')/4$, hence zero modulo two by the improved bound. The determinant term at p vanishes because $B_{\{p\}}$ and its $\Theta_{\{p\}}$ correction have the same character and cancel there; Θ_1 has no such coefficient. The remaining p^3 coefficient is a unit times

$$(a_p - 1)(a_p - p - 1)/4.$$

Here $a_p = a_p(E^k)$ is even, so $a_p - 1$ is odd. Its vanishing modulo two forces $a_p(E^k) \equiv p + 1 \pmod 8$ for every such p. Choose such a p with $p \equiv 1 \pmod 4$ and $(\ell/p) = -1$. Freshness of k permits these simultaneous conditions. The three rational two-torsion points have no rational halves: their necessary square differences include $-\ell, \ell, -4\ell$. The two-primary reduction group is exactly of order four. This contradicts the asserted congruence. This proves the required nonempty even seed. If the intrinsic odd symbol already has a unit at the empty configuration, its detection formula gives

a simple zero for E and the converse is finished. Otherwise take a least nonempty odd unit seed. Thus the remaining two-symbol address argument has two nonempty seeds.

Here the finite graph argument has a definite arithmetic order. The moving theta prime contributes only a bounded number of quadratic label coordinates. The root characters at the old level were fixed before M_θ; growing Kummer precision introduces no growing-order character on the variable primes. Thus the elementary label groups, their projections and their finite addition tables have bounded size. Choose their abstract type on the same ultrafilter as δ_o. For each fixed weight cutoff W, restrict first to the ultrafilter-large set of stages where the preceding budget and precision conditions hold for every $|U| \le W$, with $R = 2$. On those stages the coefficient reading, factorization, contraction and isolation rules hold on all configurations of at most $W + 1$ vertices, with the stated nonempty-output restriction. They therefore define actual symbol truth tables on the compatible typed graph data. There are finitely many such tables; select the one occurring on a set in the same ultrafilter. Two selected tables agree on their common restriction by taking the finite intersection of their valid-stage sets. Their union defines the odd limit symbol on compatible finite typed graphs of every size; it is not a total valuation symbol at any one finite stage. Every seed or contraction identity uses only finitely many such graphs and is inherited at a cutoff containing all of them. The bounded-weight witness remains a witness in these tables. No stage common to all cutoffs is selected. After fixing k, include its tests and choose the common projection data on the same filter, without redefining the intrinsic odd minimum. Stabilize the finite alphabet and its two projections in this way. Repeated packets of total label zero contain anchors for both seeds. Product-label contractions include all external incidences. Grounded Laplacian minors provide both the singular and varying address networks, including the singular-matrix case. Take the product of the two highest forms in the ordinary polynomial ring before Boolean reduction. An exponent two assigns one singular program to each factor; an exponent one assigns a varying program to one factor. The total degree forces the intended allocation and leaves coefficient one. Thus one finite graph gives units at every nonzero address for both symbols. Every contracted output is nonempty.

For completeness the address networks can be given by explicit matrices over $\mathbb{F}_2$. At a target $x_* \neq 0$, choose a dual basis λ_i with $\lambda_i(x_*) = 1$. Use three primary sites per group, with basis (e_i, u_i, v_i) and e_i the all-one physical vector. Give the e_i orthonormal pairing and on the remaining coordinates use

$$H(T, B) = \begin{pmatrix} 0 & T \\ T^{\mathsf{t}} & B \end{pmatrix}.$$

For the full grounded matrix Q, the columns $h_i = Qe_i$ are supported on group i. Attach them to terminals activated by λ_i. The row-sum identities then hold on every active group restriction. To attach a further column c to a pair activated by α, β, also attach it to an $\alpha + \beta$ terminal when $\alpha \neq \beta$. Its total added column is zero at every address, and that balancing terminal is absent at the target.

Cauchy–Binet applied to incidence matrices gives the constant coefficient $\det Q$ and pair-contraction coefficient $c_a^{\mathsf{t}} \operatorname{adj}(Q) c_b$. The nonzero minors are precisely forests with the indicated deleted roots. This proves the formula even for singular Q. Cycles have zero iterated difference: contract a spanning path and the closing edge becomes internal, changing neither product labels nor external incidences. A component of primary auxiliaries alone contracts to an isolated neutral vertex and vanishes.

For the singular program take $T = I + P$, with P cyclic, and let B have one diagonal entry one. A proper group restriction has invertible T; on all groups the radical is $\sum_i u_i$ and has dimension one. Thus $\operatorname{adj}(Q) = rr^{\mathsf{t}}$ there. A column dual to u_1 has pairing one with r, while the anchors have pairing zero. The program has constant zero and exactly the chosen pair difference in degree one at

x_*; elsewhere its constant is one. The same computation covers a single group.

For the varying program take $T = I + L$, where L is the consecutive lower shift, and $B_\varepsilon = \varepsilon E_{bb}$. All restrictions are invertible. The change in the upper-left inverse block is measured by $((T_J^{-1})_{b1})^2$, which is one exactly when the full path is present. Hence the difference of the two programs has the chosen degree-one difference only at x_* and degree at least two elsewhere; both have constant one everywhere.

Now expand the product of the two final evaluations and sum over the binary choices of the varying programs. A term using only a varying program's constants cancels. A singular program costs at least one degree in each factor at its target, and a surviving varying program costs at least one. The sum of these minimum costs is the total available degree of the two top forms. Thus every surviving term has exactly the intended cost at the intended address; higher and wrong-address contractions vanish. The remaining coefficient is the selected nonzero coefficient of the ordinary product of top forms, hence one. This proves that one choice works at all nonzero addresses. It also explains why Boolean reduction must follow, rather than precede, the top-form product.

For each chosen b, this construction has a finite budget W_b. Enlarge it to bound every graph product and squarefree output in its finite auxiliary tests, as well as every operator set. Contractions only decrease the number of vertices. Thus every final h_x lies in the tested range of (F.7); this enlargement changes no fixed constant. Only now take one actual theta stage in the finite intersection of the selected table sets through W_b+1, the valid-stage set for the full budget $|U| \le W_b$ with $R = 2$, and the required precision and finite exclusion sets. In particular (F.6) holds through the threshold T used for that budget. Realize its prescribed forward residue bits by distinct primes using CRT and Dirichlet; quadratic reciprocity supplies the reverse bits. There is no uniform bound on W_b, no intersection over all budgets, and no product-surjectivity assumption on the alphabets. The resulting one family satisfies (3.9), with constants fixed before b. The theta prime ℓ_0 is a coefficient prime and an exclusion, not an active relaxed Selmer prime or a moving arithmetic prime r_i.

G.5 One coordinate in the final contradiction

For this fixed K, (3.4), (F.7), and the even bound just proved give

$$2j_{\mathrm{rc}}(h_x, k) \le 4\omega(h_x) + C_J, \qquad C_J = 2C_o + C_{e,\mathrm{final}} - v_2 L_*(h_*) + C_{H,\mathrm{final}}$$

at every $x \neq 0$. The finite support comparison uses the same class, duality map, bases and coefficient stages. Its integral correction

$$\eta = \prod_q (1 + v_q(1 + g_q)), \qquad l_q/e_q^2 = 1 + 2v_q, \qquad \widetilde{u} = \eta u_{\mathrm{rel}}$$

is formed before characters. The actual integral determinant triangle gives $\widetilde{u}_x = \epsilon_x u_{\mathrm{prim},x}$ with $\epsilon_x \in \Lambda^\times$, rather than independently chosen units in $\mathcal{O}^\times$. Therefore the central identity is

$$\begin{aligned} v_2\widetilde{u}_x(0) = {} & 2j_{\mathrm{rc}}(h_x, k) - 4\omega(h_x) - s(h_x) - s(h_x k) \\ & - \sum_{p \mid k} t_p + \epsilon'_x, \qquad 0 \le t_p \le 2, \quad |\epsilon'_x| \le C_{\mathrm{cal}}. \end{aligned} \tag{F.8}$$

The nonnegative finite Sha terms give one upper bound $v_2\widetilde{u}_x(0) \le B$ for all $x \neq 0$, independent of b. At the unknown base no finiteness of Sha is assumed. The retained central class and corank-one argument instead give $\widetilde{u}_0(0) = 0$ if the base derivative vanishes.

The fixed-field construction of Section 9 has

$$c_{\mathrm{det}} = a + 4(h_K + 2c_E + 2t_K), \quad L_T(M) = 16(d_K + 5)M,$$
$$f_x\widetilde{u}_x = d_x, \qquad c = v_2 f_0(0) \le c_{\mathrm{det}}.$$

Choose $M > \max(c_{\det}, B + 2c_{\det}, 0)$, then $b > (L_T(M) + 2)(2^M - 1)$ large enough for the address construction, then its finite graph budget and actual theta stage M_θ. The residual comparison tests coefficients of f from $-L_T(M)$ through zero and only the constant coefficient of d. It gives $x \neq 0$ with

$$v_2(\widetilde{u}_x(0) - \widetilde{u}_0(0)) \geq M - 2c > B.$$

The estimate uses both $v_{\mathcal{O}}(f_x) \leq c$ and $\widetilde{u}_x \in \Lambda[1/2]$; neither a bound on the whole numerator nor a finite-Artin inverse of t is asserted. This contradicts (F.8) under the supposed base zero. The Rankin factorization and $L(E^k, 1) \neq 0$ give the asserted simple zero of E.

This completes the proof of the restricted converse, Theorem 5.3, from the displayed classical inputs and the arithmetic, coefficient and determinant constructions. The conversion from that converse to pole witnesses, the integer model, the natural compiler and finally the ordinary rational value is the separate argument in the main text.

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